API-STD-672

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Packaged, Integrally Geared Centrifugal Air Compressors for Petroleum, Chemical, and Gas Industry Services

API STANDARD 672
THIRD EDITION, SEPTEMBER 1996



American Petroleum Institute

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Packaged, Integrally Geared Centrifugal Air Compressors for Petroleum, Chemical, and Gas Industry Services

Manufacturing, Distribution and Marketing Department

API STANDARD 672
THIRD EDITION, SEPTEMBER 1996



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FOREWORD

This standard is based on the accumulated knowledge and experience of manufacturers and users of packaged, integrally geared centrifugal air compressors. The objective of this standard is to provide a purchase specification to facilitate the manufacture and procurement of these compressors for use in petroleum, chemical, and gas industry services.

The primary purpose of this standard is to establish minimum mechanical requirements. This limitation in scope is one of charter as opposed to interest and concern. Energy conservation is of concern and has become increasingly important in all aspects of equipment design, application, and operation. Thus, innovative energy-conserving approaches should be aggressively pursued by the manufacturer and the user during these steps. Alternative approaches that may result in improved energy utilization should be thoroughly investigated and brought forth. This is especially true of new equipment proposals, since the evaluation of purchase options will be based increasingly on total life costs as opposed to acquisition cost alone. Equipment manufacturers, in particular, are encouraged to suggest alternatives to those specified when such approaches achieve improved energy effectiveness and reduced total life costs without sacrifice of safety or reliability.

This standard requires the purchaser to specify certain details and features. Although it is recognized that the purchaser may desire to modify, delete, or amplify sections of this standard, it is strongly recommended that such modifications, deletions, and amplifications be made by supplementing this standard, rather than by rewriting or incorporating sections thereof into another complete standard.

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Asbestos is specified or referenced for certain components of the equipment described in some API standards. It has been of extreme usefulness in minimizing fire hazards associated with petroleum processing. It has also been a universal sealing material, compatible with most refining fluid services.

Certain serious adverse health effects are associated with asbestos, among them the serious and often fatal diseases of lung cancer, asbestosis, and mesothelioma (a cancer of the chest and abdominal linings). The degree of exposure to asbestos varies with the product and the work practices involved.

Consult the most recent edition of the Occupational Safety and Health Administration (OSHA), U.S. Department of Labor, Occupational Safety and Health Standard for Asbestos, Tremolite, Anthophyllite, and Actinolite, 29 *Code of Federal Regulations* Section 1910.1001; the U.S. Environmental Protection Agency, National Emission Standard for Asbestos, 40 *Code of Federal Regulations* Sections 61.140 through 61.156; and the U.S. Environmental Protection Agency (EPA) rule on labeling requirements and phased banning of asbestos products (Sections 763.160-179).

There are currently in use and under development a number of substitute materials to replace asbestos in certain applications. Manufacturers and users are encouraged to develop and use effective substitute materials that can meet the specifications for, and operating requirements of, the equipment to which they would apply.

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Packaged, Integrally Geared Centrifugal Air Compressors for Petroleum, Chemical, and Gas Industry Services

SECTION 1—GENERAL

1.1 Scope

This standard establishes the minimum requirements for constant-speed, packaged, integrally geared centrifugal air compressors, including their accessories. It may be applied for gas services other than air which are nonhazardous and nontoxic. This standard is not applicable to machines that develop a pressure rise of less than 0.35 bar (5.0 psi) above atmospheric pressure, which are classed as fans or blowers.

Note: A bullet (•) at the beginning of a paragraph indicates that either a decision by the purchaser is required or further information is to be provided by the purchaser. This information should be indicated on the data sheets (see Appendix A); otherwise, it should be stated in the quotation request or in the order.

1.2 Alternative Designs

The vendor may offer alternative designs. Equivalent metric dimensions, fasteners, and flanges may be substituted as mutually agreed upon by the purchaser and the vendor.

1.3 Conflicting Requirements

In case of conflict between this standard and the inquiry or order, the information included in the order shall govern.

1.4 Definition of Terms

Terms used in this standard are defined in 1.4.1 through 1.4.32.

1.4.1 The *alarm point* is a preset value of a parameter at which an alarm is actuated to warn of a condition that requires corrective action.

1.4.2 *Axially (horizontally)* split refers to casing joints that are parallel to the shaft centerline.

1.4.3 The *bearing housing* refers to all bearing enclosures, including the gear casing.

1.4.4 *Bull gear* refers to the low-speed rotor.

1.4.5 *Delivered flow* refers to the flow rate determined at the compressor discharge or after the discharge of the aftercooler when specified.

Note: When the flow is measured before the compressor inlet, it must be adjusted for the effects of aftercooler pressure drop, compressor seal losses, and interstage condensate removal.

1.4.6 A *diaphragm coupling* consists of one or more flexible elements that are attached to the outside diameter of one flange and transfer torque through the diaphragm to its inside-diameter attachment (a spacer or another flange).

1.4.7 A *disk coupling* consists of several flexible elements that are alternately attached with bolts to the opposite flanges.

1.4.8 *Flexible-element coupling* describes both disk and diaphragm couplings. A flexible-element coupling obtains its flexibility from the flexing of thin disks or diaphragms.

1.4.9 *Hydrodynamic bearings* are bearings that use the principles of hydrodynamic lubrication. Their surfaces are oriented so that relative motion forms an oil wedge to support the load without journal-to-bearing contact.

1.4.10 *Inlet volume* flow refers to the flow rate determined at the conditions of pressure, temperature, compressibility, and gas composition, including moisture, at the compressor inlet. Actual volume flow may be used to refer to flow at a number of locations and should, therefore, not be used interchangeably with inlet volume flow.

Note: To determine inlet volume flow, allowance must be made for pressure drop through the inlet filter/silencer and other equipment in vendor's scope of supply.

1.4.11 *Local* refers to a device mounted on or near the equipment or console.

1.4.12 *Material certificate of compliance* is a document by which the vendor certifies that the material represented has been produced and tested in accordance with the requirements of the basic material specification shown on the certificate.

1.4.13 *Maximum allowable temperature* is the maximum continuous temperature for which the manufacturer has designed the equipment (or any part to which the term is referred) when handling the specified fluid at the specified pressure.

1.4.14 *Maximum allowable working pressure* is the maximum continuous pressure for which the manufacturer has designed the equipment (or any part to which the term is referred) when handling the specified fluid at the specified temperature.

1.4.15 A *panel* is an enclosure used to mount, display, and protect gauges, switches, and other instruments.

1.4.16 *Pinion* refers to a high-speed rotor or rotors.

1.4.17 The *pressure casing* is the composite of all stationary pressure-containing parts of the unit, including all nozzles and other attached parts.

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1.4.18 *Pressure rise* to surge is the difference between the discharge pressure at the rated operating point and that at the surge point when the unit is operating at rated inlet conditions.

1.4.19 *Radially split* refers to casing joints that are transverse to the shaft centerline.

1.4.20 The *rated operating point* is the point at which the vendor compares the tested rated operating point to certify that the performance is within the tolerances stated in this standard.

1.4.21 *Rated speed* is the speed corresponding to the requirements of the rated operating point.

1.4.22 *Remote* refers to a device located away from the equipment or console, typically in a control house.

1.4.23 The *shutdown point* is a preset value of a parameter at which automatic or manual shutdown of the system is required.

1.4.24 *Standard flow* is the flow rate expressed in volume flow rate. ISO standard flow rate is cubic meters per hour or minute (m^3/h or m^3/min) at an absolute pressure of 1.013 bar, a temperature of 0°C and a relative humidity of 0 percent R.H. (Dry). U.S. customary units are standard cubic feet per minute (SCFM) or million standard cubic feet per day (MMSCFD) at an absolute pressure of 14.7 pounds per square inch, a temperature of 60°F and a relative humidity of 0 percent R.H. (Dry).

1.4.25 *Standby service* refers to a normally idle or idling piece of equipment that is capable of immediate automatic or manual start-up and continuous operation.

1.4.26 The *tested rated operating point* is the test point corrected to the site conditions that is equal to or greater than the rated operating point without violating the tolerances stated in this standard.

Note: Items that refer to the rated operating point will refer to the tested operating point after test (see Figure 1).

1.4.27 *Total indicated runout (TIR)*, also known as *total indicator reading*, is the runout of a diameter or face determined by measurement with a dial indicator. The indicator reading implies an out-of-squareness equal to the reading or an eccentricity equal to half the reading.

1.4.28 *Trip speed (revolutions per minute)* is the speed at which the independent emergency overspeed device operates to shut down a variable-speed prime mover (see Table 1).

1.4.29 *Turndown* is the percentage of change in capacity (referred to rated capacity) between the rated capacity and the surge point capacity at rated discharge pressure when the unit is operating at rated inlet conditions.

1.4.30 *Unit responsibility* refers to the responsibility for coordinating the technical aspects of the equipment and all auxiliary systems included in the scope of the order. It includes responsibility for reviewing such factors as the power requirements, speed, rotation, general arrangement, couplings, dynamics, noise, lubrication, sealing system, material test reports, instrumentation, piping, and testing of components.

1.4.31 *Vendor* is the agency that manufactures, sells, and provides service support for the equipment.

1.4.32 The use of the word *design* in any term (such as *design power*, *design pressure*, *design temperature*, or *design speed*) should be avoided in the purchaser's specifications. This terminology should be used only by the equipment designer and the manufacturer.

1.5 Referenced Publications

1.5.1 This standard makes reference to American standards. Other international or national standards may be used as mutually agreed between purchaser and vendor provided it can be shown that these other standards meet or exceed the American standards referenced.

1.5.2 The editions of the following standards, codes, and specifications that are in effect at the time of publication of this standard shall, to the extent specified herein, form a part of this standard. The applicability of changes in standards, codes, and specifications that occur after the inquiry shall be mutually agreed upon by the purchaser and the vendor.

AGMA[1]

2000	*Gear Classification and Inspection Handbook*
6011	*Specification for High Speed Helical Gear Units*

API

RP 520	*Sizing, Selection and Installation of Pressure-Relieving Devices in Refineries*, Part I, "Sizing and Selection," and Part II, "Installation"
Std 526	*Flanged Steel Safety-Relief Valves*
Std 541	*Form-Wound Squirrel-Cage Induction Motors—250 Horsepower and Larger*
Std 546	*Form-Wound Brushless Synchronous Motors—500 Horsepower and Larger*
Std 600	*Steel Gate Valves—Flanged and Buttwelding Ends*
Std 602	*Compact Steel Gate Valves*
Std 606	*Compact Carbon Steel Gate Valves—Extended Body*
Std 611	*General-Purpose Steam Turbines for Refinery Service*

[1]American Gear Manufacturers Association, 1500 King Street, Alexandria, Virginia 22314.

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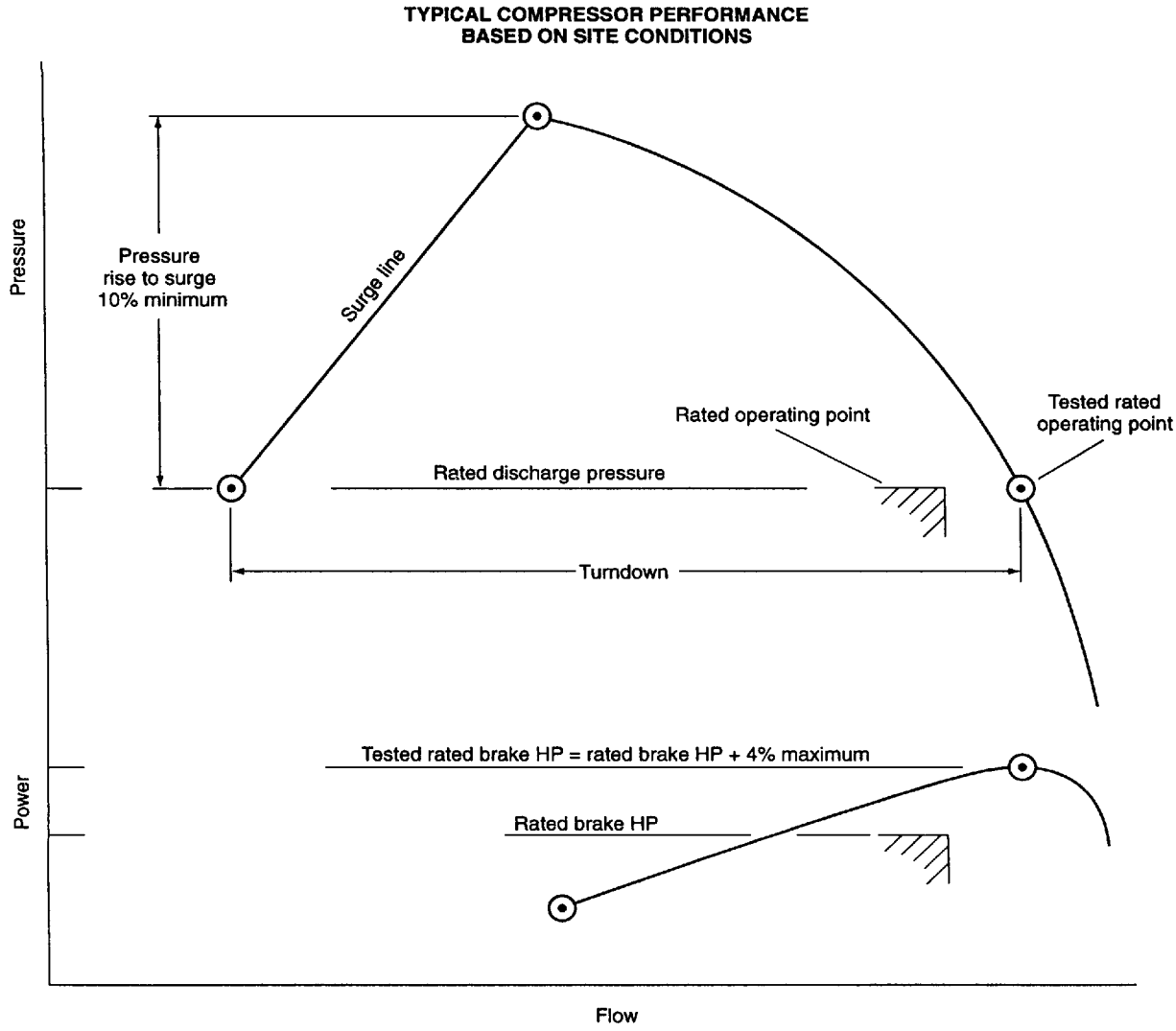


Figure 1—Illustration of Terms

Std 614 *Lubrication, Shaft-Sealing, and Control-Oil Systems for Special-Purpose Applications*
Std 670 *Vibration, Axial-Position, and Bearing-Temperature Monitoring Systems*
Std 671 *Special-Purpose Couplings for Refinery Service*
Std 677 *General-Purpose Gear Units for Refinery Service*
RP 684 *Tutorial on Rotor Dynamics, Torsional Analysis, and Rotor Balancing*

ASME[2]
B1.1 *Unified Inch Screw Threads (UN and UNR Thread Form)*
B1.20.1 *General Purpose (Inch) Pipe Threads*
B4.1 *Preferred Limits and Fits for Cylindrical Parts*
B16.1 *Cast Iron Pipe Flanges and Flanged Fittings*
B16.5 *Pipe Flanges and Flanged Fittings*
B16.11 *Forged Steel Fittings, Socket-Welding and Threaded*
B16.42 *Ductile Iron Pipe Flanges and Flanged Fittings, Class 150 and 300*
B16.47 *Large-Diameter Steel Flanges; NPS 26 through NPS 60*
B17.1 *Keys and Keyseats*
B31.3 *Chemical Plant and Petroleum Refinery Piping*
PTC 10 *Compressors and Exhausters*
Boiler and Pressure Vessel Code, Section II, "Material Specifications"; Section V, "Nondestructive Examination";

[2]American Society for Mechanical Engineers, 345 East 47th Street, New York, New York 10017.

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Table 1—Driver Trip Speeds

Driver	Trip Speed (Percent of Rated Speed)
Steam Turbine	
NEMA Class A[a]	115
NEMA Class B, C, D[a]	110
Gas turbine	115
Variable-speed motor	110
Reciprocating engine	110

[a]Indicates governor class as specified in NEMA SM 23.

Section VIII, "Pressure Vessels"; and Section IX, "Welding and Brazing Qualifications."

Y14.2M *Line Conventions and Lettering*

ASTM[3]

A 27 *Specification for Carbon Steel Castings for General Application*

A 48 *Specification for Gray Iron Castings*

A 53 *Specification for Welded and Seamless Zinc-Coated Black and Hot-Dipped Steel Pipe*

A 105 *Specification for Carbon Steel Forgings for Piping Components*

A 106 *Specification for Seamless Carbon Steel Pipe* for *High-Temperature Service*

A 108 *Specification for Standard Quality Cold-Finished Carbon Steel Bars*

A 120 *Zinc-Coated (Galvanized) Welded and Seamless Steel Pipe for Ordinary Uses*

A 181 *Specification for Carbon Steel Forgings for General Purpose Piping*

A 182 *Specification for Forged or Rolled Alloy Steel Pipe Flanges, Forged Fittings and Valves, and Parts for High- Temperature Services*

A 192 *Specification for Seamless Carbon Steel Boiler Tubes for High-Pressure Service*

A 193 *Specification for Alloy-Steel and Stainless* Steel *Bolting Materials for High-Temperature Service*

A 194 *Specification for Carbon and Alloy Steel Nuts for Bolts for High-Pressure and High-Temperature Service*

A 234 *Specification for Piping Fittings of Wrought Carbon Steel and Alloy Steel for Moderate and Elevated Temperatures*

A 269 *Specification for Seamless and Welded Austenitic Stainless Steel Tubing for General Service*

A 275 *Method for Magnetic Particle Examination of Steel Forgings*

A 278 *Specification for Gray Iron Castings for Pressure-Containing Parts for Temperatures up to 650°F*

A 283 *Specification for Low and Intermediate Tensile Strength Carbon Steel Plates*

A 307 *Specification for Carbon Steel Bolts and Studs 60,000 PSI Tensile Strength*

A 312 *Specification for Seamless and Welded Austenitic Stainless Steel Pipe*

A 320 *Specification for Alloy-Steel Bolting Materials for Low-Temperature Service*

A 338 *Specification for Malleable Iron Flanges, Pipe Fittings, and Valve Parts for Railroad, Marine, and Other Heavy Duty Services at Temperatures up to 650°F (345°C)*

A 355 *Specification for Steel Bars, Alloys, for Nitriding*

A 395 *Specification for Ferritic Ductile Iron Pressure-Retaining Castings for Use at Elevated Temperatures*

A 403 *Specification for Wrought Austenitic Stainless Steel Piping Fittings*

A 449 *Specification for Quenched and Tempered Steel Bolts and Studs*

A 498 *Specification for Seamless and Welded Carbon, Ferritic, and Austenitic Alloy Steel Heat-Exchanger Tubes with Integral Fins*

A 515 *Specification for Carbon Steel Pressure Vessels Plates for Intermediate and Higher-Temperature Service*

A 516 *Specification for Carbon Steel Pressure Vessel Plates for Moderate and Lower-Temperature Service*

A 536 *Specification for Ductile Iron Castings*

A 564 *Specification for Hot-Rolled and Cold-Finished Age-Hardening Stainless and Heat-Resisting Steel Bars, Wire, and Shapes*

A 574 *Specification for Alloy Steel Socket-Head Cap Screws*

B 26 *Specification for Aluminum-Alloy Sand Castings*

B 209 *Specification for Aluminum and Aluminum-Alloy Sheet and Plate*

E 94 *Standard Guide for Radiograph Testing*

E 125 *Standard Reference Photographs for Magnetic Particle Indication on Ferrous Castings*

E 142 *Standard Method for Controlling Quality of Radiographic Testing*

E 186 *Standard Reference Radiographs for Heavy Walled [(2 to 4.5 in.) (51 to 114 mm)] Steel Castings*

E 446 *Standard Reference Radiographs for Steel Castings up to 2 in. (51 mm) in Thickness*

E 527 *Standard Practice for Numbering Metals and Alloys*

E 709 *Standard Guide for Magnetic Particle Examination*

[3]American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.

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AWS[4]
- *D1.1* *Structural Welding Code-Steel*

IEEE[5]
- 841 *Standard for Petroleum and Chemical Industry Severe Duty TEFC Squirrel-Cage Induction Motors Up to and Including 500 HP*

ISA[6]
- S5.1 *Instrumentation Symbols and Identification*

ISO[7]
- 228-1 *Pipe Threads Where Pressure Tight Joints Are Not Made on the Threads—Part 1: Designations, Dimensions, and Tolerances.*
- 3448 *Standard Industrial Liquid Lubricants—ISO Viscosity Classification*

NACE[8]
- *Corrosion Engineers' Reference Book*

NEMA[9]
- 250 *Enclosures for Electrical Equipment (1000 Volts Maximum)*
- ICS 6 *Enclosures for Industrial Controls and Systems*
- SM 23 *Steam Turbines for Mechanical Drive Service*

NFPA[10]
- 70 *National Electrical Code*
- 496 *Standard for Purged and Pressurized Enclosures for Electrical Equipment*

OSHA[11]
- 29 *Code of Federal Regulations*, Section 1910

SAE[12]
- AMS-5353B *Steel Castings, Investment, Corrosion Resistant 16Cr 1.8Ni 0.08N, As Cast*
- J429 *Mechanical and Material Requirements for Externally Threaded Fasteners*

SSPC[13]
- SP6 *Surface Preparation Specification*

TEMA[14]
- *Standards of Tubular Exchanger Manufacturers Association*

1.5.3 The purchaser and the vendor shall mutually determine the measures that must be taken to comply with any governmental codes, regulations, ordinances, or rules that are applicable to the equipment.

1.5.4 It is the vendor's responsibility to invoke all applicable specifications to each subvendor.

SECTION 2—BASIC DESIGN

2.1 General

2.1.1 SERVICE LIFE

The equipment (including auxiliaries) covered by this standard shall be designed and constructed for a minimum service life of 20 years and at least 3 years of uninterrupted operation. It is recognized that this is a design criterion.

2.1.2 UNIT RESPONSIBILITY

The vendor shall assume unit responsibility for all equipment and auxiliary systems included in the scope of the order.

[4]American Welding Society, 550 N.W. LeJeune Road, Miami, Florida 33135.
[5]Institute of Electrical and Electronics Engineers, 345 East 47th Street, New York, New York 10017-2394.
[6]Instrument Society of America, 67 Alexander Drive, P.O. Box 12277, Research Triangle Park, North Carolina 27709.
[7]International Organization for Standardization. ISO publications are available from the American National Standards Institute, 11 West 42nd Street, New York, New York 10036.
[8]National Association of Corrosion Engineers, P.O. Box 218340, Houston, Texas 77218.
[9]National Electrical Manufacturers Association, 2101 L Street N.W., Washington, D.C.
[10]National Fire Protection Association, 1 Batterymarch Park, Quincy, Massachusetts 02269.
[11]Occupational Safety and Health Administration, U.S. Department of Labor, Washington, D.C. 20402.
[12]Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, Pennsylvania 15096.
[13]Steel Structures Painting Council, 4400 Fifth Avenue, Pittsburg, Pennsylvania 15213-2683.
[14]Tubular Exchange Manufacturers Association, 25 North Broadway, Tarrytown, New York 10591.

•2.1.3 RATED OPERATING POINT

The purchaser will specify the equipment's rated operating point.

•2.1.4 SOUND PRESSURE LEVEL

Control of the sound pressure level (SPL) of all equipment furnished shall be a joint effort of the purchaser and the vendor. The equipment furnished by the vendor shall conform to the maximum allowable SPL specified by the purchaser.

2.1.5 PACKAGED EQUIPMENT

The vendor shall provide, as a minimum, the equipment listed below (referred to herein as a *package*), packaged to meet the specified operating conditions:

a. A centrifugal compressor with integral speed-increasing gear unit.
b. Intercoolers, moisture separators, and V-notched gate drain valves.
c. An inlet throttle device (a valve or variable-inlet guide vanes).
• d. A driver (motor or turbine as specified).
e. Couplings and guards.
f. A lube-oil system.
g. A vibration monitoring system.
h. Controls and instrumentation.
i. An instrument and control panel.
j. A common baseplate.

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2.1.6 SHIPPED LOOSE EQUIPMENT

The vendor shall provide the accessory equipment listed below to meet the specified operating conditions. The equipment shall be either packaged or included within the scope of supply and shipped loose.

a. An aftercooler with moisture separator with a V-notched gate drain valve.
b. A discharge check valve (steel).
c. A discharge blowoff valve (steel).
d. An air intake filter-silencer.
e. A blowoff silencer.

• 2.1.7 STANDBY SERVICE

When standby service is specified, the vendor shall provide all necessary controls and protective systems to allow automatic and manual startup.

2.1.8 MECHANICAL DESIGN INTEGRITY

Equipment shall be designed to run to the trip speed and maximum allowable working pressure without damage.

• 2.1.9 ENVIRONMENTAL CONDITIONS

The purchaser will specify whether the installation is indoors (heated or unheated) or outdoors (with or without a roof), as well as the weather and environmental conditions in which the equipment must operate (including maximum and minimum temperatures, unusual humidity, and dusty or corrosive conditions).

2.1.10 COOLING WATER SYSTEMS

Unless otherwise specified, cooling water systems shall be designed for the following conditions:

Velocity over heat exchange surfaces	1.5–2.5 m/s	5–8 ft/s
Maximum allowable working pressure (MAWP)[a]	≥6.9 bar	≥100 psig
Test pressure (≥1.5 × MAWP)	≥10.4 bar	≥150 psig
Maximum pressure drop	1 bar	15 psi
Maximum inlet temperature	30°C	90°F
Maximum outlet temperature	50°C	120°F
Maximum temperature rise	20K	30°F
Minimum temperature rise	10K	20°F
Fouling factor on water side	$0.35 m^2 K/kW$	$0.002 hr ft^2 °F/Btu$
Shell corrosion allowance	3 mm	0.125 in.

Note: The vendor shall notify the purchaser if the criteria for minimum temperature rise and velocity over heat exchanger surfaces result in a conflict. The criterion for velocity over heat exchange surfaces is intended to minimize water-side fouling; the criterion for minimum temperature rise is intended to minimize the use of cooling water. The purchaser will approve the final selection. m = meter; / = per; s = second; ft = foot; psig = pounds per square inch; kW = kilowatt; hr = hour; Btu = British thermal unit; mm = millimeter; in. = inch.

[a]Gauge pressure.

Provision shall be made for complete venting and draining of the system or systems.

2.1.11 OIL RESERVOIRS AND HOUSINGS

Oil reservoirs and housings that enclose moving lubricated parts (such as bearings, shaft seals, highly polished parts), instruments, and control elements shall be designed to minimize contamination by moisture, dust, and other foreign matter during periods of operation and idleness.

2.1.12 PACKAGE ARRANGEMENT

The arrangement of the package (including piping coolers, pumps, and controls) shall provide adequate clearance areas and safe access for operation and maintenance.

• 2.1.13 MOTORS AND ELECTRICAL COMPONENTS

Motors, electrical components, and electrical installations shall be suitable for the area classification (class, group, and division or zone) specified by the purchaser and shall meet the requirements of NFPA 70, Articles 500, 501, 502, 504 and 505 as well as local codes specified and furnished by the purchaser.

2.1.14 PERFORMANCE CRITERIA

The package shall perform on the test stand and on its permanent foundation within the specified acceptance criteria. After installation, the performance of the package shall be the joint responsibility of the purchaser and the vendor.

• 2.1.15 PRELIMINARY REVIEW

Many factors (such as piping loads, alignment at operating conditions, supporting structure, handling during shipment, and handling and assembly at the site) may adversely affect site performance. To minimize the influence of these factors, the vendor shall review and comment on the purchaser's piping and foundation drawings. When specified, the vendor's representative shall do the following:

a. Observe a check of the piping performed by parting the flanges.
b. Be present during the initial alignment check.
c. Check alignment at the operating temperature.

2.1.16 SPARE PARTS

Spare parts for the package components shall meet all the criteria of this standard.

2.1.17 NOMENCLATURE GUIDE

A guide to the compressor nomenclature of this standard can be found in Appendix B.

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2.2 Integrally Geared Compressor

2.2.1 GENERAL

2.2.1.1 All equipment shall be designed to permit rapid and economical maintenance. Major parts such as casing components and bearing housings shall be designed (shouldered or cylindrically doweled) and manufactured to ensure accurate alignment on reassembly.

2.2.1.2 The compressor's aerodynamic performance shall be such that a continuously rising compressor total head curve is developed. The compressor total head curve shall be developed from the differential pressure measurement between the compressor inlet flange and the final-stage discharge flange.

2.2.1.3 The specified rated capacity at the rated discharge pressure shall be met or exceeded without exceeding the rated power value by more than 4 percent.

2.2.1.4 When the compressor is operating at rated operating conditions, the combined performance (impellers, diffusers, intercoolers, and so forth) shall provide a minimum of 10 percent pressure rise from rated capacity to surge while operating at rated speed and at the greatest specified operating ambient temperature (see Figure 1).

2.2.1.5 The use of tapped holes in pressure parts shall be minimized. To prevent leakage in pressure sections of casings, metal equal in thickness to at least half the nominal bolt diameter, in addition to the allowance for corrosion, shall be left around and below the bottom of drilled and tapped holes. The depth of the tapped holes shall be at least one and one-half times the stud diameter.

2.2.1.6 Studded connections shall be furnished with studs and nuts installed. Blind stud holes should be drilled only deep enough to allow a preferred tap depth of one and one-half times the major diameter of the stud; the first one and one-half threads at both ends of each stud shall be removed.

2.2.1.7 Bolting shall be furnished as specified in 2.2.1.7.1 through 2.2.1.7.5.

2.2.1.7.1 The details of threading shall conform to ASME B1.1.

2.2.1.7.2 Studs are preferred to cap screws.

2.2.1.7.3 Adequate clearance shall be provided at bolting locations to permit the use of socket or box wrenches.

2.2.1.7.4 Internal socket-type, slotted-nut, or spanner-type bolting shall not be used unless specifically approved by the purchaser.

2.2.1.7.5 Stud ASTM Grade marking shall be located on the nut end of the exposed stud end.

2.2.1.8 Jackscrews, guide rods, and cylindrical casing-alignment dowels shall be provided to facilitate disassembly and reassembly. When jackscrews are used as a means of parting contacting faces, one of the faces shall be relieved (counterbored or recessed) to prevent a leaking joint or improper fit caused by marring of the face. Guide rods shall be of sufficient length to prevent damage to the internals or casing studs by the casing during disassembly and reassembly. Lifting lugs or eyebolts shall be provided for lifting only the top half of the casing.

2.2.1.9 The machined finish of the mounting surface shall be 3–6 micrometers (125–250 microinches) arithmetic average roughness (R_a). Hold-down or foundation bolt holes shall be drilled perpendicular to the mounting surface or surfaces and spot faced to a diameter three times that of the hole.

2.2.2 PRESSURE CASINGS

2.2.2.1 The hoop-stress values used in casing design for any given material shall not exceed the maximum allowable values given for that material at the maximum allowable temperature in Section II, Part D of the ASME *Boiler and Pressure Vessel Code*. For cast materials, the factor specified in Section VIII, Division 1 of the ASME *Code* shall be applied. The thickness of the casing shall be suitable for the maximum working and test pressure and shall include a corrosion allowance of at least 3 millimeters (0.125 inch).

2.2.2.2 Casing supports and alignment bolts shall be rigid enough to permit the machine to be moved by the use of its lateral and axial jackscrews.

2.2.2.3 The equipment feet shall be provided with vertical jackscrews and shall be drilled with pilot holes that are accessible for use in final doweling.

2.2.3 CASING CONNECTIONS

• **2.2.3.1** The first-stage inlet and final-stage outlet connections shall be flanged or machined and studded, and oriented as specified. All interstage and final-stage outlet connections shall be suitable for the maximum allowable working pressure.

2.2.3.2 Connections welded to the casing shall meet the material requirements of the casing, including impact values, rather than the requirements of the connected piping. (see 2.2.10.4.6). All welding of connections shall be done before hydrostatic testing (see 4.3.2).

2.2.3.3 Casing openings for pipe connections shall be at least NPS 1/2 and shall be flanged or machined and studded. Where flanged or machined and studded openings are impractical, threaded openings in sizes NPS 1/2 through 1 1/2 are permissible. These threaded openings shall be installed as specified in 2.2.3.3.1 through 2.2.3.3.6.

2.2.3.3.1 Pipe nipples, preferably not more than 150 millimeters (6 inches) long, shall be screwed into threaded openings.

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2.2.3.3.2 Pipe nipples shall be a minimum of Schedule 160 seamless for NPS 1 and smaller and a minimum of Schedule 80 seamless for NPS 1½ and larger.

2.2.3.3.3 Pipe nipples shall be provided with welding-neck or socket-weld flanges.

2.2.3.3.4 Nipple and flange materials shall meet the requirements of 2.2.3.2.

2.2.3.3.5 Tapped openings and bosses for pipe threads shall conform to ASME B16.5.

2.2.3.3.6 Pipe threads shall be taper threads conforming to ASME B1.20.1.

2.2.3.4 Openings for NPS 1¼, 2½, 3½, 5, 7, and 9 shall not be used.

2.2.3.5 Tapped openings not connected to piping shall be plugged with solid, round-head steel plugs furnished in accordance with ASME B16.11. As a minimum, these plugs shall meet the material requirements of the casing. Plugs that may later require removal shall be of corrosion-resistant material. A lubricant that meets the proper temperature specification shall be used on all threaded connections. Tape shall not be applied to threads of plugs inserted into oil passages. Plastic plugs are not permitted.

2.2.3.6 Flanges shall conform to ASME B16.1, B16.5, B16.42, or B16.47 as applicable except as specified in 2.2.3.6.1 through 2.2.3.6.3. Class 200 and 400 flanges shall not be used.

2.2.3.6.1 Cast iron flanges shall be flat faced and shall have a minimum thickness of Class 250 per ASME B16.1 for NPS 8 and smaller.

2.2.3.6.2 Flat-faced flanges with full raised-face thickness are acceptable on casings other than cast iron.

2.2.3.6.3 Flanges that are thicker or have a larger outside diameter than required in ASME B16.5 or B16.47 are acceptable.

2.2.3.7 Connections other than those covered by 2.2.3.6 require the purchaser's approval. Unless otherwise specified, mating parts for these nonstandard flanges shall be furnished by the vendor.

2.2.3.8 Machined and studded connections shall conform to the facing and drilling requirements of ASME B16.1, B16.5, B16.42, or B16.47. Studs and nuts shall be furnished installed. The first one and one-half threads at both ends of each stud shall be removed.

2.2.3.9 All of the purchaser's connections shall be accessible for disassembly without moving either the integrally geared compressor or the driver.

2.2.4 EXTERNAL FORCES AND MOMENTS

2.2.4.1 The maximum allowable forces and moments that may be imposed on the package by the purchaser's piping, without the use of expansion joints, shall be stated in the proposal.

2.2.4.2 The maximum allowable forces and moments shall be shown on the outline drawing.

2.2.5 ROTATING ELEMENTS

2.2.5.1 Shafts

2.2.5.1.1 Shafts shall be forged or hot-rolled alloy steel.

2.2.5.1.2 Shafts shall be machined throughout their entire length. They shall have a surface finish of 0.8 micrometer (32 microinches) arithmetic average roughness (R_a) or better at bearing surfaces.

2.2.5.1.3 The rotor shaft sensing areas to be observed by radial vibration probes shall be concentric with the bearing journals. All shaft sensing areas shall be free from stencil and scribe marks or any other surface discontinuity, such as an oil hole or keyway, for a minimum of one probe tip diameter on each side of the probe. These areas shall not be metallized, sleeved, or plated. The final surface finish shall be a maximum of 0.8 micrometer (32 microinches) R_a, preferably obtained by honing or burnishing. These areas shall be demagnetized or otherwise treated so that the combined total electrical and mechanical runout shall not exceed 25 percent of the maximum allowed peak-to-peak vibration amplitude or 6 micrometer (0.25 mil), whichever is greater.

2.2.5.1.4 Chrome plating of the shaft at the journal area is unacceptable.

2.2.5.1.5 Keyways shall have fillet radii conforming to ASME B17.1.

2.2.5.2 Impellers

2.2.5.2.1 Impellers shall be stainless steel.

2.2.5.2.2 The vendor's proposal shall describe in detail the type of construction and the method of attachment to the shaft.

2.2.5.3 Gears

2.2.5.3.1 As a minimum, gears shall be manufactured to the tolerances specified in AGMA 2000, Quality Number 13.

2.2.5.3.2 The gear unit shall be rated in accordance with AGMA 6011 using minimum service factors of 1.4 for motor driven units and 1.6 for steam-turbine-driven units. The rating shall be based on the driver nameplate rating, including any service factor.

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2.2.5.3.3 Bull-gear and pinion hardness combinations shall be in accordance with the values recommended in AGMA 6011. Brinell hardness numbers of 275 and 320 or greater for bull gear and pinion, respectively, are preferred. The calculated values of gear-rated power, based on both tooth surface durability and tooth bending strength, shall be included in the vendor's proposal.

2.2.5.3.4 The tooth portion of pinions shall be integrally forged with their shaft.

2.2.5.3.5 The bull gear may be integral with or separate from its shaft. A separate gear shall be of forged or fabricated construction using a forged-steel rim and shall be assembled on the shaft with an interference fit.

2.2.6 SEALS AND SEALING SYSTEM

2.2.6.1 Air and oil shaft seals shall be provided to achieve the following:

a. Contain compressed air inside the compressor casings.
b. Prevent oil from entering the compressor casings and contaminating the compressed air.
c. Prevent oil from leaking out of the gear casing into the atmosphere.
d. Prevent atmospheric air from entering any gear or compressor casing that could allow contamination of the oil system or compressed air by dirt or moisture

An atmospheric space between the air and oil seals is preferred.

2.2.6.2 Unless otherwise specified, the sealing system shall be furnished complete with piping, switches, pressure gauges, filters, and necessary start-up interlocks. Each item shall be piped and valved to permit replacement while the compressor is operating. This system, including air consumption, shall be fully described in the proposal.

2.2.6.3 Seal operation shall be suitable for all specified operating conditions, including suction throttling, startup, shutdown, standby, and momentary surge. The type of standby operation shall be agreed upon by the purchaser and the vendor.

2.2.7 DYNAMICS

2.2.7.1 Critical Speeds

2.2.7.1.1 For a definition of critical speeds, refer to Appendix C.

2.2.7.1.2 Resonances of structural support systems may adversely affect the rotor vibration amplitude. Therefore, resonances of structural support systems that are within the vendor's scope of supply and that affect the rotor vibration amplitude shall not occur within the specified separation margins (see C.2.4, Appendix C) unless the resonances are critically damped.

2.2.7.1.3 The vendor shall determine that the drive-train (turbine, gear, motor, and the like) critical speeds (rotor lateral, system torsional, blading modes, and the like) will not excite any critical speed of the machinery being supplied and that the entire train is suitable for the rated speed, any starting-speed detent (hold-point) requirements of the train, and, when the driver is not a motor, any speed from rated speed to the trip speed. A list of all undesirable speeds from zero to trip shall be submitted to the purchaser for his review and included in the instruction manual for his guidance (see Appendix G, Item 34d).

2.2.7.2 Lateral Analysis

2.2.7.2.1 With the purchaser's approval, the vendor's standard critical speed values that have been analytically derived and proven by testing of previously manufactured compressors of the same frame size are acceptable.

2.2.7.2.2 When specified, the vendor shall provide a damped unbalanced response analysis for each machine to assure acceptable amplitudes of vibration at any speed from zero to trip.

2.2.7.2.3 When a damped unbalanced response analysis is required, it shall be conducted and confirmed by test stand data in accordance with Appendix C.

2.2.7.3 Torsional Analysis

•**2.2.7.3.1** When specified, the vendor shall perform a torsional vibration analysis of the complete coupled train and shall be responsible for directing the modifications necessary to meet the requirements of 2.2.7.3.2 through 2.2.7.3.5.

2.2.7.3.2 Excitations of undamped torsional natural frequencies may come from many sources, which should be considered in the analysis. These sources may include but are not limited to the following:

a. Gear phenomena such as unbalance and pitch line runout.
b. Startup conditions such as speed detents and other torsional oscillations.
c. Torsional transients such as start-ups of synchronous electric motors and transients due to generator phase-to-phase fault or phase-to-ground fault.
d. Torsional excitation resulting from drivers.
e. One and two times line frequency.
f. Running speeds.

2.2.7.3.3 The undamped torsional natural frequencies of the complete train shall be at least 10 percent above or 10 percent below any possible excitation frequency.

2.2.7.3.4 Torsional criticals at two or more times running speeds shall preferably be avoided or, in systems in which corresponding excitation frequencies occur, shall be shown

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to have no adverse effect. In addition to multiples of running speeds, torsional excitations that are not a function of running speeds or that are nonsynchronous in nature shall be considered in the torsional analysis when applicable and shall be shown to have no adverse effect. Identification of these frequencies shall be the mutual responsibility of the purchaser and the vendor.

2.2.7.3.5 When torsional resonances are calculated to fall within the margin specified in 2.2.7.3.3 (and the purchaser and the vendor have agreed that all efforts to remove the critical from within the limiting frequency range have been exhausted), a stress analysis shall be performed to demonstrate that the resonances have no adverse effect on the complete train. The acceptable criteria for this analysis shall be mutually agreed upon by the purchaser and vendor.

2.2.7.3.6 The vendor shall perform a transient torsional vibration analysis for synchronous-motor-driven units. The acceptance criteria for this analysis shall be mutually agreed upon by the purchaser and the vendor.

2.2.7.4 Vibration and Balancing

2.2.7.4.1 Major parts of the rotating elements, such as the shaft and impellers, shall be dynamically balanced. When a bare shaft with a single keyway is dynamically balanced, the keyway shall be filled with a fully crowned half-key. The initial balance correction to the bare shaft shall be recorded. A shaft with keyways 180 degrees apart but not in the same transverse plane shall also be filled as described in the preceding.

2.2.7.4.2 The rotating elements shall be multiplane dynamically balanced during assembly. This shall be accomplished after the addition of each major component. Balancing correction shall be applied only to the elements added. Balancing of impellers by welding is prohibited. Minor correction of other components may be required during the final trim balancing of the completely assembled element. In the sequential balancing process, any half-keys used in the balancing of the bare shaft (see 2.2.7.4.1) shall continue to be used until they are replaced with the final key and mating element. On rotors with single keyways, the keyway shall be filled with a fully crowned half-key. The weight of all half-keys used during final balancing of the assembled element shall be recorded on the residual unbalance work sheet (see Appendix D). The maximum allowable residual unbalance per plane (journal) shall be calculated as follows:

In SI units:

$$U_{max} = 6350W/N \text{ for } N \leq 25{,}000 \text{ rpm} \quad (1a)$$

$$U_{max} = 6350W/25{,}000 \text{ for } N > 25{,}000 \text{ rpm} \quad (1b)$$

In customary units:

$$U_{max} = 4W/N \text{ for } N \leq 25{,}000 \text{ rpm}$$

$$U_{max} = 4W/25{,}000 \text{ for } N > 25{,}000 \text{ rpm}$$

Where:

U_{max} = residual unbalance, in gram-millimeters (ounce-inches).
W = journal static weight load, in kilograms (pounds).
N = rated speed, in revolutions per minute (rpm).

Note: Balance tolerance above 25,000 rpm is based on an eccentricity of 0.25 micrometer (10 microinches) for each journal static weight load. Unbalance readings are measured at each journal-bearing position with no compensation to actual balance planes.

When spare rotating elements are supplied, they shall be dynamically balanced to the same tolerances as the main rotating elements.

• **2.2.7.4.3** When specified, after the final balancing of each assembled rotating element has been completed, a residual unbalance check shall be performed and recorded in accordance with the residual unbalance work sheet (see Appendix D).

2.2.7.4.4 During the shop test of the machine, assembled with the balanced rotors, operating at its rated speed, the peak-to-peak amplitude of unfiltered vibration in any plane, measured on the shaft adjacent and relative to each radial bearing, shall not exceed the following value or 50 micrometers (2.0 mils), whichever is less:

In SI units:

$$A = 25.4\sqrt{\frac{12,000}{N}} \quad (2)$$

In customary units,

$$A = \sqrt{\frac{12,000}{N}}$$

Where:

A = amplitude of unfiltered vibration, in micrometers (mil) true peak to peak.
N = rated speed, in revolutions per minute.

At any speed greater than the rated speed, up to and including the trip speed of the driver, the vibration shall not exceed 150 percent of the maximum value recorded at the rated speed. (For motor driven compressors, *trip speed* and rated *operating speed* are synonymous.)

Note: These limits are not to be confused with the limits specified in Section C.3 of Appendix C for shop verification of unbalanced response.

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2.2.7.4.5 Electrical and mechanical runout shall be determined and recorded by rolling the rotor in V blocks at the journal centerline while measuring runout with a noncontacting vibration probe and a dial indicator at the centerline of the probe location and one probe-tip diameter to either side.

2.2.7.4.6 Accurate records of electrical and mechanical runout for the full 360 degrees at each probe location shall be included in the mechanical test report.

2.2.7.4.7 If the vendor can demonstrate that electrical or mechanical runout is present, the demonstrated runout may be vectorially subtracted from the vibration signal measured on the factory test, up to a maximum of 25 percent of the allowable vibration level calculated from Equation 2 (2.2.7.4.4) or 6 micrometers (0.25 mil), whichever is greater.

2.2.8 BEARINGS AND BEARING HOUSINGS

2.2.8.1 General

2.2.8.1.1 Hydrodynamic radial and thrust bearings shall be provided.

2.2.8.1.2 Bearings shall be designed to prevent their installation backwards and/or upside down.

2.2.8.2 Radial Bearings

2.2.8.2.1 Radial bearings shall be designed for ease of assembly, precision bored and of the sleeve or pad type with steel-backed, babbitted replaceable liners, pads, or shells. These bearings shall be equipped with antirotation pins and shall be positively secured in the axial direction.

2.2.8.2.2 The bearing design shall suppress hydrodynamic instabilities and provide sufficient damping over the entire range of allowable bearing clearances to limit rotor vibration to the maximum specified amplitudes (see 2.2.7.4.4) while the equipment is operating loaded or unloaded at the rated operating speed.

2.2.8.3 Thrust Bearings

2.2.8.3.1 Thrust bearings shall be selected at no more than 50 percent of the bearing manufacturer's ultimate load rating. The ultimate load rating is the load that will produce the minimum acceptable oil-film thickness without inducing failure during continuous service or the load that will not exceed the creep-initiation or yield strength of the babbitt at the location of maximum temperature on the pad, whichever is less. In sizing thrust bearings, consideration shall be given to the following for each specific application:

a. The shaft speed.
b. The temperature of the bearing babbitt.
c. The deflection of the bearing pad.
d. The minimum oil-film thickness.
e. The feed rate, viscosity, and supply temperature of the oil.
f. The design configuration of the bearing.
g. The babbitt alloy.
h. The turbulence of the oil-film.

In addition to thrust from the rotor and any internal gear reactions due to the most extreme allowable conditions, the axial force transmitted through the flexible coupling shall be considered a part of the duty of any thrust bearing.

2.2.8.3.2 When gear-type couplings are applied, the external thrust force shall be calculated from the following formula:

In SI units:

$$F = \frac{(0.25)(9{,}550)\,P_r}{N_r\,D} \qquad (3)$$

In customary units:

$$F = \frac{(0.25)(63{,}000)\,P_r}{N_r\,D}$$

Where:

F = external force, in kilonewtons (pounds).
Pr = rated power, in kilowatts (horsepower).
Nr = rated speed, in revolutions per minute.
D = shaft diameter at the coupling, in millimeters (inches).

Note: Shaft diameter is an approximation of the coupling pitch radius.

2.2.8.3.3 Thrust forces for flexible-element couplings shall be calculated on the basis of the maximum allowable deflection permitted by the coupling manufacturer.

2.2.8.3.4 If two or more rotor thrust forces are to be carried by one thrust bearing (such as in a gear box), the resultant of the forces shall be used provided the directions of the forces make them numerically additive; otherwise, the largest of the forces shall be used.

2.2.8.3.5 Thrust bearings shall be steel backed, babbitted, and arranged for continuous pressurized lubrication. Integral thrust collars are preferred. When replaceable collars are furnished (for assembly and maintenance purposes), they shall be positively locked to the shaft to prevent fretting. The faces of the collar shall have a surface finish of not more than 0.4 micrometer (16 microinches) R_a, and the axial total indicator runout of either face shall not exceed 12 micrometers (0.0005 inch).

2.2.8.3.6 Thrust bearings shall be arranged to allow axial positioning of each rotor relative to the casing and setting of the bearings' clearance or preload.

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2.2.8.4 Bearing Housings

2.2.8.4.1 Bearing housings shall be arranged to minimize foaming. The drain system shall be adequate to maintain the oil and foam level below shaft end seals. The rise in oil temperature through the bearing and housings shall not exceed 30°C (50°F) under the most adverse specified operating conditions. The bearing outlet oil temperature shall not exceed 80°C (180°F). When the inlet oil temperature exceeds 50°C (120°F), special consideration shall be given to bearing design, oil flow, and allowable temperature rise. Oil outlets from thrust bearings shall be tangential and in the upper half of the control ring or, if control rings are not used, in the thrust-bearing cartridge. Oil connections on bearing housings shall be in accordance with 2.2.3.

2.2.8.4.2 Bearing housings shall be equipped with replaceable labyrinth-type end seals and deflectors where the shaft passes through the housing; lip-type seals shall not be used. The seals and deflectors shall be made of nonsparking materials. The design of the seals and deflectors shall effectively retain oil in the housing and prevent entry of foreign material into the housing.

2.2.8.4.3 Axially split bearing housings shall have a metal-to-metal split joint whose halves are located by means of cylindrical dowels.

• **2.2.8.4.4** When specified, thrust bearings and radial bearings shall be fitted with bearing-metal temperature sensors installed in accordance with API Standard 670.

2.2.9 LUBRICATION

2.2.9.1 Unless otherwise specified, bearings and bearing housings shall be arranged for hydrocarbon oil lubrication.

2.2.9.2 A pressurized oil system shall be furnished to supply oil at a suitable pressure or pressures, as applicable, to the following:

a. The bearings of the integrally geared compressor.
b. The spray nozzles for the gear teeth.
• c. The bearings of the driver when specified.

2.2.9.3 Where oil is supplied from a common system to two or more components of the package (such as a compressor, a gear, and a motor), the oil's characteristics will be specified on the data sheets by the purchaser on the basis of mutual agreement with all vendors supplying equipment served by the common oil system.

Note: The usual lubricant employed in a common oil system is a hydrocarbon oil that corresponds to ISO Grade 32, as specified in ISO 3448.

2.2.9.4 Any points that require grease lubrication shall have suitable extension lines to permit access during operation.

2.2.9.5 Pressure lubrication systems shall, as a minimum, consist of main and standby positive displacement oil pumps with suction strainers, supply-and-return system, an oil cooler, twin full-flow filters, a low lube-oil pressure shutdown switch, and other necessary instruments. (Refer to Figure 2 for a typical schematic drawing of a lube oil system.) The requirements of 2.2.9.5.1 through 2.2.9.5.9 shall apply.

2.2.9.5.1 Oil-containing pressure components shall be steel.

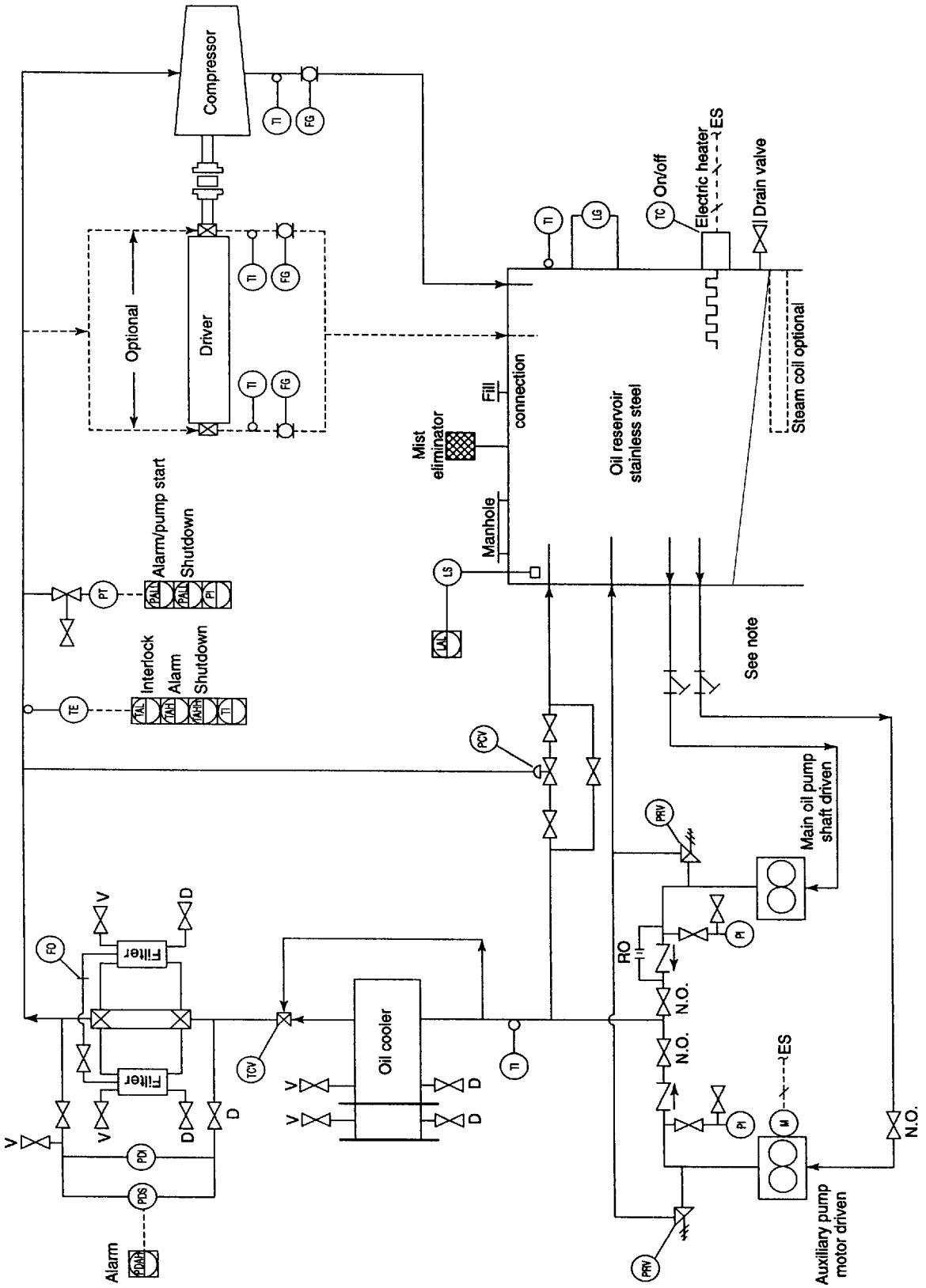
2.2.9.5.2 The main oil pump shall be shaft driven. Unless otherwise specified, the standby pump shall be automatically controlled and separately driven by an electric motor. Each pump shall be capable of supplying a minimum of 120 percent of the normal oil flow required by the equipment, at the relief-valve set pressure. The required pump shaft power shall not exceed the driver nameplate rating with the pump delivering lube oil at the relief-valve set pressure and with the oil at the maximum viscosity expected at the vendor's minimum allowable oil temperature. This temperature shall be stated in the vendor's proposal. Oil pumps shall be sized so that they can each deliver the required capacity when pumping lube oil at the highest temperature and corresponding minimum viscosity.

2.2.9.5.3 Individual external relief valves shall be provided for each positive displacement pump. These valves shall function only to protect the pumps from overpressure. Relief valves for all operating equipment shall meet the limiting relief valve requirements defined in API Recommended Practice 520, Parts I and II. Relief valves shall be set to operate at not more than the maximum allowable working pressure but not less than 110 percent of the rated pressure or the rated pressure plus 1.7 bar (25 pounds per square inch), whichever is greater. The vendor shall determine the sizes and set pressures of all relief valves related to the equipment.

2.2.9.5.4 A separate direct-acting back-pressure control valve with manual bypass shall be provided. The control valve shall be sized to maintain system pressure with both oil pumps operating.

2.2.9.5.5 An oil cooler shall be provided to maintain the oil supply temperature at or below 50°C (120°F). Unless otherwise specified, the cooler shall be of a water-cooled, shell-and-tube type. Shell-and-tube coolers shall have water on the tube side. A removable-bundle design is required for coolers with more than 0.45 square meter (5 square feet) of surface unless otherwise specified. Removable-bundle coolers shall be in accordance with TEMA Class C and shall be constructed with a removable channel cover. Tubes shall not have an outside diameter of less than 15 millimeters (5/8 inch), and the tube wall shall not have a thickness of less than 18 BWG [1.25 millimeters (0.049 inch)]. Unless otherwise specified, cooler shells, channels, and cov-

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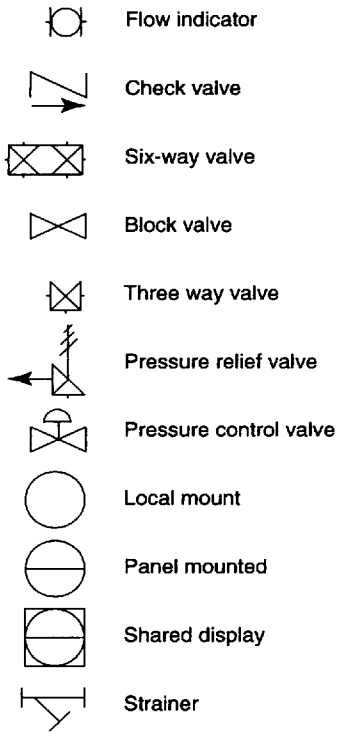


Note: A common suction line may be used but must be sized for the capability of two pumps.

Figure 2—Typical Pressure Lubrication System

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Symbols



Abbreviations

D	Drain
ES	Electronic supply
FG	Flow glass
FO	Flow restriction orifice
LG	Level gauge
LS	Level switch
LAL	Level alarm—low
N.O.	Normally open
PCV	Pressure control valve
PRV	Pressure relief valve
PDI	Pressure differential indicator
PDS	Pressure differential switch
PDAH	Pressure differential alarm—high
PI	Pressure indicator
PT	Pressure transmitter
PAL	Pressure alarm—low
PALL	Pressure alarm—low, low
RO	Restriction orifice
TE	Temperature element
TAL	Temperature alarm—low
TAH	Temperature alarm—high
TI	Temperature indicator
TC	Temperature controller
TCV	Temperature control valve
V	Vent

Note: The abbreviations and symbols in the key are used in the schematics presented in Figure 2. Some of the abbreviations and symbols listed are from ISA S5.1.

Figure 2—Typical Pressure Lubrication System (Continued)

ers shall be of steel; tube sheets shall be of brass; and tubes shall be of inhibited admiralty. U-bend tubes are not permitted. Each cooler shall be sized to accommodate the total cooling load. To prevent the oil from being contaminated if the cooler fails, the oil-side operating pressure shall be higher than the water-side operating pressure. Coolers shall be equipped with vent and drain connections on their oil and water sides. Internal oil coolers are not acceptable. A three-port oil temperature control valve with a manual override or bypass and flanged steel construction shall be provided in the oil circuit to maintain a constant oil temperature to the equipment.

•**2.2.9.5.6** Twin full-flow filters with replaceable elements and filtration of 10 microns nominal or finer shall be supplied. The filters shall be located downstream of the coolers and shall be supplied complete with a separate or integral continuous-flow transfer valve that provides tight shutoff to the idle filter. The system shall be designed to permit cartridge replacement and repressurizing during operation. The filter cases and heads shall be suitable for operation at a pressure not less than the pump relief valve setting. Filters that have covers weighing more than 15 kilograms (35 pounds) shall have cover lifters. Filters shall not be equipped with a relief valve or an automatic bypass. Filter cartridge materials shall be corrosion resistant. Metal-mesh or sintered-metal filter elements are not acceptable. Flow shall be from the outside toward the center of the filter cartridge. Stacked filter cartridge designs are prohibited. The pressure drop for clean filter elements shall not exceed 15 percent of the total allowable dirty pressure drop, or 0.34 bar (5 pounds per square inch) at an operating temperature of 38°C (100°F) and normal flow. Cartridges shall have a minimum collapsing differential pressure of 5.0 bar (70 pounds per square inch). The filters shall be equipped with a vent and clean- and dirty-side drain connections. Where a specific filter element is desired, the purchaser will specify the make and model number of the element.

Note: Micron particle size implies the diameter of a spherical bead; thus, a 10-micron particle is a sphere with a diameter of 10 microns. Within the element's recommended maximum pressure drop, 10 microns nominal implies that the efficiency of the filter on particles that are 10 microns or

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larger in diameter will be no less than 90 percent for the life of the element. Absolute micron particle ratings are different. A micron-absolute filter rating implies that no particles of the rating size or larger will pass; for example, a filter rating may be 10 microns nominal and 15 microns absolute.

• **2.2.9.5.7** When specified, a removable steam-heating element external to the oil reservoir or a thermostatically controlled electric immersion heater with a sheath of AISI[14] Standard Type 300 stainless steel shall be provided for heating the charge capacity of oil before start-up in cold weather. The heating device shall have sufficient capacity to heat the oil in the reservoir from the specified minimum site ambient temperature to the manufacturer's required start-up temperature within 12 hours. If an electric immersion heater is used, the watt density shall not exceed 2.0 watts per square centimeter (15 watts per square inch).

2.2.9.5.8 An austenitic stainless steel oil reservoir shall be supplied with the following characteristics and appendages:

a. The capacity to avoid frequent refilling, to provide adequate allowance for system rundown, and to settle moisture and foreign matter adequately.
b. Provisions to eliminate air and to minimize flotation of foreign matter to the pump suction.
c. Separate fill and vent connections, a level indicator with an armored gauge glass, and a breather suitable for outdoor use.
d. Sloped bottoms and connections for complete drainage.
e. Cleanout openings large enough to provide access for thorough cleaning
f. A retention time of at least 3 minutes, based on normal flow at pump normal capacity and total volume below the minimum operating level.
g. An electric heating element located below the pump suction loss level to assure the element operates submerged.

2.2.9.5.9 The oil system drain lines (as well as the reservoir vent and mist eliminator or both) shall be large enough to prevent emission of visible oil vapors above the vent. They shall also be large enough to provide adequate drainage with the air flow from the sealing system into the oil system, with seal clearances that are at least twice the design clearance. Drain eductor systems are not required.

2.2.10 MATERIALS

Note: This section is also applicable to package components other than the compressor.

2.2.10.1 General

2.2.10.1.1 Materials of construction shall be the manufacturer's standard for the specified operating conditions, except as required or prohibited by this standard (see 3.5 for requirements for auxiliary piping materials). The metallurgy of all major components shall be clearly stated in the vendor's proposal.

2.2.10.1.2 Appendix E lists material specifications that, when used with appropriate heat-treatment or impact-testing requirements or both, will ensure the supply of materials generally considered to be acceptable for major component parts.

2.2.10.1.3 Materials shall be identified in the proposal with their applicable ASTM, AISI, ASME, or SAE numbers, including the material grade (see Appendix E). When no such designation is available, the vendor's material specification giving physical properties, chemical composition, and test requirements shall be included in the proposal.

2.2.10.1.4 The vendor shall specify the ASTM optional tests and inspection procedures that may be necessary to ensure that materials are satisfactory for the service. Such tests and inspections shall be listed in the proposal. The purchaser may consider specifying additional tests and inspections, especially for materials used in critical components.

2.2.10.1.5 External parts that are subject to rotary or sliding motions (such as control linkage joints and adjusting mechanisms) shall be of corrosion-resistant materials suitable for the site environment and of sufficient hardness to resist wear.

2.2.10.1.6 Minor parts that are not identified (such as nuts, springs, washers, gaskets, and keys) shall have corrosion resistance at least equal to that of specified parts in the same environment.

• **2.2.10.1.7** The purchaser will specify any corrosive agents present in the process fluids and in the environment, including constituents that may cause stress corrosion cracking.

2.2.10.1.8 If parts exposed to conditions that promote intergranular corrosion are to be fabricated, hard faced, overlaid, or repaired by welding, they shall be made of low-carbon or stabilized grades of austenitic stainless steel.

Note: Overlays or hard surfaces that contain more than 0.10 percent carbon can sensitize both low-carbon and stabilized grades of austenitic stainless steel unless a buffer layer that is not sensitive to intergranular corrosion is applied.

2.2.10.1.9 Where mating parts such as studs and nuts of AISI Standard Type 300 stainless steel or materials with similar galling tendencies are used, they shall be lubricated with an antiseizure compound of the proper temperature specification.

Note: Torque loading values will differ considerably with and without an antiseizure compound.

2.2.10.1.10 When dissimilar materials with significantly different electrical potentials are placed in contact in the

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presence of an electrolytic solution, galvanic couples that can result in serious corrosion of the less noble material may be created. If such conditions exist, the purchaser and the vendor should select materials in accordance with the NACE *Corrosion Engineer's Reference Book.*

2.2.10.1.11 The use of ASTM A 515 steel is prohibited. Low-carbon steels can be notch sensitive and susceptible to brittle fracture at ambient or low temperatures. Therefore, only fully killed, normalized steels made to fine-grain practice are acceptable.

2.2.10.1.12 The minimum quality bolting material for pressure joints shall be carbon steel (ASTM A 307, Grade B) for cast iron casings and high-temperature alloy steel (ASTM A 193, Grade B7) for steel casings. Nuts shall conform to ASTM A 194, Grade 2H (or ASTM A 307, Grade B, case hardened, where space is limited). For temperatures below –30°C (–20°F), low-temperature bolting material in accordance with ASTM A 320 shall be used.

2.2.10.2 Castings

2.2.10.2.1 Castings shall be sound and free from porosity, hot tears, shrink holes, blow holes, cracks, scale, blisters, and similar injurious defects. Surfaces of castings shall be cleaned by sandblasting, shotblasting, chemical cleaning, or any other standard method. Mold-parting fins and remains of gates and risers shall be chipped, filed, or ground flush.

2.2.10.2.2 The use of chaplets in pressure castings shall be held to a minimum. The chaplets shall be clean and corrosion free (plating permitted) and of a composition compatible with the casting.

2.2.10.2.3 Ferrous castings shall not be repaired by welding, peening, plugging, burning in, or impregnating except as specified in 2.2.10.2.3.1 and 2.2.10.2.3.2.

2.2.10.2.3.1 Weldable grades of steel castings, may be repaired by welding, using a qualified welding procedure based on the requirements of Section VIII, Division 1, and Section IX of the ASME *Code*.

2.2.10.2.3.2 Cast gray iron or nodular iron may be repaired by plugging within the limits specified in ASTM A 278, A 395, or A 536. The holes drilled for plugs shall be carefully examined, using liquid penetrant, to ensure that all defective material has been removed. All repairs that are not covered by ASTM specifications shall be subject to the purchaser's approval.

2.2.10.2.4 Fully enclosed cored voids, including voids closed by plugging, are prohibited.

2.2.10.2.5 Nodular iron castings shall be produced in accordance with ASTM A 395.

2.2.10.3 Forgings

Unless otherwise agreed upon by the purchaser and the vendor, the forging material shall be selected from those listed in Appendix E.

2.2.10.4 Welding

2.2.10.4.1 Welding of piping and pressure-containing parts, as well as any dissimilar-metal welds and weld repairs, shall be performed and inspected by operators and procedures qualified in accordance with Section VIII, Division 1, and Section IX of the ASME *Code*. No weld repairs are permitted after final machining.

2.2.10.4.2 The vendor shall be responsible for the review of all repairs and repair welds to ensure that they are properly heat treated and nondestructively examined for soundness and compliance with the applicable qualified procedures. Repair welds shall be nondestructively tested by the same method used to detect the original flaw. As a minimum, the inspection shall be by the magnetic particle method in accordance with 4.2.2.4 for magnetic material and by the liquid penetrant method in accordance with 4.2.2.5 for nonmagnetic material.

2.2.10.4.3 Unless otherwise specified, documentation of major defects shall be submitted to the purchaser prior to any repairs being conducted and shall include the following:

a. Extent of the repair.
b. Location.
c. Size.
d. Welding procedure specification.
e. Detailed photographs of the defect prior to any preparatory work and after preparation but prior to the actual repair. If the location of the defect cannot be clearly defined by photographic means, the location shall be indicated on a sketch or drawing of the affected component.

Note: Repairs shall be considered major if any of the following conditions apply:

a Castings leak during hydrostatic testing.
b. The depth of the repair cavity prepared for welding exceeds 20 percent of the wall thickness or 25 millimeters (1 inch), whichever is smaller.
c. The surface area of the repair cavity exceeds 65 square centimeters (10 square inches).
d. The repair cavity penetrates the section thickness.
e. Repairs are to any rotating components.

2.2.10.4.4 Unless otherwise specified, all welding other than that covered by Section VIII, Division 1 of the ASME *Code* and ASME B31.3, such as welding on baseplates, non-pressure ducting, lagging, and control panels, shall be performed in accordance with AWS D1.1.

2.2.10.4.5 Pressure-containing casings made of wrought materials or combinations of wrought and cast materials shall conform to the conditions specified in 2.2.10.4.5.1 through 2.2.10.4.5.4.

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2.2.10.4.5.1 Plate edges shall be inspected by magnetic particle or liquid penetrant examination as required by Section VIII, Division 1, UG-93(d)(3), of the ASME *Code*.

2.2.10.4.5.2 Accessible surfaces of welds shall be inspected by magnetic particle or liquid penetrant examination after back chipping or gouging and again after post-weld heat treatment.

2.2.10.4.5.3 Pressure-containing welds, including welds of the case to horizontal- and vertical-joint flanges, shall be full-penetration welds.

2.2.10.4.5.4 Casings fabricated from materials that, according to Section VIII, Division 1 of the ASME *Code*, require post-weld heat treatment shall be heat treated regardless of thickness.

2.2.10.4.6 Connections welded to pressure casings shall be installed as specified in 2.2.10.4.6.1 through 2.2.10.4.6.5.

• **2.2.10.4.6.1** In addition to the requirements of 2.2.10.4.1, the purchaser may specify that 100-percent radiography, magnetic particle inspection, or liquid penetrant inspection of welds is required.

2.2.10.4.6.2 Auxiliary piping welded to chromium-molybdenum alloy steel or 12-percent chrome steel components shall be of the same material, except that chromium-molybdenum alloy steel pipe may be substituted for 12-percent chrome steel pipe.

2.2.10.4.6.3 When heat treatment is required, piping welds shall be made before the component is heat treated.

• **2.2.10.4.6.4** When specified, proposed connection designs shall be submitted to the purchaser for approval before fabrication. The drawings shall show weld designs, size, materials, and pre- and post-weld heat treatments.

2.2.10.4.6.5 All welds of materials not exempted according to Section VIII, Division 1, UCS-36 or UHA-32 of the ASME *Code* shall be heat treated in accordance with the methods described in Section VIII, Division 1, UW-40 of the ASME *Code*.

2.2.10.5 Low Temperature

2.2.10.5.1 To avoid brittle fracture during operation, maintenance, transportation, erection, and testing, good design practice shall be followed in the selection of fabrication methods, welding procedures, and materials for vendor-furnished steel pressure-retaining parts that may be subject to temperature below the ductile-brittle transition point.

2.2.10.5.2 All pressure-retaining steels applied at a specified minimum design metal temperature below –30°C (–20°F) require a Charpy V-notch impact test of the base metal and the weld joint unless they are exempt in accordance with the requirements of paragraph UHA-51 in Section VIII, Division 1 of the ASME *Code*. Impact test results shall meet the requirements of paragraph UG-84 of the ASME *Code*.

2.2.10.5.3 Carbon and low-alloy steel, pressure-retaining parts applied at a specified minimum design metal temperature between –30°C (–20°F) and 40°C (100°F) shall require impact testing in accordance with 2.2.10.5.3.1 and 2.2.10.5.3.2.

2.2.10.5.3.1 Impact-testing is not required for parts with a governing thickness of 25 millimeters (1 inch) or less.

2.2.10.5.3.2 Impact-testing exemptions for parts with governing thickness greater than 25 millimeters (1 inch) shall be established in accordance with paragraph UCS-66 in Section VIII, Division 1 of the ASME *Code*. Curve B shall be used for all carbon and low-alloy steel materials (including castings) which are not specifically listed for curves A, C, or D. Minimum design metal temperature without impact testing may be reduced as shown in Figure UCS-66.1. If the material is not exempt, Charpy V-notch impact test results shall meet the minimum impact energy requirements of paragraph UG-84 of the ASME *Code*.

Note 1: The governing thickness relates to pressure containment and excludes the following:

a. Structural support sections such as feet or lugs.

b. Sections with increased thickness required for rigidity to mitigate shaft deflection.

c. Structural sections required for attachment or inclusion of mechanical features such as jackets or seal chambers.

Note 2: Governing thickness considers only one-fourth of the nominal flange thickness, including parting flange thickness for axially split casings (in recognition that the predominant flange stress is not a membrane stress).

2.2.10.5.3.3 Unless otherwise specified, the minimum site ambient temperature shall be used to establish impact test requirements.

2.2.11 NAMEPLATES AND ROTATION ARROWS

2.2.11.1 A nameplate shall be securely attached at a readily visible location on the equipment and on any other major piece of auxiliary equipment

2.2.11.2 Rotation arrows shall be cast in or attached to each major item of rotating equipment at a readily visible location. Nameplates and rotation arrows (if attached) shall be of AISI Standard Type 300 stainless steel or of nickel-copper alloy (Monel or its equivalent). Attachment pins shall be of the same material. Welding is not permitted.

2.2.11.3 As a minimum, the following data shall be clearly stamped on the compressor nameplate:

a. Vendor's name.
b. Serial number.
c. Size and type.
d. Rated capacity.
e. Rated discharge pressure.
f. Maximum allowable working pressure.
g. Rated input speed.
h. Maximum allowable discharge temperature.
i. Purchaser's item number.

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SECTION 3—ACCESSORIES

3.1 Drivers

•**3.1.1** The type of driver will be specified. The driver shall be sized to meet the maximum specified operating conditions, including gear and coupling losses, and shall be in accordance with applicable specifications as stated in the inquiry and order. The driver shall be suitable for satisfactory operation under the utility and site conditions specified by the purchaser.

•**3.1.2** Anticipated process variations that may affect the sizing of the driver (such as changes in the pressure, temperature, relative humidity of the air, cooling water temperature, or properties of the fluid handled as well as special plant start-up conditions) will be specified by the purchaser.

•**3.1.3** The starting conditions for the driven equipment will be specified, and the starting method shall be mutually agreed upon by the purchaser and the vendor. The driver's starting-torque capabilities shall exceed the speed-torque requirements of the driven equipment.

3.1.4 Motor drives shall conform to API standard 541 or 546, as applicable. Motors that are below the power scope of API Standard 541 or 546 shall be in accordance with IEEE 841.

3.1.5 Electric motor drivers shall be sized in accordance with the more restrictive of the criteria given in 3.1.5.1 and 3.1.5.2.

3.1.5.1 The motor nameplate rating (exclusive of the service factor) shall be at least 110 percent of the maximum power required for all of the specified operating conditions. Consideration should be given to starting the compressor at the normal suction pressure. Equipment driven by induction motors shall be rated at the actual motor speed for the rated load condition.

•**3.1.5.2** The product of motor nameplate rating and service factor shall be no less than the power required (including losses from shaft-driven oil pump, coupling, and gear) when the compressor is operated unthrottled (inlet throttle device wide open) at the specified low-ambient operating conditions. The purchaser will specify on the data sheet the inlet air temperature and the inlet cooling water temperature to be used by the vendor in calculating the maximum unthrottled power.

Note: The specified inlet temperature is not necessarily the minimum ambient temperature.

•**3.1.6** The motor's starting-torque requirements shall be met at a specified reduced voltage, and the motor shall accelerate to full speed within a period of time agreed upon by the purchaser and the vendor.

Note: For most applications, the starting voltage is typically 80 percent of the normal voltage, and the time required to accelerate to full speed is generally less than 15 seconds.

3.1.7 Unless otherwise specified, steam turbine drivers shall conform to API Standard 611.

3.1.8 Steam turbines shall be sized in accordance with the more restrictive of the criteria given in 3.1.8.1 and 3.1.8.2.

3.1.8.1 Steam turbine drivers shall be sized to deliver continuously 110 percent of the maximum power required for all of the specified operating conditions while operating at a corresponding speed with the specified steam conditions.

•**3.1.8.2** The turbine nameplate rating shall not be less than the power required (including losses from shaft-driven oil pump, coupling, and gear) when the compressor is operated un-throttled (inlet throttle device wide open) at the specified low ambient operating conditions. The purchaser will specify on the data sheet the minimum inlet air temperature and the minimum inlet water temperature to be used by the vendor in calculating the maximum unthrottled power.

Note: The specified inlet temperature is not necessarily the minimum ambient temperature.

•**3.1.9** The steam turbine shall be equipped with a NEMA Class D constant speed governor as specified in NEMA SM 23. The purchaser will specify whether the governor is to be hydraulic or electronic.

3.1.10 Unless otherwise specified, external gears shall conform to API Standard 677.

3.1.11 For drivers that weigh more than 250 Kilograms (500 pounds), the equipment feet shall be provided with vertical jackscrews.

3.2 Couplings and Guards

3.2.1 Unless otherwise specified, flexible-element couplings and guards between drivers and driven equipment shall meet the requirements of 3.2.1.1 through 3.2.1.12.

3.2.1.1 The coupling shall be of the forged-steel, nonlubricated, flexible-element spacer type. The purchaser and the vendor shall mutually agree upon the make, model, type, and mounting arrangement of the coupling.

3.2.1.2 The coupling spacer shall be of sufficient length to permit removal of the coupling hubs without removal of the rotating elements.

3.2.1.3 The removable guard shall be of a sufficiently heavy and rigid design to avoid contact with the coupling or the shaft as a result of bodily contact. The guard shall be

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designed to prevent drawing oil out of adjacent bearing housings. The coupling guard shall be furnished and shall be in accordance with the Occupational Safety and Health Administration's requirements for coupling guards or that of applicable national, industrial or statutory bodies.

3.2.1.4 Limited-end-float couplings with maximum coupling-end floats as specified in Table 2 shall be supplied with horizontal-sleeve-bearing motors to prevent the motor rotor from rubbing stationary motor parts.

Note: Flexible element couplings with axial elastic-centering forces are usually satisfactory without these precautions.

Table 2—Maximum Coupling End Floats

Minimum Motor Rotor End Float		Maximum Coupling End Float	
(mm)	(in.)	(mm)	(in.)
6	0.250	2	0.09
13	0.500	5	0.19

Note; mm = millimeters; in. = inches.

3.2.1.5 For a flexible-element spacer-type coupling, the shaft thermal movements for all operating conditions shall be compatible with optimum deflection positions of the coupling. The shaft end clearances for assembly shall be specified by the vendor. The flexible elements shall be stainless steel or suitably coated to prevent corrosion.

3.2.1.6 Where necessary, the driver half of the coupling shall be furnished with an idling adapter to permit uncoupled operation.

3.2.1.7 Coupling hubs shall be mounted on the shafts with a cylindrical fit. Cylindrical fits shall conform to ASME B4.1, Class FN-1, shrink fit between shaft and coupling hub.

3.2.1.8 Coupling hubs shall be provided with tapped puller holes to aid removal.

3.2.1.9 The maximum coupling operating torque load shall be limited to 80 percent of the manufacturer's published rating. Couplings bored larger than the manufacturer's nominal rating shall be subject to the purchaser's approval.

3.2.1.10 Each coupling-to-shaft juncture shall be designed and manufactured to be capable of transmitting power at least equal to the power rating of the coupling.

•**3.2.1.11** When specified by the purchaser, couplings, coupling-to-shaft juncture, and guards shall conform to API Standard 671.

3.2.1.12 Information on shafts, keyway dimensions (if any), and shaft end movements due to end play and thermal effects shall be furnished to the vendor supplying the couplings.

3.3 Baseplate

3.3.1 The equipment feet and mating baseplate shall have machined surfaces.

3.3.2 Mounting pads shall be provided for the integrally geared compressor and all drive train components. The pads shall be larger than the foot of the mounted equipment to allow leveling of the baseplate without removal of the equipment. The pads shall be fully machined flat and parallel. Corresponding surfaces shall be in the same plane within 50 micrometers per meter (0.002 inch per foot) of distance between the pads. This requirement shall be met by supporting and clamping the baseplate at the foundation bolt holes only.

3.3.3 The baseplate shall not be drilled for equipment to be mounted by others. A baseplate shall be supplied with leveling screws. Baseplates that are to be grouted shall have 50-millimeter-radius (2-inch-radius) outside corners (in the plan view). Mounting surfaces that are not to be grouted shall be coated with a rust preventive immediately after machining.

3.3.4 Anchor bolt holes shall be drilled perpendicular to the mounting surfaces and spot faced to a diameter three times that of the hole.

3.3.5 Anchor bolts will be furnished by the purchaser.

3.3.6 Anchor bolts shall not be used to fasten machinery to the baseplate.

3.3.7 The vendor shall furnish stainless steel (AISI Standard Type 300) shim packs 3 millimeters to 6 millimeters (0.125 inch to 0.250 inch) thick under the driver feet and the mounting pads. All shim packs shall straddle the hold-down bolts and vertical jackscrews and be at least 5 millimeters (0.250 inch) larger on all sides than the footprint of the equipment.

Fasteners for attaching the components to the baseplate and jackscrews for leveling the baseplate shall be supplied by the vendor.

3.3.8 The integrally geared compressor shall be doweled, and the feet of the driver shall be drilled with pilot holes for use in final doweling.

3.3.9 When the supported driver weighs more then 250 kilograms (500 pounds), the mounting plates shall be furnished with axial and lateral jackscrews the same size as or larger than the vertical jackscrews. Vertical jackscrews in the equipment feet shall be arranged to prevent marring of shimming surfaces. The lugs holding these jackscrews shall

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be attached to the mounting plates so that the lugs do not interfere with the installation or removal of the equipment, jackscrews, or shims. If the equipment is too heavy to use jackscrews, other means shall be provided.

•**3.3.10** When specified, the baseplate shall be suitable for column mounting (that is, of sufficient rigidity to be supported at specified points) without continuous grouting under structural members. The baseplate design shall be mutually agreed upon by the purchaser and the vendor.

3.3.11 The baseplate shall be provided with lifting lugs for at least a four-point lift. Lifting the baseplate complete with all equipment mounted shall not permanently distort or otherwise damage the baseplate or the machinery mounted on it.

3.3.12 The bottom of the baseplate between structural members shall be open. When the baseplate is installed on a concrete foundation, accessibility shall be provided for grouting under all load-carrying structural members.

3.3.13 The mounting pads on the bottom of the baseplate shall be in one plane to permit use of a single-level foundation. When required by vendor design, subsoleplates shall be provided by the vendor.

3.3.14 The baseplate mounting pads shall be machined after the baseplate is fabricated.

3.3.15 Unless otherwise specified, the vendor shall commercially sand blast, in accordance with SSPC SP6, all grout contact surfaces of the baseplate, and coat those surfaces with inorganic zinc silicate in preparation for epoxy grouting.

3.4 Controls and Instrumentation

3.4.1 GENERAL

3.4.1.1 Unless otherwise specified, a microprocessor based control and instrumentation system suitable for outdoor installation shall be provided.

•**3.4.1.2** When specified, the microprocessor shall be capable of communicating with the purchaser's distributed control system (DCS).

3.4.1.3 All conduit shall be designed and installed so that it can be easily removed without damage and located so that it does not hamper removal of bearings, seals, or equipment internals.

3.4.1.4 Piping without breakout points or rigid conduit shall not be routed over the cases of horizontally split rotating machinery and shall not be routed over or in front of removable heads on vessels and exchangers, or where the piping impairs the functionality of inspection openings or panel doors.

3.4.2 CONTROL SYSTEMS

•**3.4.2.1** The purchaser will specify which of the following compressor capacity control modes shall be furnished by the vendor:

a. Capacity modulation (inlet throttle device or variable inlet guide vanes) used when constant discharge pressure to surge is required and when the system air demand is relatively constant.
b. Two-step (load-unload or intermittent) operation used when large variations in system air demand are expected and constant pressure is not mandatory.
c. Automatic dual control-capacity modulation plus intermittent mode control for smaller air demands.
d. An automatic start and automatic stop control.
e. Constant discharge pressure mode (blowoff to the atmosphere without inlet throttling).
f. Mass-flow control (used when the mass-flow rate is critical to the process).

3.4.2.2 When more than one mode is specified, a change to any mode shall be accomplished by a selector switch. If two compressors are to be operated in parallel, the control system proposed shall include all the necessary controls to permit the operation of both compressors on the same control mode or one on a separate control mode.

3.4.2.3 A compressor surge recognition and protection system shall be furnished.

3.4.2.4 An automatic driver-overload control system shall be included to permit continuous operation at minimum ambient air and water temperatures without exceeding the nameplate rating (excluding service factor, if any).

3.4.2.5 Manual override shall be provided to allow manual operation of the inlet throttle device and discharge blowoff valve. The system shall provide bumpless transfer from manual to automatic for smooth mode transfer.

3.4.2.6 To reduce driver load during startup of a motor-driven compressor, automatic unloading of the compressor by closing the inlet throttle device and opening the discharge blowoff valve shall be provided by the vendor. (It should be noted that an auxiliary source of control air or nitrogen may be required for initial startup.)

The control system shall provide a "soft" shutdown (or unloaded condition) in which the inlet valve is closed and the unloading valve is opened prior to terminating the power source to the driver except for an emergency stop. This feature allows for less severe surging when stopping the unit.

The control system shall also provide warning to the operator that a hot-start condition exists for the motor driver because the unit was shut down and an adequate cool-down time period has not occurred for restart of the driver.

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3.4.2.7 The system shall have the capability of recording data at multiple intervals just prior to an alarm or trip as an aid for troubleshooting surge and other compressor operational problems.

3.4.2.8 The vendor's proposal shall fully describe the design and operation of the control system, including surge recognition and protection and the electric power and operating air requirements.

3.4.3 INSTRUMENT AND CONTROL PANELS

3.4.3.1 A panel from which startup and shutdown can be accomplished shall be provided and shall include the following:

a. Components for control systems as defined in 3.4.2, exclusive of the inlet throttle device or variable inlet guide vanes and discharge blowoff valve.
b. A control mode selector keypad (or switch) (see 3.4.2.2).
c. Manual override and adjustment of control valves using the keypad (or switch) (see 3.4.2.5).
d. Digital-readout pressure measurements.
e. Digital-readout temperature measurements.
f. A display for annunciation (see 3.4.5.2).
g. Control devices for alarms and shutdowns.
h. A lamp alarm light and reset test push button.
i. The capability for starting and stopping the package from the control panel.
j. Vibration measurement and readout instruments (3.4.4.6).
k. Self-diagnostics to check that the microprocessor and all instruments are functioning properly.
l. Logging of the compressor's cumulative operating time.
m Logging of the total number of compressor starts.
n. Automatic shut off of the auxiliary oil pump after an adequate cool down of the oil system has been achieved.

• **3.4.3.2** The panel shall be fully enclosed. The panel enclosure shall be watertight, Type 4X per NEMA ICS 6 and NEMA 250 as a minimum, shall have a display visible in darkness/direct sunlight, and shall be mounted on the package baseplate. The purchaser will specify the area classification and the type of enclosure required. Some area classifications may require purging or alternative enclosures. Purging required to meet the area classification shall be provided in accordance with NFPA 496. The panel shall include the following:

a. Shielding of the devices in the panel from radio-frequency (RF) interference.
b. Cooling for devices within the panel if the temperature inside the panel exceeds the electronic hardware temperature rating [typically of concern for ambient conditions above 38°C (100°F)].
c. An interior panel heater for units when required by the ambient condition.
d. Appropriate pneumatic filter/moisture separators to allow devices such as current to pneumatic transducers to operate reliably (typically 5-micron or better filtration).

3.4.3.3 Panels shall be completely assembled, requiring only connection to the purchaser's external piping and wiring circuits. When more than one wiring point is required on a unit for control or instrumentation, the wiring to each switch or instrument shall be provided from a single terminal box with terminal posts mounted on the unit (or its base). Wiring shall be installed in metal conduits or enclosures. All leads and posts on terminal strips, switches, and instruments shall be tagged for identification.

3.4.4 INSTRUMENTATION

3.4.4.1 Temperature Gauges

3.4.4.1.1 Dial-type temperature gauges shall be heavy-duty and corrosion resistant. They shall be at least 125 millimeters (5 inches) in diameter and bimetallic type or liquid filled. Black printing on a white background is standard for gauges.

3.4.4.1.2 The sensing elements of temperature gauges shall be in the flowing fluid.

Note: This is particularly important for lines that may run partially full.

3.4.4.2 Thermowells

Temperature gauges that are located in pressurized or flooded lines shall be furnished with at least NPS 3/4 AISI Standard 300 stainless steel separable solid-bar thermowells.

3.4.4.3 Thermocouples and Resistance Temperature Detectors

Where practical, the design and location of thermocouples and resistance temperature detectors shall permit replacement while the unit is operating. The lead wires of thermocouples and resistance temperature detectors shall be installed as continuous leads between the thermowell or detector and the terminal box. Conduit runs from thermocouple heads to a pull box or boxes located on the baseplate shall be provided.

3.4.4.4 Pressure Gauges

Pressure gauges (not including built-in instrument air gauges) shall be furnished with AISI Standard 316 stainless steel bourdon tubes and stainless steel movements, 110-millimeter (4-1/2-inch) dials, and NPS 1/2 male alloy steel connections. Black printing on a white background is standard for gauges. When specified, liquid-filled gauges shall be furnished in locations subject to vibration. Gauge ranges

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shall preferably be selected so that the normal operating pressure is at the middle of the gauge's range. In no case, however, shall the maximum reading on the dial be less than the applicable relief valve setting plus 10 percent. Each pressure gauge shall be provided with a device such as a disk insert or blowout back designed to relieve excess case pressure.

3.4.4.5 Pressure Transducers

Transducers shall have local zeroing and span adjustment features to aid in calibration.

3.4.4.6 Vibration and Position Detectors

•**3.4.4.6.1** Each high-speed rotor and, when specified, the driver bearings, shall be provided with a vibration-monitoring system consisting of the following: (a) single radially oriented noncontacting shaft vibration sensing probe, (b) an oscillator-demodulator, and (c) a readout instrument. For single-impeller rotors, the probe shall be located at the bearing adjacent to the impeller. The vendor shall include with his proposal a statement listing whether x and y radial probes can be mounted adjacent to each impeller. The monitoring instrument shall be mounted in the instrument and control panel.

•**3.4.4.6.2** When specified, vibration and axial position transducers shall be supplied, installed, and calibrated in accordance with API Standard 670.

•**3.4.4.6.3** When specified, vibration and axial position monitors shall be supplied, installed, and calibrated in accordance with API Standard 670.

•**3.4.4.6.4** When specified, a bearing-temperature monitor shall be supplied and calibrated in accordance with API standard 670.

3.4.4.7 Solenoid Valves

Solenoid-operated valves shall be used only in clean, dry, instrument-air service; shall have Class F insulation or better; and shall have a continuous service rating. When required for other services, the solenoid shall act as a pilot valve to pneumatic valves, hydraulic valves, and the like.

3.4.4.8 Relief Valves

3.4.4.8.1 The vendor shall furnish the relief valves that are to be installed on equipment or in piping that the vendor is supplying. Other relief valves will be furnished by the purchaser. Relief valves for all operating equipment shall meet the limiting relief-valve requirements defined in API Recommended Practice 520, Parts I and II, and in API Standard 526. The vendor shall determine the size and set pressure of all relief valves related to the equipment. The vendor's quotation shall list all relief valves and shall clearly indicate those to be furnished by the vendor. Relief-valve settings, including accumulation, shall take into consideration all possible types of equipment failure and the protection of piping systems.

3.4.4.8.2 Unless otherwise specified, relief valves shall have steel bodies.

•**3.4.4.8.3** When specified, thermal relief valves shall be provided for components that may be blocked in by isolation valves.

3.4.4.9 Flow indicators

3.4.4.9.1 Flow indicators shall be furnished in the atmospheric oil-drain return lines.

3.4.4.9.2 Unless otherwise specified, the flow indicator shall be the bull's-eye type, and shall have a steel body.

3.4.4.9.3 To facilitate viewing of the flow of oil through the line, each flow indicator should be installed with its bull's-eye glass in a vertical plane. The diameter of the bull's-eye shall be at least one half the inside diameter of the oil pipe and shall clearly show the minimum oil flow.

3.4.5 ALARMS AND SHUTDOWNS

3.4.5.1 General

Switches, control devices, and annunciator display units shall be furnished as specified by purchaser and mounted by the vendor and an shall include those listed in Table 3 as a minimum. The alarm setting shall precede the shutdown setting. Program logic shall distinguish between a shutdown device and alarm device such that failure of a shutdown device will not allow operation of the compressor until the device problem is corrected; whereas, failure of an alarm device will cause an alarm condition but will allow continued operation of the compressor.

•3.4.5.2 Annunciator

The vendor shall furnish a first-out annunciator. The annunciator shall contain approximately 25 percent spare points and, when specified, shall be arranged for purging. Connections shall be provided for actuation of a remote signal when any function alarms or trips. The sequence of operation shall be as specified in 3.4.5.2.1 through 3.4.5.2.7.

3.4.5.2.1 Alarm indication shall consist of the flashing of a light and the sounding of an audible device.

3.4.5.2.2 The alarm condition shall be acknowledged by operating an alarm-silencing button via the keypad or a switch common to all alarm functions.

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Table 3—Conditions Requiring Alarms and Shutdowns

Condition	Alarm	Shutdown
High vibration of compressor	X	X
High last-stage air temperature (inlet)	X	X
Low lube-oil pressure	X	X
High oil-supply temperature	X	X
High oil filter differential pressure	X	
Low sealing-system pressure	X	[d]
Operation of the standby oil pump	X	
Low-lube level in reservoir[a]	X	
High inlet-air filter differential pressure	X	
High vibration of driver[b]	X	X
Panel purge[c]	X	X
Surge recognition	X	
Permissive start contact	[d]	

[a]With oil heater cutout.
[b]If specified.
[c]If required.
[d]Separate pilot-light indication.

3.4.5.2.3 When the alarm is acknowledged, the audible device shall be silenced, but the light shall remain steadily lit as long as the alarm condition exists. The annunciator shall be capable of indicating a new alarm (with a flashing light and sounding horn) if another function reaches an alarm condition, even if the previous alarm condition has been acknowledged but still exists.

3.4.5.2.4 When measurement levels return to normal, the annunciation lights and system shall return to normal condition after being acknowledged and reset through a keypad or switch.

If more than one alarm or trip have occurred at the same time, the reset operation must be performed for each alarm or trip condition until all alarms and trips have been acknowledged and reset; at which time the panel will return to normal.

3.4.5.2.5 The control system shall maintain a chronological record of the alarms and trips. The panel shall have the capability of storing all operational parameters related to the chronological alarm or trips in a battery-backed nonvolatile memory which can be printed out for review.

3.4.5.2.6 Alarm and shutdown set points shall have default values set by the vendor. These values can be changed by the operator and will be stored in a nonerasable memory. A user-defined password or key will be required to change the setpoint.

3.4.5.2.7 Connections shall be provided for a common remote alarm and a common remote shutdown indication when any of the switches or locally displayed compressor alarms operate.

3.4.5.3 Alarm and Trip Devices

3.4.5.3.1 Each alarm device and each shutdown device shall be furnished in a separate housing located to facilitate inspection and maintenance. Where switches are specified, hermetically sealed, single-pole, double-throw switches with a minimum capacity of 5 amperes at 120 volts AC shall be used. Mercury switches shall not be used.

3.4.5.3.2 Unless otherwise specified, electric switches that open (deenergize) to alarm and close (energize) to trip shall be furnished by the vendor.

3.4.5.3.3 Alarm and trip switch settings shall not be adjustable from outside the housing. Alarm and trip switches shall be arranged to permit testing of the control circuit, including when possible the actuating element, without interfering with normal operation of the equipment. The vendor shall provide a clearly visible light on the panel to indicate when trip circuits are in a test bypass mode. Unless otherwise specified, shutdown systems shall be provided with switches or another suitable means to permit testing without shutting down the unit.

3.4.5.3.4 Pressure-sensing elements shall be of AISI Standard 300 stainless steel. Low-pressure alarms, which are activated by falling pressure, shall be equipped with a valved bleed or vent connection to allow controlled depressurizing so that the operator can note the alarm set pressure on the associated pressure gauge. High-pressure alarms, which are activated by rising pressure, shall be equipped with valved test connections so that a portable test pump can be used to raise the pressure.

3.4.5.3.5 The vendor shall furnish with the proposal a complete description of the alarm and shutdown facilities to be provided.

•3.4.5.4 Replacement of Instruments and Controls

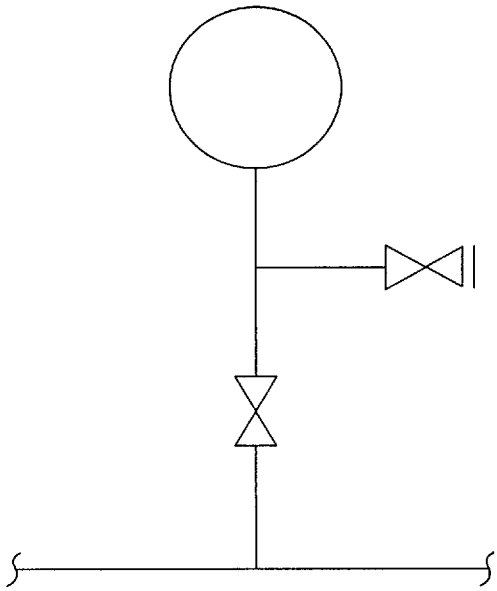
Unless otherwise specified, all instruments and controls other than shutdown sensing devices shall be installed with sufficient valving to permit their replacement while the system is in operation (see Figure 3). When shutoff valves are specified for shutdown sensing devices, the vendor shall provide a means of locking the valves in the open position.

3.4.5.5 Housing for Arcing-Type Switches

Particular attention is called to the requirements of 2.1.13 concerning the characteristics of housings for arcing-type switches outlined in the applicable codes.

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Notes:
1. If approved by the purchaser, a combination block-and-bleed guage valve may be substituted for individual block and vent-bleed valves. Block and vent-bleed valves shall be omitted for instruments in shutdown service.
2. See Figure 2 for a key to the symbols used in Figure 3.

Figure 3—Instrument Piping and Valving Details

3.4.6 ELECTRICAL SYSTEMS

•**3.4.6.1** The characteristics of electrical power supplies for motors, heaters, and instrumentation will be specified. A pilot light shall be provided on the incoming side of each supply circuit to indicate that the circuit is energized. The pilot lights shall be installed on the control panels.

•**3.4.6.2** Electrical equipment located on the unit or on any separate panel shall be suitable for the hazard classification specified. Electrical starting and supervisory controls may be either AC or DC.

3.4.6.3 Power and control wiring within the confines of the baseplate shall be resistant to oil, heat, moisture, and abrasion. Stranded conductors shall be used within the confines of the baseplate and in other areas subject to vibration. Measurement and remote-control panel wiring may be solid conductor. Thermoplastic insulation shall be used and shall be covered by a neoprene or equal sheath for abrasion protection. Wiring shall be suitable for the environmental temperatures specified.

3.4.6.4 Unless otherwise specified, all leads on terminal strips, switches, and instruments shall be permanently tagged for identification. All terminal boards in junction boxes and control panel shall have at least 20 percent spare terminal points.

3.4.6.5 To facilitate maintenance, liberal clearances shall be provided for all energized parts (such as terminal blocks and relays) on equipment. The clearances required for 600-volt service shall also be provided for lower voltages. To guard against accidental contact, enclosures shall be provided for all energized parts.

•**3.4.6.6** Electrical materials including insulation shall be corrosion resistant and nonhygroscopic insofar as is possible. When specified for tropical location, materials shall be given the treatments specified in 3.4.6.6.1 and 3.4.6.6.2.

3.4.6.6.1 Parts (such as coils and windings) shall be protected from fungus attack.

3.4.6.6.2 Unpainted surfaces shall be protected from corrosion by plating or another suitable coating.

3.4.6.7 Control, instrumentation, and power wiring (including temperature element leads) within the limits of the baseplate shall be installed in rigid metallic conduits and boxes, properly bracketed to minimize vibration, and isolated or shielded to prevent interference between voltage levels. Conduits may terminate (and in case of temperature element, heads shall terminate) with a flexible metallic conduit long enough to permit access to the unit for maintenance without removal of the conduit. If temperature element heads will be exposed to temperatures above 60°C (140°F), a 20-millimeter (3/4-inch) bronze hose with four-wall interlocking construction and joints with packed-on (heatproof) couplings shall be used.

3.4.6.8 For Division 2 location, flexible metallic conduits shall have a liquid tight thermosetting or thermoplastic outer jacket approved fittings. For Division 1 locations, an NFPA-approved connector shall be provided.

3.4.6.9 No terminal blocks shall be located in wire-ways. The terminals shall be straight-through compression type with shrouded screws (dead front) and center tapping for test purposes. Terminal block connections shall be single level (not tiered). The panel shall contain two 6-millimeter × 25-millimeter (1/4 inch × 1-inch) minimum bare soft copper, solid bar grounding buses. One shall be used for a signal ground, the other an equipment ground bus. The instrument case shall not be grounded through the steel of the panel.

3.4.6.10 Internal vibration probe or thermocouple leads exposed to lube-oil turbulence shall be sufficiently anchored to prevent fatigue failures due to excessive movement.

3.4.6.11 AC and DC circuits shall be clearly labeled, connected to separate terminal blocks, and isolated from each other.

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3.5 Piping

3.5.1 GENERAL

3.5.1.1 Piping design and joint fabrication, examination, and inspection shall be in accordance with ASME B31.3.

3.5.1.2 Auxiliary systems are defined as piping systems that are in the following services:

a. Instrument and control air.
b. Lubricating oil.
c. Cooling water.
d. Sealing fluid.
e. Drains.
f. Interstage piping.

Note: Casing connections are discussed in 2.2.3.

3.5.1.3 Piping systems shall include piping, isolating valves, control valves, relief valves, pressure reducers, orifices, temperature gauges and thermowells, pressure gauges, sight flow indicators, and all related vents and drains.

3.5.1.4 The vendor shall furnish all piping systems including mounted appurtenances located within the confines of the package. Each piping system requiring external connections shall terminate with flanged single-supply and single-return connections at the edge of the package. It is not necessary to provide flanged connections for instrument tubing systems. The purchaser will furnish only interconnecting piping between equipment groupings and off-base facilities.

3.5.1.5 The design of piping systems shall achieve the following:

a. Proper support and protection to prevent damage from vibration or from shipment, operation, and maintenance.
b. Proper flexibility and normal accessibility for operation, maintenance, and thorough cleaning.
c. Installation in a neat and orderly arrangement adapted to the contour of the package without obstructing access openings.
d. Elimination of air pockets in the oil system by the use of valved vents or nonaccumulating piping arrangements.
e. Complete drainage through low points without disassembly of piping.

3.5.1.6 Piping shall preferably be fabricated by bending and welding to minimize the use of flanges and fittings. Welded flanges are permitted only at equipment connections, at the edge of any base, and for ease of maintenance. The use of flanges at other points is permitted only with the purchaser's specific approval. Other than tees and reducers, welded fittings are permitted only to facilitate pipe layout in congested areas. Threaded connections shall be held to a minimum. Pipe bushings, unions, and couplings shall not be used.

3.5.1.7 Pipe threads shall be taper threads in accordance with ASME B1.20.1. Alternately, pipe threads in accordance with ISO 228, Part 1 are acceptable when required for compliance with local standards. Flanges shall be in accordance with ASME B16.5. For socket-welded construction, a 1.5-millimeter ($^1/_{16}$ inch) gap shall be left between the pipe end and the bottom of the socket.

3.5.1.8 Threaded connections for oil service shall be seal welded; however, seal welding is not permitted on cast iron equipment, on instruments, or on places disassembly is required for maintenance. Seal-welded joints shall be in accordance with ASME B31.3.

3.5.1.9 Connections, piping, valves, and fitting that are NPS $1^1/_4$, $2^1/_2$, $3^1/_2$, 5, 7, and 9 shall not be used.

3.5.1.10 The minimum requirement for piping material shall be as specified by Tables 4A, 4B, and 4C.

3.5.1.11 Piping wall thickness shall conform to the minimum requirement of Table 5A. Where space does not permit the use of NPS $^1/_2$, $^3/_4$, or 1 pipe, seamless tubing may be furnished in accordance with Table 5B. Stainless steel

Table 4A—Minimum Requirements for Piping Materials: Instrument and Control Sealing Air

Material	Requirements
Pipe	ASTM A 312 Type 304 or 316 stainless steel[a]
Tubing	ASTM A 269 Type 304 or 316 stainless steel
Valves	Manufacturer's standard tube end or instrument valve, stainless steel, or plated carbon steel
Pipe fittings	ASTM A 182 F304 forged Class 3000 SW, TE ASTM A 403 WP304 wrought butt weld.
Tube fittings	Stainless steel (manufacturer's standard with purchaser's approval).
Flanges	Stainless steel

[a]Seamless per ASTM A 312; schedule 10S and 40S may be electric fusion welded per ASTM A 312.

Table 4B—Minimum Requirements for Piping Materials: Cooling Water

Material	Requirements
Pipe	ASTM A 120 galvanized
Tubing	ASTM A 192 carbon steel or ASTM A 269 stainless steel
Valves	Class 200, bronze TE
Pipe fittings	Class 150 ASTM A 338 galvanized malleable iron
Tube fittings	Manufacturer's standard (with purchaser's approval
Flanges	Carbon steel

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Table 4C—Minimum Requirements for Piping Materials: Oil Piping

Material	Requirements
Pipe	ASTM A 312 Type 304 or 316 stainless steel[a b]
Tubing	ASTM A 269 Type 304 or 316 stainless steel
Valves	Class 150 flanged, Class 800 SW, TE carbon steel[c]
Pipe fittings	ASTM A 182 F304 forged Class 3000 SW, TE ASTM A 403 WP304 wrought butt weld[bd]
Tube fittings	Stainless steel (manufacturer's standard with purchaser's approval)
Flanges	Carbon steel slip-on or stainless steel weld neck or slip-on[e]

[a]Seamless per ASTM A 312; schedule 10S and 40S may be electric fusion welded per ASTM A 312.
[b]When approved by the purchaser for oil service, seamless carbon steel pipe per A 106, or A 192 may be used with Class 3000 SWE or TE forged steel pipe fittings per ASTM A 105 or wrought butt-welded fittings per ASTM A 234.
[c]Valves shall meet the applicable portions of API Standards 600, 602, or 606; gate and globe valves shall have bolted bonnets and bolted glands; valves greater than NPS 2 for ANSI Class 600 and lower shall be flanged.
[d]Threaded fittings shall be seal welded per 3.5.1.8.
[e]Carbon steel flanges per ASTM A 105 or ASTM A 181 (Class 150 and 300 only) or Type 304 stainless steel flanges per ASTM A 182 shall be used.

Table 5A—Minimum Pipe Wall Thicknesses

Material	Nominal Pipe Size (inches)	Minimum Schedule
Carbon steel	≤1 1/2	80
Carbon steel	2 to 8	40
Carbon steel	>8	20
Stainless steel	≤1	80
Stainless steel	1 1/2 to 3	40S
Stainless steel	≥4	10S

Table 5B—Minimum Tubing Wall Thicknesses

	Minimum Wall Thickness	
Nominal Tubing Size[a]	Inches	Millimeters
6.3 (1/4[b])	0.035	0.89
9.4 (3/8[b])	0.035	0.89
12.5 (1/2)	0.065	1.65
19.0 (3/4)	0.095	2.41
25.0 (1)	0.109	2.76

[a]The tubing size is the outside diameter in millimeters (inches).
[b]The sizes 1/4 and 3/8 are permitted for instrument and control air only.

fittings shall be furnished with stainless steel tubing. The make and model of fittings shall be subject to purchaser's approval.

3.5.1.12 For other than instrument and control air service, the minimum size of any connection or piping shall be NPS 1/2.

3.5.1.13 Piping systems furnished by the vendor shall be fabricated, installed in the shop, and properly supported. Bolt holes for flanged connections shall straddle lines parallel to the main horizontal or vertical centerline of the equipment.

3.5.1.14 Steel flanges mating with iron compressor flanges shall be flat faced.

3.5.1.15 Welding shall be performed by operators and procedures qualified in accordance with Section IX of the ASME *Code*.

3.5.1.16 Valves shall meet the applicable portions of API Standards 600, 602 and 606. Gate and globe valves shall have bolted bonnets and bolted glands. Valves over NPS 2 shall be flanged. Butterfly valves shall not be used unless approved by the purchaser.

3.5.1.17 Block valves shall be supplied with nominal 13 Cr stainless steel trim.

3.5.1.18 The bolting for pressure joints, valves, and piping shall be based on the actual bolting temperature as defined in ANSI B31.3. As a minimum, pressure bolting shall be ASTM A 193, Grade B7 with ASTM A 194, Grade 2H nuts per the requirements of ASME B1.1.

3.5.1.19 Pipe flange gaskets shall be nonasbestos flat gaskets up to and including ANSI Class 300 pressure ratings. For ANSI Class 600 and higher ratings, spiral-wound gaskets with nonasbestos filler, an external centering ring, and AISI Standard Type 304 or 316 windings shall be supplied.

3.5.1.20 Combination stop/check valves shall not be used.

3.5.1.21 Gaskets and packing for flanges, valves, and other components shall not contain asbestos.

3.5.2 OIL PIPING

3.5.2.1 Oil piping, tubing, and fittings (excluding slip-on flanges), shall be stainless steel (see Table 4C).

3.5.2.2 Oil drains shall be sized to run no more than half full when flowing at a velocity of 0.3 meter per second (1 foot per second) and shall be arranged to ensure good drainage (recognizing the possibility of foaming conditions). Horizontal runs shall slope continuously, at least 40 millimeters per meter (1/2 inch per foot), toward the reservoir. If possible, laterals (not more than one in any transverse plane) should enter drain headers at 45-degree angles in the direction of the flow.

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3.5.2.3 Nonconsumable backup rings and sleeve-type joints shall not be used. Pressure piping downstream of oil filters shall be free from internal obstructions or pockets that could accumulate dirt. Pipe joints downstream of the oil filter (filter to bearing housing) shall be butt-welded. Piping joints in return lines and upstream of the filter (reservoir to filter) may be socket welded. Threaded connections shall be used for instrument connections and where tubing is used.

• **3.5.2.4** Heads of oil-actuated control valves shall be vented back to the reservoir. When specified, instrument sensing lines to safety switches shall have a continuous through flow of oil.

3.5.2.5 The root pass of all butt welds on stainless steel pipe shall be made by tungsten inert-gas arc welding. Filler passes may be made by tungsten inert-gas arc welding or by the shielded metal arc process.

3.5.2.6 All piping components such as flanges, valves, control valve bodies or heads, and relief valves shall be made of steel.

3.5.3 INSTRUMENT PIPING

3.5.3.1 Instrument and control air piping tubing and fittings shall be AISI Series 300 stainless steel. Tubing thicknesses shall meet the requirements of Table 5B. Figure 3 shows instruments piping and valving details.

3.5.3.2 Bleeder valves are required between instruments and their isolating valves, except in instrument-air service, where combinations of isolating and bleeder valves may be used.

3.5.3.3 Bleed valves provided at instruments may be the manufacturer's standard bleed fitting. Where test valves are provided, bleed valves may be omitted.

3.5.3.4 Instrument valves for oil service located in sensing lines downstream of a primary service block valve may be barstock instrument valves, provided the valve is protected against accidental disassembly. Valves shall be stainless steel or carbon steel, with a corrosion resistant plating and stainless-steel stem.

3.5.3.5 Tubing valves shall be manufacturer's standard tube end or instrument valve, stainless steel, as approved by the purchaser.

3.6 Intercoolers and Aftercoolers

3.6.1 The vendor shall furnish an intercooler between each compression stage, and unless otherwise specified, an aftercooler after the final compression stage. The intercooler shall have continuous-bleed V-Notch gate valves to permit removal of liquid.

3.6.2 Intercooler and aftercooler shall be designed and constructed in accordance with Section VIII, Division 1, of the ASME *Code*.

3.6.3 Unless otherwise specified, intercoolers and aftercooler shall be of the water-cooled shell and tube type with water on the tube side. A removable-bundle design is required. Removable-bundle coolers shall be in accordance with TEMA Class C and shall be constructed with a removable channel cover. Tubes shall not have an outside diameter of less than 15 millimeters (5/8 inch), and the tube wall shall not have a thickness of less than 18 BWG, 1.25 millimeters (0.049 inch). Unless otherwise specified, cooler shells, channels, and covers shall be of steel; tube sheet shall be of brass; and tubes shall be of inhibited admiralty. U-bend tubes are not permitted. Each cooler shall be sized to accommodate the total cooling load.

Note: Some plant locations may require consideration of alternative materials to combat atmospheric corrosion.

3.6.4 Unless otherwise approved by the purchaser, intercoolers and aftercoolers shall be constructed and arranged to allow removal of tube bundles without dismantling piping or compressor components.

3.6.5 The vendor shall include in the proposal complete details of any proposed air-cooled cooler.

3.6.6 Unless otherwise specified, air-cooled heat exchangers used for intercoolers shall have automatic temperature control. This control may be accomplished by means of louvers, variable-speed fans, variable-pitch fans, bypass valves, or any combination of these. Proposed control systems will be approved by the purchaser.

3.7 Inlet Air Filter/Silencer

The vendor shall furnish a dry-type multistage, high-efficiency air intake filter-silencer suitable for mounting outdoors. This filter-silencer shall be provided with the following:

a. A differential pressure switch and gauge.
b. Filter portion designed such that the first-stage (prefilter) elements may be changed while the unit is operating.
c. Weather hood or louvers.
d. Clean pressure drop across the filter elements which shall not exceed 5.0 millibar (2 inches) water gauge.
e. Removal of a minimum of 97 percent of particle sizes 1 micron or larger over the inlet capacity range.

• Note 1: Many configurations and arrangements are available. Thus, the purchaser will specify any required specific features, such as special coatings or construction to avoid corrosion.

Note 2: The filter-silencer may be elevated some distance above the compressor for certain plant locations subject to unusual conditions such as sand storms. Inlet piping between filter-silencer and the compressor shall be of corrosion-resistant material.

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3.8 Discharge Blowoff Silencer

3.8.1 The vendor shall furnish a flanged discharge blowoff silencer.

3.8.2 Silencer construction shall be suitable for service in an unprotected location. The silencer preferably should be located immediately downstream of the discharge blowoff valve.

Note: Refer to data sheet for maximum allowable noise level requirements.

3.9 Special Tools

3.9.1 When special tools and fixtures are required to disassemble, assemble, or maintain the unit, they shall be included in the quotation and furnished as part of the initial supply of the machine. For multiple-unit installations, the requirements for quantities of special tools and fixtures shall be mutually agreed upon by the purchaser and the vendor. These or similar special tools shall be used during shop assembly and post-test disassembly of the equipment.

3.9.2 When special tools are provided, they shall be packaged in a separate, rugged metal box or boxes and shall be marked "special tools for (tag/item number)." Each tool shall be stamped or tagged to indicate its intended use.

SECTION 4—INSPECTION, TESTING, AND PREPARATION FOR SHIPMENT

4.1 General

• **4.1.1** The purchaser will specify the extent of participation in the inspection and testing, including any witnessed or observed points and the amount of advance notification required.

4.1.1.1 *Witnessed* means that a hold shall be applied to the production schedule and that the inspection or test shall be carried out with the purchaser or his representative in attendance. For mechanical running or performance tests, this requires written notification of a successful preliminary test.

4.1.1.2 *Observed* means that the purchaser shall be notified of the timing of the inspection or test; however, the inspection or test shall be performed as scheduled, and if the purchaser or his representative is not present, the vendor shall proceed to the next step. (The purchaser should expect to be in the factory longer than for a witnessed test.)

4.1.2 The vendor shall notify subvendors of the purchaser's inspection and testing requirements.

4.1.3 The purchaser's representative shall have entry to all vendor and subvendor plants where manufacturing, testing, or inspection of the equipment is in progress.

4.1.4 Equipment for the specified inspection and tests shall be provided by the vendor.

• **4.1.5** When specified, the purchaser's representative, the vendor's representative, or both shall indicate compliance in accordance with the inspector's checklist (see Appendix G) by initialing, dating, and submitting the completed checklist to the purchaser before shipment.

4.2 Inspection

4.2.1 GENERAL

4.2.1.1 The vendor shall keep the following data available for at least 20 years for examination or reproduction by the purchaser or his representative upon request:

a. Material certificates of compliance for shafts, pinions, bull gears, and impellers.

• b. Test data to verify that the requirements of the specification have been met, such as hydro tests, mechanical performance tests, and any optional tests specified by the purchaser.

c. Results of documented tests and inspections, including fully identified records of all heat treatment and nondestructive examinations.

• d. When specified, final assembly maintenance and running clearances.

4.2.1.2 Pressure-containing parts shall not be painted until the specified inspection of the parts is completed.

4.2.1.3 The purchaser's representative shall have access to the vendor's quality program for review.

4.2.2 MATERIAL INSPECTION

4.2.2.1 General

Unless otherwise specified, the criteria in 4.2.2.2 through 4.2.2.5 shall apply when radiographic, ultrasonic, magnetic particle, or liquid penetrant inspection of welds or materials is required. Cast iron may be inspected in accordance with 4.2.2.4 and 4.2.2.5. Welds, cast steel, and wrought material may be inspected in accordance with 4.2.2.2 through 4.2.2.5.

4.2.2.2 Radiography

4.2.2.2.1 Radiography shall be in accordance with ASTM E 94 and ASTM E 142.

4.2.2.2.2 The acceptance standard used for welded fabrications shall be Section VIII, Division 1, UW-51 (100 Percent) and UW-52 (Spot) of the ASME *Code*. The acceptance standard used for castings shall be Section VIII, Division 1, Appendix 7 of the ASME *Code*.

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4.2.2.3 Ultrasonic Inspection

4.2.2.3.1 Ultrasonic inspection shall be in accordance with Section V, Articles 5 and 23 of the ASME *Code*.

4.2.2.3.2 The acceptance standard used for welded fabrications shall be Section VIII, Division 1, Appendix 12 of the ASME *Code*. The acceptance standard used for castings shall be Section VIII, Division 1, Appendix 7 of the ASME *Code*.

4.2.2.4 Magnetic Particle Inspection

4.2.2.4.1 Both wet and dry methods of magnetic particle inspection shall be in accordance with ASTM E 709.

4.2.2.4.2 The acceptance standard used for welded fabrications shall be Section VIII, Division 1, Appendix 6 and Section V, Article 25 of the ASME *Code*. The acceptability of defects in castings shall be based on a comparison with the photographs in ASTM E 125. For each type of defect, the degree of severity shall not exceed the limits specified in Table 6.

Table 6—Maximum Severity of Defects in Castings

Type	Defect	Maximum Severity Level
I	Linear discontinuity	1
II	Shrinkage	2
III	Inclusions	2
IV	Chills and Chaplets	1
V	Porosity	1
VI	Welds	1

4.2.2.5 Liquid Penetrant Inspection

4.2.2.5.1 Liquid penetrant inspection shall be in accordance with Section V, Article 6 of the ASME *Code*.

4.2.2.5.2 The acceptance standard used for welded fabrications shall be Section VIII, Division 1, Appendix 8, and Section V, Article 24 of the ASME *Code*. The acceptance standard used for castings shall be Section VIII, Division 1, Appendix 7 of the ASME *Code*.

Note: Regardless of the generalized limits in 4.2.2, it shall be the vendor's responsibility tc review the design limits of the equipment in the event that more stringent requirements are necessary. Defects that exceed the limits imposed in 4.2.2 shall be removed to meet the quality standards cited, as determined by the inspection method specified.

4.2.2.6 Cast impellers are to be 100 percent radiographed and inspected. The radiographs, when compared with the standard reference radiographs within ASTM E446 for steel castings up to 50 millimeters (2 inches) in thickness or standard reference radiographs for heavy walled 50 millimeters to 100 millimeters (2 inches to 4 inches) steel castings within ASTM E186, shall show a casting quality equal to or better than Severity Level 2 for Categories A, B, and C (Types 1–4). Defects per categories D, E, and F are unacceptable. The methods of radiographic examination shall be in accordance with ASTM E 94 and ASTM E142.

4.2.2.7 Inspection of the impeller is required following overspeed testing per 4.3.3.

4.2.2.8 All bull gear and pinion teeth shall be 100 percent magnetic particle inspected in accordance with ASTM A 275. Cracks are not acceptable. *Linear indications* due to metallic inclusions larger than 1.5 millimeters (0.06 inch) located in the tooth flanks or roots shall be reported to the purchaser for disposition. Linear indications are defined as indications whose length is at least three times the width. Acceptance or rejection shall be decided on a case-by-case basis and shall be mutually agreed upon by the purchaser and the vendor.

4.2.3 MECHANICAL INSPECTION

4.2.3.1 During assembly of the equipment and before testing, each component (including cast-in passages of these components) and all piping and appurtenances shall be cleaned chemically or by another appropriate method to remove foreign materials, corrosion products, and mill scale.

4.2.3.2 Any portion of the oil system furnished shall meet the cleanliness requirements of API Standard 614.

• **4.2.3.3** When specified, the purchaser may inspect for cleanliness the equipment and all piping and appurtenances furnished by or through the vendor before heads are welded to vessels, openings in vessels or exchangers are closed, or piping is finally assembled.

• **4.2.3.4** When specified, the hardness of parts, welds, and heat-affected zones shall be verified as being within the allowable values by testing of the parts, welds, or heat-affected zones. The method, extent, documentation, and witnessing of the testing shall be mutually agreed upon by the purchaser and the vendor.

4.2.3.5 The vendor shall verify that dimensions of all rotating components and stationary gas path components fall within the drawing tolerances. Dimensional nonconformances shall be reported to the purchaser within 5 days after approval of the non-conformance by the vendor's engineering department.

4.3 Testing

4.3.1 GENERAL

4.3.1.1 The package shall be tested in accordance with 4.3.2 through 4.3.4. Other tests that may be specified by the purchaser are described in 4.3.5.

4.3.1.2 At least 6 weeks before the first scheduled running test, the vendor shall submit to the purchaser, for

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review and comment, detailed procedures for the mechanical running test and all specified running optional tests, including acceptance criteria for all monitored parameters. The oil parameters described in 4.3.4.1.4 shall be included in these test procedures.

4.3.1.3 The vendor shall notify the purchaser not less than 5 working days before the date the equipment will be ready for testing. If the testing is rescheduled, the vendor shall notify the purchaser not less than 5 working days before the new test date.

4.3.2 HYDROSTATIC TESTS

4.3.2.1 Pressure-containing parts (including auxiliaries) shall be tested hydrostatically with liquid at a minimum of one and one-half times the maximum allowable working pressure but not less than 1.5 bar (20 pounds per square inch gauge). The test liquid shall be at a higher temperature than the nil-ductility transition temperature of the material being tested.

4.3.2.2 Where applicable, tests shall be in accordance with the ASME *Code*. In the event that a discrepancy exists between the code test pressure and the test pressure in this standard, the higher pressure shall govern.

4.3.2.3 Tests shall be maintained for a sufficient period of time to permit complete examination of parts under pressure. The hydrostatic test shall be considered satisfactory when neither leaks nor seepage through the casing or casing joint is observed for a minimum of 30 minutes. Large, heavy castings may require a longer testing period to be agreed upon by the purchaser and the vendor.

4.3.2.4 The chloride content of liquids used to test austenitic stainless steel materials shall not exceed 50 parts per million. To prevent deposition of chlorides as a result of evaporative drying, all residual liquid shall be removed from tested parts at the conclusion of the test.

4.3.2.5 Following hydrotest, the parts including the air intercoolers and aftercoolers shall be cleaned and dried to prevent corrosion.

4.3.3 IMPELLER OVERSPEED TEST

Each impeller shall be subjected to an overspeed test of at least 115 percent of rated operating speed for a minimum duration of 1 minute. After the overspeed test, each impeller shall be examined by magnetic particle or liquid penetrant methods. Impeller dimensions identified by the manufacturer as critical (such as bore and outside diameter) shall be measured before and after the overspeed test. Any permanent deformation of the bore or other critical dimensions outside drawing tolerances shall be resolved to the satisfaction of the vendor and the purchaser.

4.3.4 COMBINED MECHANICAL AND PERFORMANCE TEST

4.3.4.1 The combined mechanical and performance test of the package, including the lube oil system, bearings, seals and vibration monitoring system, shall be conducted at rated operating speed for a continuous period of 4 hours.

4.3.4.1.2 A minimum of five test points shall be recorded, including surge, rated, and maximum capacity.

4.3.4.1.3 Throttling is not permitted using inlet guide vanes (or between a flow-measuring device and either the suction or discharge flange).

4.3.4.1.4 All oil pressures, viscosities, and temperatures shall be within the range of operating values recommended in the vendor's operating instructions for the specific unit being tested. Performance data shall be obtained only after bearing and lube-oil temperatures have stabilized.

4.3.4.1.5 While the equipment is operating at rated speed, sweeps shall be made for vibration amplitudes at frequencies other than synchronous. As a minimum, these sweeps shall cover a frequency from 0.25 times to 8 times the rated speed of the shaft being observed. If the amplitude of any discrete, nonsynchronous vibration exceeds 20 percent of the allowable vibration as defined in 2.2.7.4.4, the purchaser and the vendor shall mutually agree on requirements for any additional testing and on the equipment's suitability for shipment.

4.3.4.1.6 Unless otherwise agreed upon, performance shall be calculated using the test raw data, reduced to the specified site-rated conditions, including expected inlet air filter and aftercooler losses, cooling water temperatures and flows, tube side fouling factors, and all mechanical, blowdown, and condensate losses in accordance with the vendor's standard procedure.

4.3.4.2 The requirements of 4.3.4.2.1 through 4.3.4.2.4 shall be met before the combined mechanical and performance test of the package is performed.

4.3.4.2.1 All joints and connections shall be checked for tightness, and any leaks shall be corrected.

4.3.4.2.2 Oil-system components downstream of the filters shall meet the cleanliness requirements of 4.2.3.2 before any test is started.

4.3.4.2.3 Calibration records for flow, pressure, temperature, and vibration-measuring devices utilized during the test shall be submitted to the purchaser's representative for review. If unwitnessed, certified calibration records shall be included with the test logs and forwarded to the purchaser as required in 4.3.6.2.

4.3.4.2.4 All warning, protective, and control devices used during the test shall be checked, and adjustments shall be made as required.

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4.3.4.3 The requirements of 4.3.4.3, Items (a) through (e) shall be met during the combined mechanical and performance test (see Figure 1).

a. With the compressor operating at its rated discharge pressure, the delivered capacity at the tested, rated operating point shall have zero negative tolerance when compared to rated capacity (that is, -0 percent tolerance on the specified rated flowrate).
b. The required power at the tested rated operating point shall not exceed 104 percent of the value quoted for the rated operating point.
c. Overall pressure rise from the tested rated operating point to surge shall be a minimum of 10 percent.
d. Compressor and contract driver vibration levels shall be recorded at every performance data point and shall meet the criteria of 2.2.7.4.4, 2.2.7.4.7, and 4.3.4.1.5.
e. The difference between inlet- and drain-oil temperature shall not exceed 30°C (50°F).

4.3.4.4 If replacement or modification of bearings or seals or dismantling of the case to replace or modify other parts is required to correct mechanical or performance deficiencies, the initial test will not be acceptable, and the final shop tests shall be run after these replacements or corrections are made.

4.3.4.5 The requirements of 4.3.4.5.1 through 4.3.4.5.3 shall be met after the combined mechanical and performance test is completed.

4.3.4.5.1 The bearings, seals, and gearing shall be inspected. Bearings shall not exhibit any indication of distress.

• **4.3.4.5.2** When, due to the design of the integrally geared compressor, inspection of the bearings and seals requires disassembly of any pinion rotor, the purchaser shall specify either (a) to inspect the bearings one time and retest in accordance with 4.3.4 or (b) to forego inspection of the bearings and seals based upon analysis of test data.

4.3.4.5.3 The gear contact pattern shall be checked by the hard-bluing method with all pinions in place.

4.3.4.5.3.1 Unmodified profile leads shall show a minimum contact of 70 percent along the axis, 30 percent radially.

• 4.3.5 OPTIONAL TESTS

The purchaser will specify whether any of the shop tests specified in 4.3.5.1 through 4.3.5.4 shall be performed (see also 2.2.7.2.3).

• 4.3.5.1 Performance Test

The package shall be Class I tested in accordance with ASME, PTC 10. Equipment and accessories (inlet air filters, aftercoolers, and so forth) to be included in the scope of the test shall be mutually agreed upon by the purchaser and the vendor.

• 4.3.5.2 Guide Vane Test

The package shall be tested at the number of guide vane settings specified by the purchaser. Each setting shall include five test points including surge, rated, and maximum capacity.

• 4.3.5.3 Sound-Level Test

The sound-level test shall be performed in accordance with the purchaser's requirements.

• 4.3.5.4 Spare Rotating Element(s) Test

Rotating elements shall be given a combined mechanical and performance test in accordance with the requirements of this standard.

4.3.6 TEST DATA

4.3.6.1 Immediately upon completion of each witnessed mechanical and performance test, copies of the data logged and the corrected performance curves shall be given to the Purchaser's representative witnessing the tests.

4.3.6.2 If unwitnessed, a certified copy of the raw test data and the corrected performance curves shall be sent to the purchaser immediately upon the completion of the test.

4.4 Preparation for Shipment

• **4.4.1** Equipment shall be suitably prepared for the type of shipment specified, including blocking of the rotor when necessary. Blocked rotors shall be identified by means of corrosion-resistant tags attached with stainless-steel wire. The preparation shall make the equipment suitable for 6 months of outdoor storage from the time of shipment, with no disassembly required before operation except for inspection of bearings and seals. If storage for a longer period is contemplated, the purchaser will consult with the vendor regarding the recommended procedures to be followed.

4.4.2 The vendor shall provide the purchaser with the instructions necessary to preserve the integrity of the storage preparation after the equipment arrives at the job site and before startup.

4.4.3 The package shall be prepared for shipment after all testing and inspection has been completed and the equipment has been approved by the purchaser. The preparation shall include that specified in 4.4.3.1 through 4.4.3.12.

4.4.3.1 Exterior surfaces, except for machined surfaces, shall be given at least one coat of manufacturer's standard paint. The paint shall not contain lead or chromates.

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4.4.3.2 Exterior machined surfaces except for corrosion resistant material shall be coated with a suitable rust preventive.

4.4.3.3 The interior of the integrally geared compressor (and driver if turbine driven) shall be clean; free from scale, welding spatter, and foreign objects; and sprayed or flushed with a suitable rust preventive that can be removed with solvent. The rust preventive shall be applied through all openings while the unit is slow-rolled.

4.4.3.4 Internal steel areas of bearing housings and carbon steel oil systems' auxiliary equipment, such as vessels and pipe fittings, shall be coated with a suitable oil-soluble rust preventative.

4.4.3.5 Exposed shafts and shaft couplings shall be wrapped with waterproof, moldable waxed cloth or volatile-corrosion inhibitor paper. The seams shall be sealed with oilproof adhesive tape.

4.4.3.6 Bearing assemblies shall be fully protected from the entry of moisture and dirt. If vapor-phase-inhibitor crystals in bags are installed in large cavities to absorb moisture, the bags must be attached in an accessible area for ease of removal. Where applicable, bags shall be installed in wire cages attached to flanged covers, and bag locations shall be indicated by corrosion-resistant tags attached with stainless steel wire.

4.4.3.7 Flanged openings shall be provided with metal closures at least 5.0 millimeters ($^3/_{16}$ inch) thick, with elastomer gaskets and at least four full-diameter bolts. For studded openings, all nuts needed for the intended service shall be used to secure closures. Each opening shall be car sealed so that the protective cover cannot be removed without breaking the seal.

4.4.3.8 Threaded openings shall be provided with steel caps or round head steel plugs. In no case shall nonmetallic (such as plastic) caps or plugs be used.

Note: These are shipping plugs; permanent plugs are covered in 2.4.5

4.4.3.9 Openings that have been beveled for welding shall be provided with closures designed to prevent entrance of foreign materials and damage to the bevel.

4.4.3.10 Lifting points and lifting lugs shall be clearly identified on the equipment or the equipment package. The recommended lifting arrangement shall be identified on the boxed equipment.

4.4.3.11 The package shall be identified with item and serial number. Material shipped separately shall be identified with securely affixed, corrosion-resistant metal tags indicating the item and serial number of the equipment for which it is intended. In addition, crated equipment shall be shipped with duplicate packing lists, one inside and one on the outside of the shipping container.

4.4.3.12 When spare rotating elements are purchased, they shall be prepared for unheated indoor storage for a period of at least 3 years. The rotating elements shall be treated with a rust preventive and shall be housed in a vapor-barrier envelope with a slow-release volatile-corrosion inhibitor. They shall be crated for domestic or export shipment, as specified. A purchaser-approved resilient material 3 millimeters ($^1/_8$ inch) thick [not tetrafluoroethylene (TFE) or polytetrafluoroethylene (PTFE)] shall be used between the rotating elements and the cradle at the support areas.

4.4.4 Auxiliary piping connections furnished on the purchased equipment shall be impression stamped or permanently tagged to agree with the vendor's connection table or general arrangement drawing. Service and connection designations shall be indicated.

4.4.5 One copy of the vendor's standard installation instructions shall be packed and shipped with the equipment.

SECTION 5—VENDOR'S DATA

5.1 General

5.1.1 The information to be furnished by the vendor is specified in 5.2 and 5.3. The vendor shall complete and forward the Vendor Drawing and Data Requirements form (see Appendix G) to the address or addresses noted on the inquiry or order. This form shall detail the schedule for transmission of drawings, curves, and data as agreed to at the time of the order, as well as the number and type of copies required by the purchaser.

5.1.2 The data shall be identified on transmittal (cover) letters and in title blocks or title pages with the following information:

a. The purchaser's/ user's corporate name.
b. The job/project number.
c. The equipment item number and service name.
d. The inquiry or purchase order number.
e. Any other identification specified in the inquiry or purchase order.
f. The vendor's identifying proposal number, shop order number, or serial number, or other reference required to identify return correspondence completely.

5.1.3 A coordination meeting shall be held, preferably at the vendor's plant, within 4–6 weeks after the purchase commitment. Unless otherwise specified, the vendor shall

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prepare and distribute an agenda prior to this meeting. The agenda shall include review of the following items:

a. The purchase order, scope of supply, unit responsibility, and subvendor items.
b. The data sheets.
c. Applicable specifications and previously agreed-upon exceptions.
d. Schedules for transmittal of data, production, and testing.
e. The quality assurance program and procedures.
f. Inspection, expediting, and testing.
g. Schematics and bills of material for auxiliary systems.
h. The physical orientation of the equipment, piping, and auxiliary systems.
i. Coupling selections.
j. Thrust-bearing sizing and estimated loadings.
k. The rotor dynamics analysis.
l. Other technical items.

5.2 Proposals

• 5.2.1 GENERAL

The vendor shall forward the original proposal and the specified numbers of copies to the addresses specified in the inquiry documents. As a minimum, the proposal shall include the data specified in 5.2.2 through 5.2.4 as well as a specific statement that the system and all its components are in strict accordance with this standard. If the system and components are not in strict accordance, the vendor shall provide details to enable the purchaser to evaluate any proposed alternative designs. All correspondence shall be clearly identified in accordance with 5.1.2.

5.2.2 DRAWINGS

The drawings indicated on the Vendor Drawing and Data Requirements form (see Appendix G) shall be included in the proposal. As a minimum, the following data shall be furnished:

a. A general arrangement or outline drawing for each major skid or system, showing overall dimensions, maintenance clearance dimensions, overall weights, erection weights, and maximum maintenance weights (indicated for each piece). The direction of rotation and the size and location of major purchaser connections shall also be indicated.
b. Cross-sectional drawings showing the details of the proposed equipment.
c. Schematics of all auxiliary systems, including the lube-oil, control, and electrical systems. Bills of material shall be included.
d. Sketches that show methods of lifting the assembled machine or machines and major components. (This information may be included on the drawings specified in Item a of the preceding.)

5.2.3 TECHNICAL DATA

The following data shall be included in the proposal:

a. The purchaser's data sheets with complete vendor's information entered thereon and literature to fully describe details of the offering.
b. The purchaser's noise data sheet.
c. The Vendor Drawing and Data Requirements form (see Appendix G), indicating the schedule according to which the vendor agrees to transmit all the data specified as part of the contract.
d. A schedule for shipment of the equipment, in weeks after receipt of the order.
e. A list of major wearing components, showing interchangeability with the purchaser's other units.
f. A list of spare parts recommended for start-up and normal maintenance purposes.
g. A list of the special tools furnished for maintenance, as required by 3.9.1. The vendor shall identify any metric items included in the offering.
h. A statement of any special weather protection and winterization required for start-up, operation, and periods of idleness under the site conditions specified. The list shall show the protection to be furnished by the purchaser, as well as that included in the vendor's scope of supply.
i. A complete tabulation of utility requirements, such as those for steam, water electricity, air, gas, and lube oil (including the quantity of lube oil required and the supply pressure), the heat load to be removed by the oil, and the nameplate power rating and operating power requirements of auxiliary drivers. (Approximate data shall be defined and clearly identified as such.)
j. A description of the tests and inspection procedures for materials, as required by 2.2.10.1.4.
k. A description of any special requirements specified in the purchaser's inquiry and as outlined in 2.2.5.2.2, 2.2.10.1.1, 2.2.10.1.3, and 3.6.5.
l. A list of similar machines installed and operating under conditions analogous to those specified in the proposal.
m. Any start-up, shutdown, or operating restrictions required to protect the integrity of the equipment.
n. A list of all relief valves, specifying those furnished by the vendor as required by 3.4.4.8.1.
o. Allowable forces and moments as required by 2.2.4.1.
p. A description of the sealing system including air consumption as required by 2.2.6.2.
q. A description of the design and operation of the control system as required by 3.4.2.8.
r. A description of the alarm and shutdown functions as required by 3.4.5.3.5.
s. A statement of the number of radial vibration probes that can be mounted adjacent to each impeller as required by 3.4.4.6.1.
t. The calculated values of gear-rated power, based on both

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tooth surface durability and tooth bending strength as required by 2.2.5.3.3.
u. The vendor's minimum allowable oil temperature as requested in 2.2.9.5.2.

5.2.4 CURVES

5.2.4.1 The vendor shall provide complete performance curves to encompass the map of operations, with any limitations indicated thereon.

5.2.4.2 Overall performance curves shall be submitted for rated minimum and maximum specified ambient temperatures. All curves shall be marked "PREDICTED."

5.2.4.3 Curves shall include a plot of discharge pressure, and brake horsepower against delivered standard flow. Curves shall indicate surge, rated capacity, and any other specified operating points. Curves shall also be provided that show inlet guide vane effects at off-design inlet conditions.

5.3 Contract Data

5.3.1 GENERAL

5.3.1.1 The contract data to be furnished by the vendor is specified in Appendix G. Each drawing, bill of material, and data sheet shall have a title block that shows the date of certification, a reference to all identification data specified in 5.1.2, the revision number and date, and the title. Title blocks for drawings shall be in the lower right-hand corner.

5.3.1.2 The purchaser will promptly review the vendor's data when received; however, this review shall not constitute permission to deviate from any requirements in the order unless specifically agreed upon in writing. After the data have been reviewed, the vendor shall furnish certified copies in the quantities specified.

5.3.1.3 A complete list of vendor data shall be included with the first issue of the major drawings. This list shall contain titles, drawing numbers, and a schedule for transmission of all the data the vendor will furnish (see Appendix G).

5.3.2 DRAWINGS

The drawings furnished shall contain sufficient information so that with the drawings and the manuals specified in 5.3.6, the purchaser can properly install, operate, and maintain the ordered equipment. Drawings shall be clearly legible, shall be identified in accordance with 5.3.1.1, and shall be in accordance with ASME Y14.2M. As a minimum, each drawing shall include the details for that drawing listed in Appendix G.

5.3.3 TECHNICAL DATA

The data shall be submitted in accordance with Appendix G and identified in accordance with 5.3.1.1 Any comments on the drawings or revisions of specifications that necessitate a change in the data shall be noted by the vendor. These notations will result in the purchaser's issue of completed, corrected data sheets as part of the order specifications.

5.3.4 PROGRESS REPORTS

The vendor shall submit progress reports to the purchaser at the intervals specified on the Vendor Drawing and Data Requirements form (see Appendix G).

5.3.5 PARTS LISTS AND RECOMMENDED SPARES

5.3.5.1 The vendor shall submit complete parts lists for all equipment and accessories supplied. The lists shall include manufacturer's unique part numbers, materials of construction, and delivery times.

Materials shall be identified as specified in 2.2.10.1.3. Each part shall be completely identified and shown on cross-sectional or assembly-type drawings so that the purchaser may determine the interchangeability of the part with other equipment. Parts that have been modified from standard dimensions and/or finish to satisfy specific performance requirements shall be uniquely identified by part number for interchangeability and future duplication purposes. Standard purchased items shall be identified by the original manufacturer's numbers.

5.3.5.2 The vendor shall indicate on the preceding spare parts list which parts are recommended spares for start-up and which are recommended for normal maintenance (see Item f of 5.2.3).

The vendor shall forward the list to the purchaser promptly after receipt of the reviewed drawings and in time to permit order and delivery of the parts before field start-up. The transmittal letter shall be identified with the data specified in 5.1.2.

5.3.6 INSTALLATION, OPERATION, MAINTENANCE, AND TECHNICAL DATA MANUALS

5.3.6.1 General

The vendor shall provide sufficient written instructions and a list of all drawings to enable the purchaser to correctly install, operate, and maintain all of the equipment ordered. This information shall be compiled in a manual or manuals with a cover sheet that contains all reference-identifying data specified in 5.1.2, and index sheet that contains section titles, and a complete list of referenced and enclosed drawings by title and drawing number. The manual shall be prepared for the specified installation; a typical manual is not acceptable.

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API STD*672 96 ■ 0732290 0559829 552 ■

5.3.6.2 Installation Manual

Any special information required for proper installation design that is not on the drawings shall be compiled in a manual that is separate from the operating and maintenance instructions. This manual shall be forwarded at a time that is mutually agreed upon in the order but not later than the final issue of prints.

The manual shall contain information such as special alignment and grouting procedures, utility specifications (including quantities), and all other installation design data, including the drawings and data specified in 5.2.2 and 5.2.3. The manual shall also include sketches that show the location of the center of gravity and rigging provisions to permit the removal of the top half of the casings, rotors, and any subassemblies that weigh more than 136 kilograms (300 pounds).

5.3.6.3 Operating and Maintenance Manual

The manual containing operating and maintenance data shall be forwarded no more than 2 weeks after all of the specified tests have been successfully completed. This manual shall include a section that provides special instructions for operation at specified extreme environmental conditions, such as temperatures. As a minimum, the manual shall also include all of the data listed in Appendix G.

5.3.6.4 Technical Data Manual

• When specified, the vendor shall provide the purchaser with a technical data manual within 30 days of completion of shop testing. (see Appendix G for detail requirements.)

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APPENDIX A DATA SHEETS

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INTEGRALLY GEARED CENTRIFUGAL AIR COMPRESSOR DATA SHEET U.S. CUSTOMARY UNITS

JOB NO. _____ ITEM NO. _____ PAGE _____ OF _____

PURCHASE ORDER NO. _____ DATE _____

INQUIRY NO. _____ BY _____

REVISION _____ DATE _____

APPLICABLE TO: ○ PROPOSAL ○ PURCHASE ○ AS BUILT

FOR: _____ UNIT _____

SITE: _____ NO. REQUIRED _____

SERVICE: _____ DRIVER ITEM NO. _____

○ CONTINUOUS ○ INTERMITTENT ○ STANDBY (2.1.7) SPARED BY: _____

NOTE: INFORMATION TO BE COMPLETED BY: ○ BY PURCHASER ☐ BY MANUFACTURER ☐ BY PURCHASER OR MFR.

GENERAL

COMPR.MFR. _____ MODEL (SIZE AND TYPE) _____ SERIAL NO. _____

DRIVER MFR. _____ DRIVER TYPE _____ RATED HP _____ RPM _____

DRIVE SYSTEM: ○ DIRECT COUPLED ○ OTHER _____

OPERATING CONDITIONS (2.1.3)(3.1.2)

(ALL DATA ON PER UNIT BASIS)

	RATED (2.1.3)	LOW AMB.* (3.1.5.2)	MIN. AMB. (2.1.9)	OTHER
○ DELIVERED FLOW, SCFM (14.7 PSIA & 60F DRY)				
○ WEIGHT FLOW, LBS/MIN (WET)(DRY)				
○ INLET COOLING WATER TEMP				
INLET CONDITIONS: (2.1.3)(3.1.2)				
○ PRESSURE (PSIA)				
○ TEMPERATURE (°F)				
○ RELATIVE HUMIDITY (%)				
○ MOLECULAR WEIGHT				
☐ INLET VOLUME (CFM-WET)				
DISCHARGE CONDITIONS:				
○ PRESSURE (PSIA)				
☐ TEMPERATURE (°F)				
PERFORMANCE:				
☐ MAX BHP REQUIRED (ALL LOSSES INCLUDED)				
☐ BHP/100 CFM AIR DELIVERED				
☐ INPUT SPEED (RPM)				
☐ ESTIMATED SURGE, ICFM (@ ABOVE SPEED)				
○ MAX. ΔP ACROSS INLET FILTER (PSI)				
ΔP INCLUDED IN CALCULATION ☐ YES ☐ NO				
☐ AFTERCOOLER OUTLET TEMP. (°F)				
☐ PERFORMANCE CURVE NO.				
☐ % RISE TO SURGE (2.2.1.4)				
☐ % TURNDOWN (2.2.1.4)				
☐ _____				

*UNTHROTTLED PERFORMANCE FOR DRIVER SIZING

CONTROL SYSTEM (3.4.2)

CONTROL METHOD: (3.4.2.1)

○ CAPACITY MODULATION *(CONSTANT DISCH. PRESSURE)* (3.4.2.1a)
- ○ INLET THROTTLE DEVICE: ○ DAMPER;
- ○ GLOBE VALVE; ○ BUTTERFLY VALVE;
- ○ VARIABLE INLET GUIDE VANES

○ TWO STEP CONTROL *(LOAD-UNLOAD)* (3.4.2.1b)
- ○ _____ PSIG TO _____ PSIG DISCH. PRESS

○ AUTOMATIC DUAL CONTROL (3.4.2.1c)

○ AUTO. START AND STOP (3.4.2.1d):
- ○ START _____ PSIG; STOP _____ PSIG

○ CONSTANT DISCHARGE PRESS. BASE MODE *(BLOWOFF)* (3.4.2.1e)

○ MASS FLOW CONTROL (3.4.2.1f)

CONTROL SYSTEM REQUIREMENTS:

○ UNIT OPERATES IN PARALLEL (3.4.2.2)
- ○ W/ CENTRIFUGAL;
- ○ W/ ROTARY; ○ W/ RECIPROCATING

○ MICROPROCESSOR CAPABLE OF COMMUNICATION WITH PURCHASER'S DCS (3.4.1.2)

CONTROL SYST. ALTERNATES: (3.4.1.1)

○ OTHER THAN MICROPROCESSOR BASED: _____

○ SUITABLE FOR INDOOR ONLY

○ FURNISHED BY PURCHASER

INTER- & AFTER-COOLERS

AFTERCOOLER (3.6):
- ○ FURNISHED BY PURCHASER (3.6.1)
- ○ NOT NEEDED (3.6.1)
- ○ AIR-COOLED TYPE (3.6.3) BY VENDOR

○ AIR-COOLED INTERCOOLERS REQ'D (3.6.3)
- ○ FURNISHED BY PURCHASER

☐ AIR-COOLED EXCHANGER AUTOMATIC TEMPERATURE CONTROL MEANS: (3.6.6)
- ○ LOUVERS ○ VARIABLE SPEED FANS
- ○ VARIABLE PITCH FANS ○ BYPASS VALVE

☐ AIR-COOLER CONTROL MANUAL ONLY (3.6.6) BY:
- ○ LOUVERS ○ BYPASS VALVE
- ○ VARIABLE PITCH FANS

REMARKS: _____

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JOB NO. _____ ITEM NO. _____ PAGE _____ OF _____

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LOCATION, SITE DATA

LOCATION (2.1.9):

○ INDOOR ○ HEATED ○ UNDER ROOF

○ OUTDOOR ○ UNHEATED ○ PARTIAL SIDES

○ GRADE ○ MEZZANINE ○ _____

○ WINTERIZATION REQ'D ○ TROPICALIZATION REQ'D

SITE DATA:

○ ELEVATION _____ FT. ○ BAROMETER _____ PSIA (IN. HG.A)

○ RANGE OF AMBIENT TEMPERATURE (°F):

	DRY BULB	WET BULB
NORMAL		
MAXIMUM		
MINIMUM		

UNUSUAL CONDITIONS: (2.1.9)

○ DUST ○ FUMES ○ CORROSIVE CONDITIONS

○ CORROSIVES PRESENT: _____

○ CONDITIONS CAUSE STRESS CORROSION CRACKING (2.2.10.1.7)

○ OTHER _____

AREA ELECTRICAL CLASSIFICATION (2.1.13) T-CODE _____

○ CLASS _____ GROUP _____ DIVISION _____

○ LOCAL ELECTRICAL CODES: _____

SPECIFICATIONS

NOISE SPECIFICATIONS: (2.1.4)

○ MAX. ALLOWABLE SPL _____ @ 3 FT.

○ APPLICABLE SPEC. _____

ACOUSTICAL HOUSING ○ YES ○ NO

APPLICABLE SPECIFICATIONS:

API 672 AND ○ _____

○ NON-ASME WELDING IF NOT AWS D1.1: (2.2.10.4.4) _____

PAINTING:

○ MANUFACTURER'S STANDARD

○ OTHER _____

BASEPLATE GROUT: (3.3.16) ○ EPOXY ○ CEMENT ○ NONE

PREPARATION FOR GROUT SURFACES: (3.3.15)

○ MFR. STD. ○ SSPC-6 BLAST ○ BARE FOR FIELD BLAST

○ INORGANIC ZINC SILICATE COATING

○ OTHER _____

SHIPMENT: (4.4.1)(4.4.3.12)

○ DOMESTIC ○ EXPORT ○ EXPORT BOXING REQ'D

○ OUTDOOR STORAGE OVER 6 MONTHS

☐ MAT'L FOR SPARE ROTOR SUPPORT (4.4.3.12) _____

UTILITY CONDITIONS: (3.1.1)

○ **STEAM HEATING: (2.2.9.5.7)**

INLET MIN.:	_____ PSIG	_____ °F
NORM.:	_____ PSIG	_____ °F
MAX.:	_____ PSIG	_____ °F
OUTLET MIN.:	_____ PSIG	_____ °F
NORM.:	_____ PSIG	_____ °F
MAX.:	_____ PSIG	_____ °F

○ **ELECTRICITY:**

	HEATING	CONTROL	SHUTDOWN
VOLTAGE:			
HERTZ:			
PHASE:			

○ **COOLING WATER: (2.1.10)**

TEMP: INLET _____ °F MAX. RETURN _____ °F

PRESS: NORM. _____ PSIG DESIGN _____ PSIG

MIN. RETURN _____ PSIG MAX. ALLOW ΔP _____ PSI

WATER SOURCE _____

○ **AIR/NITROGEN:**

MAX. PRESS _____ PSIG MIN. PRESS _____ PSIG

GAS COMPOSITION: _____

UTILITY CONSUMPTION: (5.2.3i)

STEAM:

OIL HEATER: _____ LBS/HR. OTHER: _____ LBS/HR

ELECTRIC:

	HP	LOCKED ROTOR AMPS.	FULL LOAD AMPS.
MAIN LO PUMP:			
AUX. LO PUMP:			

OIL HEATER: _____ W; SPACE HEATER _____ W

CONTROL SYSTEM LOAD: _____ W (3.4.2.8)

COOLING WATER:

	LO COOLER	INTER COOLER	AFTER COOLER	OTHER
QUANTITY, GPM				
OUTLET TEMP, °F				
PRESS. DROP, PSI				
TOTAL CW, GPM				

AIR/NITROGEN:

	INLET PRESS. (PSIG)	QUANTITY (SCFM)
SEAL SYSTEM:		
CONTROL PANEL:		
LO RESERVOIR:		
INSTR. HOUSINGS:		
CONTROL SYSTEM:		
OTHER:		

TOTAL PURGE, SCFM: _____

REMARKS: _____

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CONSTRUCTION FEATURES

☐ **COMPRESSOR SPEEDS:**

RATED INPUT: _____ RPM TRIP: _____ RPM

BULLGEAR CRITICALS: 1st: _____ RPM 2nd: _____ RPM

PINION CRITICALS:

1st STG. PINION: 1st: _____ RPM 2nd: _____ RPM

2nd STG. PINION: 1st: _____ RPM 2nd: _____ RPM

3rd STG. PINION: 1st: _____ RPM 2nd: _____ RPM

4th STG. PINION: 1st: _____ RPM 2nd: _____ RPM

OTHER UNDESIRABLE SPEEDS: (2.2.7.1.3) _____

	STAGE SPEED	IMPELLER DIAMETER	TIP SPEED
1st STAGE:	_____ RPM	_____ IN.	_____ FPM
2nd STAGE:	_____ RPM	_____ IN.	_____ FPM
3rd STAGE:	_____ RPM	_____ IN.	_____ FPM
4th STAGE:	_____ RPM	_____ IN.	_____ FPM

☐ **IMPELLERS:**

NO. OF IMPELLERS: _____ MATERIAL: _____

TYPE: (OPEN, RADIAL, BACKWARD LEANING, ETC.) _____

TYPE CONSTRUCTION: (2.2.5.2.2) _____

METHOD OF ATTACH: (2.2.5.2.2) _____

ROTATION, VIEWED FROM INPUT SHAFT END: ☐ CW ☐ CCW

☐ **COMPRESSOR CASING:**

MODEL: _____ CASING SPLIT: _____

	STG. 1	STG. 2	STG. 3	STG. 4
MATERIAL:				
M.A.W.P. (PSIG)				
HYDRO TEST (PSIG)				
MAX OPER TEMP (°F)				
MIN OPER TEMP (°F)				

○ MIN. DESIGN METAL TEMP.: (2.2.10.5.3) _____ °F

☐ CASING HEAT TREATMENT REQUIRED (2.2.10.4.5.4)

☐ MAX. ALLOW. STRESS FOR MAT'L: (2.2.2.1) _____ PSI

☐ CODE CASTING FACTOR: (2.2.2.1) _____

WELDED CONNECTIONS - NDT REQ'D: (2.2.10.4.6.1)

☐ 100% RADIOGRAPH ☐ MAG. PARTICLE ☐ LIQ. PENETRANT

○ PURCHASER REVIEW OF CONNECTION DESIGN (2.2.10.4.6.4)

☐ **COMPRESSOR BEARINGS & BEARING HOUSES:**

BEARING HSG. MATERIAL: _____

PINION RADIAL BEARINGS: (2.2.8.2)

	STG. 1	STG. 2	STG. 3	STG. 4
BRG. TYPE:				
ALLOW LOAD: (PSI)				
ACTUAL LOAD: (PSI)				
BRG. SPAN:				

PINION THRUST BEARINGS: (2.2.8.3)

	STG. 1	STG. 2	STG. 3	STG. 4
BRG. TYPE:				
ALLOW LOAD: (PSI)				
ACTUAL LOAD: (PSI)				
THRUST COLLAR:				

☐ **INTEGRAL GEAR HOUSING:**

MATERIAL: _____ ; SPLIT: _____

☐ **BULL GEAR: (2.2.5.3)**

RATED POWER BASED ON TOOTH SURFACE DURABILITY: _____ HP

RATED POWER BASED ON TOOTH BENDING: _____ HP

○ MIN. AGMA SERVICE FACTOR: _____ ☐ ACT.S.F. _____

GEAR RIM MAT'L: _____ HARDNESS: _____

GEAR FACE WIDTH: _____ IN.; GEAR CENTER MAT'L: _____

MECHANICAL EFFICIENCY: _____ %; AGMA QLTY: _____

PITCH DIA. _____ IN.; PITCH LINE VEL. _____ FPS

☐ **PINIONS: (2.2.5.3)**	1st	2nd	3rd	4th
SERVICE FACTOR:				
MATERIAL:				
HARDN'S:(BHN)(R_c)				

☐ **BULL GEAR SHAFT:**

☐ REPLACEABLE ☐ INTEGRAL W/ GEAR

MAT'L: _____ ; HARDNESS _____ (BHN)(R_c)

BRG. SPAN: _____ IN.; WEIGHT (W/GEAR): _____ LBS

DIA. @ GEAR: _____ IN.; DIA @ CPLG: _____ IN.

SHAFT SLEEVES AT SEALS: MAT'L: _____

SHAFT LABY'S: TYPE: _____ MAT'L: _____

BULL GEAR RAD. BRG. TYPE: _____ LGTH: _____ IN.

ALLOW LOAD: _____ PSI; ACT LOAD: _____ PSI

☐ **BULL GEAR THRUST BEARINGS: (2.2.8.3)**

LOCATION: _____ ; TYPE: _____

MFR: _____ ; AREA: _____ SQ IN

THRUST COLLAR: (2.2.8.3.5) ☐ INTEGRAL ☐ REPLACEABLE

ALLOW. LOAD: _____ PSI; ACT. LOAD: _____ PSI

GAS LOAD: _____ LBS; CPLG. LOAD: _____ LBS

BEARINGS FITTED W/ TEMP. SENSORS (2.2.8.4.4)

○ PINION RAD. BRG. ○ BULLGEAR RAD. BRG.

○ THRUST BRG. _____

☐ **MAIN CONNECTIONS: (2.2.3.1)**

	SIZE	ASME RATING	FACING	POSITION
COMP. INLET:				
COMP. DISCH.:				
PKG. OUTLET:				
ATM. BLOWOFF:				
FILTER OUTLET:				

☐ **OTHER CONNECTIONS:**

	NO.	SIZE	TYPE
LUBE OIL INLET:			
LUBE OIL OUTLET:			
C.W. INLET:			
PRESS. GAUGE:			
TEMP. GAUGE:			
COND. DRAINS:			
_____			

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VIBRATION DETECTORS: (3.4.4.6)

☐ TYPE: _____ ☐ MODEL _____

○ MFR.: _____

○ NO. AT EACH PINION BEARING: _____ TOTAL NO.: _____

○ NO. AT EACH DRIVER BEARING: _____ TOTAL NO.: _____

☐ X&Y RADIAL PROBES CAN BE MOUNTED ADJACENT TO IMPELLERS FOR:

☐ 1st STG. ☐ 2nd STG. ☐ 3rd STG. ☐ 4th STG.

OSCILLATOR-DEMODULATORS:

○ MFR.: _____ ☐ MODEL _____

○ MONITOR SUPPLIED BY: _____

○ MFR.: _____ ☐ MODEL _____

○ LOCATION: _____ ENCLOSURE: _____

☐ READOUT SCALE RANGE: _____ ○ ALARM ☐ SET @ _____ MILS

○ SHUTDOWN ☐ SET @ _____ MILS ○ TIME DELAY: _____ SEC

○ PER API 670: (3.4.4.6.2&3)

BEARING-TEMPERATURE MONITOR: (3.4.4.6.4)

○ REQ'D ○ SUPPLIED BY: _____ ○ PER API 670

○ MFR.: _____ ☐ MODEL _____

AXIAL POSITION MOVEMENT DETECTOR: (3.4.4.6.2)

☐ TYPE: _____ ☐ MODEL _____

○ MFR.: _____

☐ READOUT SCALE RANGE: _____ ○ ALARM ☐ SET @ _____ MILS

○ SHUTDOWN ☐ SET @ _____ MILS ○ TIME DELAY: _____ SEC

DYNAMICS: (2.2.7)

○ CRITICAL LATERAL SPEEDS ARE PROVEN BY PRIOR UNITS (2.2.7.2.1)

○ DAMPED UNBALANCED RESPONSE ANALYSIS REQ'D. (2.2.7.2.2)

○ TORSIONAL VIBRATION ANALYSIS OF TRAIN REQ'D. (2.2.7.3.1)

○ RESIDUAL UNBALANCE WORKSHEET REQ'D. (2.2.7.4.3)

☐ REMARKS: _____

COUPLINGS: (3.2)

TYPE: ☐ DISK PAK ☐ DIAPHRAGM ○ OTHER _____

DISK MAT'L: ☐ STAINLESS STEEL ☐ COATED W/ _____

☐ MAKE: _____ ☐ MODEL: _____

○ NON-LUBE ○ LUB'D. ☐ LUBRICATION: _____

☐ SPACER LENGTH: _____ IN. ☐ LIMITED END-FLOAT REQ'D

☐ CPLG. RATING: _____ HP/100 RPM @ 1.0 S.F.; ACTUAL S.F.: _____

☐ SHAFT JUNCT. RATING: @ DRIVER: _____ HP; @ INPUT SHAFT: _____ HP

☐ MOUNTING ARR'GMT: @ INPUT SHAFT: _____ DRIVER: _____

☐ MFR. MAX. BORE: _____ IN.; PROPOSED BORE: _____ IN (3.2.1.9)

DRIVER HALF-CPLG. MTD BY: ☐ DRIVER MFR ☐ COMP. VENDOR

☐ IDLING ADAPTER FOR DRIVER HALF-CPLG. REQ'D (3.2.1.6)

PIPING REQUIREMENTS:

☐ RECOMMENDED STRAIGHT RUN OF PIPE DIA. BEFORE SUCTION: _____

○ VENDOR TO OBSERVE FLANGE PARTING (2.1.15ba)

○ THROUGH STUDS REQUIRED FOR PIPING FLANGES (3.5.1.18)

MISCELLANEOUS:

○ VENDOR PRESENT DURING INITIAL ALIGN. CHECK (2.1.15b)

○ VENDOR CHECK ALIGN. AT OPERATING TEMP. (2.1.15c)

○ BASE DESIGNED FOR COLUMN MOUNTING (3.3.10)

○ THERMAL RELIEF VALVES PROVIDED BY VENDOR (3.4.4.8.3)

○ FOR WATER COOLED EXCHANGERS

○ FOR _____

○ PURCHASER WILL PREPARE COORDINATION MEETING AGENDA (5.1.3)

○ SHOP INSPECTIONS & TESTS: (4.1.1)

○ ADVANCE NOTIFICATION REQ'D: _____ DAYS (4.1.1)

	OBS'D	WIT.
● SHOP INSPECTION: (4.1.3)		
● HYDROSTATIC: (4.3.2)	○	○
● 4-HOUR COMBINED TEST: (4.3.4)	○	○
○ ASME, PTC-10 TEST: (4.3.5.1)	○	○
INCLUDES ○ AIR FILTER; ○ AFTERCOOLER; ○ _____		
○ GUIDE VANE TEST: (4.3.5.2)	○	○
○ AT _____ NON-100% POSITIONS		
○ SOUND-LEVEL TEST: (4.3.5.3)	○	○
○ SPARE ROTOR TEST: (4.3.5.4)	○	○
○ SPARE ROTOR MECH ONLY:	○	○
● IMPELLER OVERSPEED TEST: (4.3.3)	○	○
● RADIOGRAPH OF IMPELLERS: (4.2.2.6)	○	○
○ RESID. UNBALANCE CHECK: (2.2.7.4.3)	○	○
● OIL SYSTEM CLEAN: (4.3.4.2.2)	○	○
● CONTROL SYSTEM CHECK: (4.3.4.2.4)	○	○
● BRG., SEAL, GEAR CHECK: (4.3.4.5.1)	○	○
● GEAR CONTACT CHECK: (4.3.4.5.3)	○	○
● CLEANLINESS CHECK - VESSELS: (4.2.3.3.)	○	○
● CLEANLINESS CHECK - PIPING: (4.2.3.3.)	○	○
○ HARDNESS CHECK OF PINIONS: (4.2.3.4)	○	○
○ OF BULL-GEAR:	○	○
○ OF WELD REPAIRS:	○	○
○ NDE OF MAJOR REPAIRS: (2.2.10.4.2)	○	○
○ GEAR TOOTH MAG-PART.: (4.2.2.8)	○	○
○ FINAL INSP. PRIOR TO PAINT:	○	○
○ INSP. OF PREP. FOR SHIPMENT: (4.4)	○	○
○ _____	○	○
○ _____	○	○

○ PRIOR DOCUMENTATION ON MAJOR REPAIRS NOT REQ'D (2.2.10.4.3)

○ RETAIN FINAL ASSEMBLY CLEARANCES (4.2.1.1d)

○ SUBMITTAL OF INSPECTOR'S CHECKLIST (4.1.5)

SIGNED BY REP FOR: ○ PURCHASER ○ VENDOR

IF DESIGN REQ'S DISASSEMBLY OF PINION FOR BRG. INSPECTION,

○ FORGO BEARING INSPECTION BASED ON TEST DATA; OR

○ INSPECT BEARING AND RETEST. (4.3.4.5.2)

☐ WEIGHT, LBS.

INTEG. GEAR/COMP.: _____ DRIVER: _____

GEAR UPPER CASE: _____ BULL-GEAR: _____

1st STG. PINION: _____ 2nd STG. PINION: _____

INTERCOOLER: _____ BUNDLE: _____

AFTERCOOLER: _____ BUNDLE: _____

BASE: _____ CONTROL PANEL: _____

MAX. FOR MAINTENANCE (IDENTIFY): _____

TOTAL SHIPPING WT.: _____

☐ SPACE REQUIREMENTS (FEET & INCHES)

	L	W	H
COMPLETE UNIT:			
CONTROL PANEL: (IF SEP.)			
INLET FILTER - SILENCER:			
AFTERCOOLER(IF FURN.):			
OTHER:			

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LUBE OIL SYSTEM (2.2.9)

BASIC SYSTEM REQ'MTS. - NORMAL OIL FLOW (2.2.9.5.2)

LUBE OIL TO:	GPM	PSIG	SSU @ 100°F
☐ COMP./GEAR:			
☐ DRIVER:			
☐ EXT. GEAR:			

☐ OIL SYSTEM PRESSURES:

SUPPLY: _____ PSIG PUMP RV SETTING: _____ PSIG

SYSTEM DESIGN: _____ PSIG HYDROTEST: _____ PSIG

OIL COOLER: (2.2.9.5.5)

	SHELL SIDE	TUBESIDE
☐ OPERATING PRESSURE: (PSIG)		
☐ MAX ALLOW WORK PRESS: (PSIG)		
☐ MAX ALLOW TEMP: °F		
◯ FOULING FACTOR:		

☐ SURFACE AREA: _____ FT²; DUTY: _____ BTU/HR

◯ REMOVABLE BUNDLE TO BE FURNISHED

☐ ASME CODE STAMPED ◯ DESIGNED TO TEMA _____

◯ TUBES: NO. _____; O.D. _____ IN.; LENGTH: _____ IN

WALL THICKNESS: _____ IN. ☐ AVG. ☐ MIN.

◯ MATERIALS:

CHANNELS/HEADS: _____ SHELL: _____

TUBES: _____ TUBE SHEETS: _____

CHANNEL COVERS: _____ TUBE SUPTS: _____

OIL FILTERS: (2.2.9.5.6)

◯ MICRON RATING: _____ ◯ NOMINAL ◯ ABSOLUTE

◯ ΔP (PSI): CLEAN: _____; DIRTY: _____; COLLAPSE _____

◯ ELEMENT: MAKE: _____; MODEL: _____

☐ NO. ELEMENTS: _____; ◯ MEDIA: _____

◯ CORE MAT'L: _____; ◯ HSG. MAT'L: _____

☐ HSG. MAWP: _____ PSIG; ☐ MAX. ALLOW TEMP. _____ °F

OIL HEATER: (2.2.9.5.7)

◯ STEAM HEATER REQ'D; ☐ ELECTRIC HEATER REQUIRED;

☐ RATING: _____ BTU/HR

☐ WATT DENSITY: _____ W/IN²

OIL RESERVOIR: (2.2.9.5.8)

◯ RETENTION TIME: _____ MIN.; ☐ CAPACITY: _____ GAL.

☐ FREE SURFACE AREA: _____ FT² ☐ INT'L BAFFLES

LUBRICANT: (2.2.9.3) ◯ SYNTHETIC ◯ HYDROCARBON

◯ DESCRIPTION: _____

☐ MIN. ALLOW. OIL TEMP: (2.2.9.5.2) _____ °F, _____ SSU

◯ SYSTEM COMPONENT SUPPLIERS:

	MFR.	MODEL
MAIN PUMP:		
STANDBY PUMP:		
ELECTRIC MOTOR(S):		
STEAM TURBINE(S):		
OIL COOLER(S):		
OIL FILTERS:		
ACCUMULATOR(S):		
SUCTION STRAINERS:		
CHECK VALVES:		
TRANSFER VALVE(S):		
PUMP COUPLING:		
PUMP RELIEF VALVES:		
ELECTRIC HEATER:		
_____		

PUMPS: (2.2.9.5.2)	MAIN	STANDBY
◯ HORIZONTAL:		
◯ VERTICAL:		
◯ SUBMERGED:		
◯ MOTOR DRIVEN:		
◯ TURBINE DRIVEN:		
◯ SHAFT DRIVEN:		
◯ CENTRIFUGAL:		
◯ ROTARY:		
◯ FLANGE CONNECTED:		
☐ GPM (RATED):		
☐ @ PSIG DISCH.:		
☐ SHP @ MAX. SSU:		
☐ DRIVER HP:		
◯ CASING MATERIAL:		
☐ SPEED:		
◯ COUPLING:		
◯ OSHA GUARD:		
◯ MECHANICAL SEAL:		

STANDBY PUMP CONTROL RESET:

◯ MANUAL ◯ AUTOMATIC ◯ HOA SELECTOR SWITCH

SILENCERS

INLET AIR FILTER/SILENCER: (3.7)

☐ MFR.: _____; MODEL: _____

☐ DESCRIPTION: _____

☐ PIPING CONNECTION: _____

☐ CLEAN ΔP, AS QUOTED: _____ PSIG

◯ CORROSION PROTECTION: _____

◯ FILTER WILL BE REMOTE MOUNTED BY PURCHASER AT A DISTANCE _____ FT. FROM COMPRESSOR.

◯ FILTER WILL BE ELEVATED _____ FT ABOVE GRADE.

DISCHARGE BLOWOFF SILENCER: (3.7)

☐ MFR: _____; MODEL: _____

☐ DESCRIPTION: _____

☐ FLANGE CONNECTION: _____

MOUNTING: ◯ HORIZONTAL ◯ VERTICAL

SUPPORTED BY: ◯ PIPING ◯ OTHER: _____

☐ SPL (dBA) AT DISCHARGE OF SILENCER: _____

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CONTROLS AND INSTRUMENTATION (3.4)

LOCAL CONTROL PANEL: (3.4.3)

◯ ELECTRICAL AREA CLASSIFICATION: CL _____, GR _____, DIV _____ (3.4.3.2)

PANEL ENCLOSURE REQUIREMENT: (3.4.3.2)

☐ NEMA TYPE 4X ENCLOSURE MATERIAL: _____

☐ NEMA TYPE 7 (*INDOOR EXPLOSION-PROOF FOR HAZARDOUS GAS AREAS*) REQ'D

PANEL FEATURES (3.4.3.2)

◯ VIBRATION ISOLATORS ☐ STRIP HEATER ☐ INTERNAL COOLING

◯ WEATHERHOOD ☐ PURGE CONNECTIONS ◯ OTHER _____

PURGE REQUIREMENT: (3.4.3.2)

☐ NONE ◯ INSTRUMENT AIR ◯ NITROGEN

☐ TYPE X - *REDUCES THE CLASSIFICATION FROM DIV.1 TO NONHAZARDOUS.*

☐ TYPE Y - *REDUCES THE CLASSIFICATION FROM DIV.1 TO DIV.2.*

☐ TYPE Z - *REDUCES THE CLASSIFICATION FROM DIV.2 TO NONHAZARDOUS.*

◯ TROPICALIZATION REQUIRED (3.4.6.6)

☐ **INSTRUMENT SUPPLIERS:**

	MFR:	SIZE & TYPE:
PRESSURE GAUGES:		
TEMPERATURE GAUGES:		
LEVEL GAUGES:		
DIFF. PRESSURE GAUGES:		
PRESSURE SWITCHES:		
TEMPERATURE SWITCHES:		
LEVEL SWITCHES:		
PRESSURE TRANSMITTERS:		
TEMPERATURE TRANSMITTERS:		
LEVEL TRANSMITTERS:		
CONTROL VALVES:		
PRESSURE RELIEF VALVES:		
THERMAL RELIEF VALVES:		
TEMPERATURE CONTROL VALVES:		
SIGHT FLOW INDICATORS:		
PURGE FLOW INDICATORS:		
SOLENOID VALVES:		
ANNUNCIATOR:		
TUBE FITTINGS:		
_____		
_____		
_____		
_____		

SWITCH CLOSURES: (3.4.5.3.2)

ALARM CONTACTS SHALL: ◯ OPEN ◯ CLOSE TO SOUND ALARM AND BE NORMALLY ◯ ENERGIZED ◯ DE-ENERGIZED

SHUTDOWN CONTACTS SHALL: ◯ OPEN ◯ CLOSE TO TRIP AND BE NORMALLY ◯ ENERGIZED ◯ DE-ENERGIZED

(NOTE: *NORMAL* CONDITION IS WHEN COMPRESSOR IS IN OPERATION.)

◯ SHUTDOWN SYSTEMS ARE *NOT* TO BE PROVIDED WITH A MEANS TO PERMIT TESTING WITHOUT SHUTTING DOWN THE UNIT. (3.4.5.3.3)

◯ NON-SHUTDOWN DEVICES ARE *NOT* REQUIRED TO HAVE VALVING TO PERMIT REPLACEMENT DURING OPERATION. (3.4.5.4.1)

◯ ISOLATION VALVES *ARE* REQUIRED FOR SHUTDOWN SENSING DEVICES. (3.4.5.4.1)

MISCELLANEOUS INSTRUMENTATION:

◯ THROUGH FLOWING INSTRUMENT SENSING LINE REQUIRED. (3.5.2.4)

◯ LIQUID FILLED GAUGES ARE REQUIRED FOR AREAS SUBJECT TO VIBRATION. (3.4.4.4.1)

◯ RELIEF VALVES MAY HAVE BODIES IN MATERIALS OTHER THAN STEEL. (3.4.4.8.2.) ☐ RV BODY MATERIAL: _____

◯ THERMAL RELIEF VALVES REQUIRED FOR COMPONENTS THAT CAN BE ISOLATED. (3.4.4.9.1)

◯ FLOW INDICATOR TYPE/MATERIAL IF OTHER THAN BULLS EYE TYPE WITH STEEL BODY: (3.4.4.9.1) _____

◯ PURGE REQUIRED FOR ANNUNCIATOR. (3.4.5.2) NFPA 496 PURGE TYPE: ◯ X ◯ Y ◯ Z ◯ CONNECTION ONLY

◯ COMBINATION BLOCK & BLEED VALVES MAY BE SUBSTITUTED. (3.5.3.1)

◯ _____

◯ _____

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INSTRUMENTATION SCOPE OF SUPPLY

	ELEMENT										INDICATOR							
	PROV. BY		TYPE			LOCATION			INSTALL BY		PROV. BY		LOCATION					
	VENDOR	PURCHASER	DIRECT READOUT	SWITCH	TRANSMITTER (1)	VENDOR PKG.	LOCAL PANEL	PURCH. PIPING	VENDOR	PURCHASER	VENDOR	PURCHASER	VENDOR PKG.	PURCH. PIPING	LOCAL PANEL	ALARM	SHUTDOWN	REPEAT SIGNAL (2)
PRESSURE:																		
COMPRESSOR SUCTION _____ STAGE																		
COMPRESSOR DISCH. _____ STAGE																		
LUBE OIL DISCHARGE																		
LUBE OIL FILTER ΔP																		
LUBE OIL SUPPLY																		
AIR FILTER/SILENCER ΔP																		
TEMPERATURE:																		
COMPRESSOR SUCTION _____ STAGE																		
COMPRESSOR DISCHARGE _____ STAGE																		
OIL COOLER INLET & OUTLET																		
COMP. PINION JOURNAL BRG.																		
BULL GEAR JOURNAL BRG.																		
BULL GEAR THRUST BRG.																		
DRIVER JOURNAL BRG.																		
DRIVER THRUST BRG.																		
RESERVOIR																		
LEVEL:																		
LUBE OIL RESERVOIR																		
SEPARATOR																		
VIBRATION:																		
RADIAL VIBRATION EACH STAGE																		
RADIAL VIBRATION BULL GEAR SHAFT																		
AXIAL POSITION BULL GEAR SHAFT																		
AXIAL POSITION _____ STAGE PINION																		
RADIAL VIBRATION ON DRIVER																		
AXIAL POSITION ON DRIVER SHAFT																		
ACCELEROMETER ON GEAR BOX																		
FLOW:																		
OIL RETURN																		
SEAL GAS																		
MISCELLANEOUS:																		
STANDBY L.O. PUMP RUNNING																		
PANEL PURGE FAILURE																		
ANNUNCIATOR PURGE FAILURE																		
SURGE RECOGNITION																		
OIL HEATER ON																		
COMMON REMOTE ALARM INDICATION																		
COMMON REMOTE SHUTDOWN INDICATION																		

NOTES: 1) TRANSMITTERS SUPPLIED BY VENDOR SHALL INCLUDE SENSING ELEMENT.
2) SUPPLY "REPEAT SIGNAL" FOR CONTROL ROOM ALSO.

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(INTER-) (AFTER) COOLER(S) (3.6)

SERVICE OF UNIT: _____ ITEM NO.: _____

SIZE: _____ TYPE: _____ ☐ HOR ☐ VERT CONNECTED IN: ☐ PARALLEL ☐ SERIES

SURF/UNIT (GROSS/EFF.): _____ SQ.FT.: SHELLS/UNIT: _____ SURF./SHELL (GROSS/EFF.) _____ SQ.FT.

PERFORMANCE OF ONE UNIT

	SHELL SIDE		TUBE SIDE	
○ FLUID NAME				
☐ FLUID QUANTITY, TOTAL (LB/HR)				
VAPOR - IN/OUT				
LIQUID - IN/OUT				
☐ TEMPERATURE - IN/OUT (°F)				
☐ SPECIFIC GRAVITY				
☐ VISCOSITY, LIQUID (CP) (SSU)				
☐ SPECIFIC HEAT (BTU/LB-°F)				
☐ THERMAL CONDUCTIVITY (BTU-FT/HR-SQ.FT.-°F)				
☐ LATENT HEAT (BTU/LB @ °F)				
☐ INLET PRESSURE (PSIG)				
◯ VELOCITY (FT/SEC)				
◯ PRESSURE DROP - ALLOW./CALC. (PSI)				
◯ FOULING RESISTANCE - MINIMUM				

☐ HEAT EXCHANGED _____ BTU/HR MTD - CORRECTED _____ °F

☐ TRANSFER RATE (BTU/HR-SQ.FT.-°F): SERVICE _____ CLEAN _____

☐ CONSTRUCTION OF ONE SHELL

	SHELL SIDE	TUBE SIDE
DESIGN/TEST PRESS. (PSIG)		
DESIGN TEMPERATURE (°F)		
NO. PASSES PER SHELL		
CORROSION ALLOWANCE (INCH)		
NOZZLES: SIZE & RATING — _____ INLET		
_____ OUTLET		
_____ VENT-DRAIN		

SKETCH: BUNDLE, NOZZLE ORIENTATIONS

TUBE NO. _____; OD _____ (IN.); THK (MIN.)(AVG.) _____ (IN.); LENGTH _____ (FT); PITCH _____ (IN.) ◁30 △60 ☐90 ◇45

TUBE TYPE: _____; MATERIAL _____

SHELL MAT'L. _____; ID _____ (IN); OD _____ (IN); SHELL COVER MAT'L _____ (INTEG.) (REMOV.)

CHANNEL OR BONNET MAT'L. _____ CHANNEL COVER MAT'L _____

TUBESHEET - STATIONARY MAT'L. _____ TUBESHEET - FLOATING MAT'L _____

FLOATING HEAD COVER MAT'L. _____ IMPINGEMENT PROTECTION _____

BAFFLES - CROSS MAT'L. _____ TYPE _____ % CUT (DIAM.) (AREA) _____ SPACING: C/C: _____ INLET _____ (IN)

BAFFLES - LONG MAT'L. _____ SEAL TYPE _____

SUPPORTS - TUBE _____ U-BEND _____ TYPE: _____

BYPASS SEAL ARRANGEMENT _____ TUBE - TUBESHEET JOINT _____

GASKETS - SHELL SIDE _____ - TUBE SIDE _____

- FLOATING HEAD _____

ASME SEC. VIII CODE REQUIREMENTS: ☐ DESIGN & TEST ☐ STAMP ☐ NOT APPLIC. TEMA CLASS _____

WEIGHT/SHELL _____ (LBS); FILLED W/ WATER _____ (LBS); BUNDLE _____ (LBS)

REMARKS: _____

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NEMA FRAME INDUCTION MOTORS TO IEEE 841

MANUFACTURER: _____ MODEL: _____ SERIAL NO.: _____ NEMA FRAME: _____

DRIVEN EQUIPMENT TYPE: _____ DRIVEN EQUIPMENT ITEM NO.: _____ MOTOR ITEM NO.: _____

○ OPERATING CONDITIONS

SITE DATA: (3.1.1)

ELECTRICAL SUPPLY: VOLTS _____, PHASE _____, HERTZ _____

ELECTRICAL AREA CLASSIFICATION: ○ NON-HAZARDOUS

○ CLASS _____, GROUP _____, DIVISION _____

ATMOSPHERIC MIXTURE: _____

IGNITION TEMPERATURE: _____ °(F)(C); TEMP CODE: _____

ALTITUDE: ○ LESS THAN 3300 FT. ○ _____ FT

AMBIENT TEMPERATURE: ○ BETWEEN -25 & 40°C ○ _____ °C

UNUSUAL CONDITIONS: _____

DRIVE SYSTEM:

○ DIRECT CONNECTED ○ EXTERNAL GEAR

○ OTHER _____

STARTING: (3.1.3)(3.1.6)

○ FULL VOLTAGE ○ REDUCED VOLTAGE _____ %

○ LOADED ○ UNLOADED

○ VOLTAGE DIP _____ %

☐ PERFORMANCE

NO LOAD CURRENT (AMPS) _____

FULL LOAD TORQUE (FT-LBS) _____

STARTS PER HOUR: _____ HOT, _____ COLD

ACCELERATION TIME: _____ SECONDS

LOAD	CURRENT-AMP	EFFICIENCY	POWER FACTOR
FULL			
75%			
50%			
LOCKED ROTOR			

☐ CONSTRUCTION FEATURES

☐ NAMEPLATE: _____ HP, RPM _____, S.F. 1.15

NEMA TORQUE DESIGN: ○ A ○ B ○ C ○ D

☐ NEMA LOCKED ROTOR kVA CODE LETER: _____

EFFICIENCY: ○ STANDARD ○ HIGH ○ PREMIUM

NOISE DESIGN: ○ STANDARD ○ LOW NOISE

○ MAX. SOUND PRESSURE LEVEL (dBA) @ 3 FT: _____

☐ EXPECTED SPL (dBA) @ 3 FT: _____

ENCLOSURE: ☐ TEFC ☐ TENV ☐ EXPLOSION PROOF

MOUNTING: ☐ HORIZONTAL ☐ VERTICAL

☐ FOOT MOUNTED ☐ FLANGE MOUNTED

☐ SHAFT UP ☐ SHAFT DOWN

MAIN TERMINAL BOX MOUNTING LOCATION: ☐ F-1 ☐ F-2

FAN: ☐ REVERSIBLE ☐ UNI-DIRECTIONAL ● NON-SPARKING

BEARING TYPE: ☐ BALL ☐ ROLLER ☐ SLEEVE

BEARING LUBRICATION: ☐ GREASE ☐ RING OIL ○ OIL MIST

GREASE FITTING: ☐ PLUGGED ○ ALEMITE ○ OTHER _____

BEARING SHIELDING: ☐ SINGLE ☐ DOUBLE ○ SEALED FOR LIFE

MOTOR ROTATION: (FACING END OPPOSITE SHAFT EXTENSION)

☐ CW ☐ CCW ☐ BI-DIRECTIONAL

INSULATION CLASS: ○ B ● F ○ OTHER: _____

● NON-HYGROSCOPIC ○ TROPICALIZED

☐ TEMPERATURE RISE: *(DEFAULT IS 80°C ABOVE 40°C BY RES. @ 1.0)*

_____ °C ABOVE _____ °C BY: _____ @ _____ S.F.

☐ MOTOR TO BE "THERMALLY PROTECTED"

☐ MOTOR TO BE "OVER TEMP PROT- "

☐ TYPE "1 - "WINDING - RUNNING & LOCKED-ROTOR PROTECTED"

☐ TYPE "2 - "WINDING - RUNNING PROTECTED"

☐ TYPE "3 - "WINDING - PROTECTED, NON-SPECIFIC"

○ SPACEHEATER REQ'D. ☐ RATED AT: _____ WATTS

☐ VOLTS _____, PHASE _____, HERTZ _____

☐ MAX SHEATH TEMPERATURE: _____ °C

○ SEPARATE JUNCTION BOX FOR SPACE HEATER LEADS

MOTOR THRUST LOAD: ○ _____ LBS ○ NONE

DIRECTION OF THRUST: ○ TOWARD COUPLING ○ AWAY FROM COUPLING

☐ MOTOR THRUST RATING: _____ LBS

TESTING

IEEE TESTING: ○ OBSERVED ○ WITNESSED ○ SUBMIT CERT'D. RESULTS

○ SPECIAL TESTING: _____

MISCELLANEOUS

PAINTING: ○ IEEE 841 STD. ○ OTHER: _____

○ _____

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ALLOWABLE PIPING FORCES & MOVEMENTS: (2.2.4)

	COMPRESSOR INLET		COMPRESSOR DISCHARGE		PACKAGE OUTLET	
	FORCE (LBS)	MOMENT (FT.-LBS)	FORCE (LBS)	MOMENT (FT.-LBS)	FORCE (LBS)	MOMENT (FT.-LBS)
AXIAL						
VERT.						
TRANS.						

ADDITIONAL DATA:

8.26.96 SHT 10 OF 10 USDATASH.XLS

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APPLICABLE TO: ○ PROPOSAL ○ PURCHASE ○ AS BUILT

FOR: _____ UNIT _____

SITE: _____ NO. REQUIRED _____

SERVICE: _____ DRIVER ITEM NO. _____

○ CONTINUOUS ○ INTERMITTENT ○ STANDBY (2.1.7) SPARED BY: _____

NOTE: INFORMATION TO BE COMPLETED: ○ BY PURCHASER ☐ BY MANUFACTURER ☐ BY PURCHASER OR MFR.

GENERAL

COMPR.MFR. _____ MODEL (SIZE AND TYPE) _____ SERIAL NO. _____

DRIVER MFR. _____ DRIVER TYPE _____ RATED HP _____ RPM _____

DRIVE SYSTEM ○ DIRECT COUPLED ○ _____

OPERATING CONDITIONS (2.1.3)(3.1.2)

(ALL DATA ON PER UNIT BASIS)

	RATED (2.1.3)	LOW AMB.* (3.1.5.2)	MIN. AMB. (2.1.9)	OTHER
○ DELIVERED FLOW, M³/H (1.013 BARA & 0°C DRY)				
○ WEIGHT FLOW, KG/H (WET)(DRY)				
○ INLET COOLING WATER TEMP (°C)				
INLET CONDITIONS: (2.1.3)(3.1.2)				
○ PRESSURE (BAR)(kPA ABS)				
○ TEMPERATURE (°C)				
○ RELATIVE HUMIDITY (%)				
○ MOLECULAR WEIGHT				
☐ INLET VOLUME (M³/H-WET)				
DISCHARGE CONDITIONS:				
○ PRESSURE (BAR)(kPA ABS)				
☐ TEMPERATURE (°C)				
PERFORMANCE:				
☐ MAX kW REQUIRED (ALL LOSSES INCLUDED)				
☐ kW/100 M³/H AIR DELIVERED				
☐ INPUT SPEED (RPM)				
☐ ESTIMATED SURGE, IM³/H (@ ABOVE SPEED)				
○ MAX. ΔP ACROSS INLET FILTER (BAR)(kPA)				
ΔP INCLUDED IN CALCULATION ☐ YES ☐ NO				
☐ AFTERCOOLER OUTLET TEMP. (°C)				
☐ PERFORMANCE CURVE NO.				
☐ % RISE TO SURGE (2.2.1.4)				
☐ % TURNDOWN (2.2.1.4)				
☐ _____				

*UNTHROTTLED PERFORMANCE FOR DRIVER SIZING

CONTROL SYSTEM (3.4.2)

CONTROL METHOD (3.4.2.1):

○ CAPACITY MODULATION *(CONSTANT DISCH. PRESSURE)* (3.4.2.1a):
- ○ INLET THROTTLE DEVICE: ○ DAMPER;
- ○ GLOBE VALVE; ○ BUTTERFLY VALVE;
- ○ VARIABLE INLET GUIDE VANES

○ TWO STEP CONTROL *(LOAD-UNLOAD)* (3.4.2.1b):
- ○ _____ TO _____ DISCH. P (BARG)(kPA)

○ AUTOMATIC DUAL CONTROL (3.4.2.1c)

○ AUTO. START AND STOP (3.4.2.1d):
- ○ START _____; STOP _____ (BARG)(kPA)

○ CONSTANT DISCHARGE PRESS. BASE MODE *(BLOWOFF)* (3.4.2.1e)

○ MASS FLOW CONTROL (3.4.2.1f)

CONTROL SYSTEM REQUIREMENTS:

○ UNIT OPERATES IN PARALLEL (3.4.2.2)
- ○ W/ CENTRIFUGAL;
- ○ W/ ROTARY; ○ W/ RECIPROCATING

○ MICROPROCESSOR CAPABLE OF COMMUNICATION WITH PURCHASER'S DCS (3.4.1.2)

CONTROL SYST. ALTERNATES (3.4.1.1)

○ OTHER THAN MICROPROCESSOR BASED: _____

○ SUITABLE FOR INDOOR ONLY

○ FURNISHED BY PURCHASER

INTER- & AFTER-COOLERS

AFTERCOOLER (3.6.2):

○ FURNISHED BY PURCHASER (3.6.1)

○ NOT NEEDED (3.6.1)

○ AIR-COOLED TYPE (3.6.3) BY VENDOR

○ AIR-COOLED INTERCOOLERS REQ'D. (3.6.3)
- ○ FURNISHED BY PURCHASER

☐ AIR-COOLED EXCHANGER AUTOMATIC TEMPERATURE CONTROL MEANS: (3.6.6)
- ○ LOUVERS ○ VARIABLE SPEED FANS
- ○ VARIABLE PITCH FANS ○ BYPASS VALVE

☐ AIR-COOLER CONTROL MANUAL ONLY (3.6.6) BY:
- ○ LOUVERS ○ BYPASS VALVE
- ○ VARIABLE PITCH FANS

REMARKS: _____

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○ LOCATION, SITE DATA

LOCATION (2.1.9):

○ INDOOR ○ HEATED ○ UNDER ROOF

○ OUTDOOR ○ UNHEATED ○ PARTIAL SIDES

○ GRADE ○ MEZZANINE ○ _____

○ WINTERIZATION REQ'D. ○ TROPICALIZATION REQ'D.

SITE DATA:

○ ELEVATION _____ M. ○ BAROMETER _____ (BARA)(kPA ABS)

○ RANGE OF AMBIENT TEMPERATURE (°C):

	DRY BULB	WET BULB
NORMAL		
MAXIMUM		
MINIMUM		

UNUSUAL CONDITIONS: (2.1.9)

○ DUST ○ FUMES ○ CORROSIVE CONDITIONS

○ CORROSIVES PRESENT: _____

○ CONDITIONS CAUSE STRESS CORROSION CRACKING (2.2.10.1.7)

○ OTHER _____

AREA ELECTRICAL CLASSIFICATION (2.1.13) ZONE _____ T-CODE _____

○ CLASS _____ GROUP _____ DIVISION _____

○ LOCAL ELECTRICAL CODES: _____

○ UTILITY CONDITIONS: (3.1.1)

○ **STEAM HEATING: (2.2.9.5.7)**

INLET MIN.:	_____ (BARG)(kPA)	_____ °C
NORM.:	_____ (BARG)(kPA)	_____ °C
MAX.:	_____ (BARG)(kPA)	_____ °C
OUTLET MIN.:	_____ (BARG)(kPA)	_____ °C
NORM.:	_____ (BARG)(kPA)	_____ °C
MAX.:	_____ (BARG)(kPA)	_____ °C

○ **ELECTRICITY:**

	HEATING	CONTROL	SHUTDOWN
VOLTAGE:			
HERTZ:			
PHASE:			

○ **COOLING WATER: (2.1.10)**

TEMP: INLET _____ °C MAX. RETURN _____ °C

PRESS: NORM. _____ (BARG)(kPA) DESIGN _____ (BARG)(kPA)

MIN. RETURN _____ (BARG)(kPA) MAX. ALLOW ΔP _____ (BAR)(kPA)

WATER SOURCE _____

○ **AIR/NITROGEN:**

MAX. PRESS. _____ (BARG)(kPA) MIN. PRESS _____ (BARG)(kPA)

GAS COMPOSITION: _____

REMARKS: _____

○ SPECIFICATIONS

NOISE SPECIFICATIONS (2.1.4):

○ MAX. ALLOWABLE SPL _____ @ 1M.

○ APPLICABLE SPEC. _____

ACOUSTIC HOUSING: ○ YES ○ NO

APPLICABLE SPECIFICATIONS:

API 672 AND ○ _____

○ NON-ASME WELDING IF NOT AWS D1.1: (2.2.10.4.4) _____

PAINTING:

○ MANUFACTURER'S STANDARD

○ OTHER _____

BASEPLATE GROUT: (3.3.16) ○ EPOXY ○ CEMENT ○ NONE

PREPARATION FOR GROUT SURFACES (3.3.16):

○ MFR. STD. ○ SSPC-6 BLAST ○ BARE FOR FIELD BLAST

○ INORGANIC ZINC SILICATE COATING

○ OTHER _____

SHIPMENT: (4.4.1)(4.4.3.12)

○ DOMESTIC ○ EXPORT ○ EXPORT BOXING REQ'D

○ OUTDOOR STORAGE OVER 6 MONTHS

☐ MAT'L FOR SPARE ROTOR SUPPORT (4.4.3.12) _____

☐ UTILITY CONSUMPTION: (5.2.3l)

STEAM:

OIL HEATER: _____ KG/H. OTHER: _____ KG/H.

ELECTRIC:

	kW	LOCKED ROTOR AMPS.	FULL LOAD AMPS.
MAIN LO PUMP:			
AUX. LO PUMP:			

OIL HEATER: _____ W; SPACE HEATER _____ W

CONTROL SYSTEM LOAD: _____ W (3.4.2.8)

COOLING WATER:

	LO COOLER	INTER COOLER	AFTER COOLER	OTHER
QUANTITY, M³/H				
OUTLET TEMP, °C				
PRESS. DROP, (BAR)(kPA)				
TOTAL CW, M³/H				

AIR/NITROGEN:

	INLET PRESS. (BARG)(kPA)	QUANTITY (SM³/H)
SEAL SYSTEM:		
CONTROL PANEL:		
LO RESERVOIR:		
INSTR. HOUSINGS:		
CONTROL SYSTEM:		
OTHER:		

TOTAL PURGE, SM³/H: _____

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CONSTRUCTION FEATURES

☐ **COMPRESSOR SPEEDS:**

RATED INPUT: _____ RPM TRIP: _____ RPM

BULLGEAR CRITICALS: 1st: _____ RPM 2nd: _____ RPM

PINION CRITICALS:

1st STG. PINION: 1st: _____ RPM 2nd: _____ RPM

2nd STG. PINION: 1st: _____ RPM 2nd: _____ RPM

3rd STG. PINION: 1st: _____ RPM 2nd: _____ RPM

4th STG. PINION: 1st: _____ RPM 2nd: _____ RPM

OTHER UNDESIRABLE SPEEDS: (2.2.7.1.3) _____

	STAGE SPEED	IMPELLER DIAMETER	TIP SPEED
1st STAGE:	_____ RPM	_____ MM.	_____ MPS
2nd STAGE:	_____ RPM	_____ MM.	_____ MPS
3rd STAGE:	_____ RPM	_____ MM.	_____ MPS
4th STAGE:	_____ RPM	_____ MM.	_____ MPS

☐ **IMPELLERS:**

NO. OF IMPELLERS _____; MATERIAL _____

TYPE: (OPEN, RADIAL, BACKWARD LEANING, ETC.) _____

TYPE CONSTRUCTION: (2.2.5.2.2) _____

METHOD OF ATTACH: (2.2.5.2.2) _____

ROTATION, VIEWED FROM INPUT SHAFT END: ☐ CW ☐ CCW

☐ **COMPRESSOR CASING:**

MODEL: _____; CASING SPLIT: _____

	STG. 1	STG. 2	STG. 3	STG. 4
MATERIAL:				
M.A.W.P. (BAR)(kPA)				
HYDRO TEST (BAR)(kPA)				
MAX OPER TEMP (°C)				
MIN OPER TEMP (°C)				

○ MIN. DESIGN METAL TEMP.: (2.2.10.5.3) _____ °C

☐ CASING HEAT TREATMENT REQUIRED (2.2.10.4.5.4)

☐ MAX. ALLOW. STRESS FOR MAT'L: (2.2.2.1) _____ N/MM²

☐ CODE CASTING FACTOR: (2.2.2.1) _____

WELDING CONNECTIONS - NDT REQ'D: (2.2.10.4.6.1)

☐ 100% RADIOGRAPH ☐ MAG. PARTICLE ☐ LIQ. PENETRANT

○ PURCHASER REVIEW OF CONNECTION DESIGN (2.2.10.4.6.4)

☐ **COMPRESSOR BEARINGS & BEARING HOUSES:**

BEARING HSG. MATERIAL:

PINION RADIAL BEARINGS: (2.2.8.2)

	STG. 1	STG. 2	STG. 3	STG. 4
BRG. TYPE:				
ALLOW LOAD: N				
ACTUAL LOAD: N				
BRG. SPAN: MM				

PINION THRUST BEARINGS: (2.2.8.2)

	STG. 1	STG. 2	STG. 3	STG. 4
BRG. TYPE:				
ALLOW LOAD: N				
ACTUAL LOAD: N				
TRUST COLLAR:				

☐ **INTEGRAL GEAR HOUSING:**

MATERIAL: _____; SPLIT: _____

☐ **BULL GEAR: (2.2.5.3)**

RATED POWER BASED ON TOOTH SURFACE DURABILITY: _____ kW

RATED POWER BASED ON TOOTH BENDING: _____ kW

○ MIN. AGMA SERVICE FACTOR: _____ ☐ ACT.S.F. _____

GEAR RIM MAT'L: _____ HARDNESS: _____

GEAR FACE WIDTH: _____ MM; GEAR CENTER MAT'L: _____

MECHANICAL EFFICIENCY: _____ %; AGMA QLTY: _____

PITCH DIA. _____ MM; PITCH LINE VEL. _____ MPS

☐ **PINIONS: (2.2.5.3)**	1st	2nd	3rd	4th
SERVICE FACTOR:				
MATERIAL:				
HARDN'S:(BHN)(R_c)				

☐ **BULL GEAR SHAFT:**

☐ REPLACEABLE ☐ INTEGRAL W/ GEAR

MAT'L: _____ HARDNESS: _____ (BHN)(R_c)

BRG. SPAN: _____ MM; WEIGHT (W/GEAR): _____ KG

DIA. @ GEAR: _____ MM; DIA @ CPLG: _____ MM

SHAFT SLEEVES AT SEALS: MAT'L: _____

SHAFT LABY'S: TYPE: _____; MAT'L: _____

BULL GEAR RAD. BRG. TYPE: _____; LENGTH: _____ MM

ALLOW LOAD: _____ N/MM²; ACT LOAD: _____ N/MM²

☐ **BULL GEAR THRUST BEARINGS: (2.2.8.3)**

LOCATION: _____; TYPE: _____

MFR: _____; AREA: _____ MM²

THRUST COLLAR: (2.2.8.3.5) ☐ INTEGRAL ☐ REPLACEABLE

ALLOW. LOAD: _____ N/MM²; ACT. LOAD: _____ N/MM²

GAS LOAD: _____ N; CPLG. LOAD: _____ N

BEARINGS FITTED W/ TEMP. SENSORS (2.2.8.4.4)

○ PINION RAD. BRG. ○ BULLGEAR RAD. BRG.

○ THRUST BRG. _____

☐ **MAIN CONNECTIONS: (2.2.3.1)**

	SIZE	ASME RATING	FACING	POSITION
COMP. INLET:				
COMP. DISCH.:				
PKG. OUTLET:				
ATM. BLOWOFF:				
FILTER OUTLET:				

☐ **OTHER CONNECTIONS:**

	NO.	SIZE	TYPE
LUBE OIL INLET:			
LUBE OIL OUTLET:			
C.W. INLET:			
PRESS. GAUGE:			
TEMP. GAUGE:			
COND. DRAINS:			

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VIBRATION DETECTORS: (3.4.4.6)

☐ TYPE: _____; ☐ MODEL: _____

○ MFR.: _____

○ NO. AT EACH PINION BEARING: _____ TOTAL NO.: _____

○ NO. AT EACH DRIVER BEARING: _____ TOTAL NO.: _____

☐ X&Y RADIAL PROBES CAN BE MOUNTED ADJACENT TO IMPELLERS FOR:

☐ 1st STG. ☐ 2nd STG. ☐ 3rd STG. ☐ 4th STG.

OSCILLATOR-DEMODULATORS:

○ MFR.: _____ ☐ MODEL: _____

○ MONITOR SUPPLIED BY: _____

○ MFR.: _____ ☐ MODEL: _____

○ LOCATION: _____ ENCLOSURE: _____

☐ READOUT SCALE RANGE: _____ ○ ALARM ☐ SET @ _____ μ

○ SHUTDOWN ☐ SET @ _____ μ ○ TIME DELAY: _____ SEC

○ PER API 670: (3.4.4.6.2&3)

BEARING-TEMPERATURE MONITOR: (3.4.4.6.4)

○ REQ'D. ○ SUPPLIED BY: _____ ○ PER API 670

○ MFR: _____ ☐ MODEL: _____

AXIAL POSITION MOVEMENT DETECTOR: (3.4.4.6.2)

☐ TYPE: _____ ☐ MODEL: _____

○ MFR: _____

☐ READOUT SCALE RANGE: _____ ○ ALARM ☐ SET @ _____ μ

○ SHUTDOWN ☐ SET @ _____ μ ○ TIME DELAY: _____ SEC

DYNAMICS: (2.2.7)

○ CRITICAL LATERAL SPEEDS ARE PROVEN BY PRIOR UNITS (2.2.7.2.1)

○ DAMPED UNBALANCED RESPONSE ANALYSIS REQ'D (2.2.7.2.2)

○ TORSIONAL VIBRATION ANALYSIS OF TRAIN REQ'D (2.2.7.3.1)

○ RESIDUAL UNBALANCE WORKSHEET REQ'D (2.2.7.4.3)

☐ REMARKS: _____

COUPLINGS: (3.2)

TYPE: ☐ DISK PAK ☐ DIAPHRAGM ○ OTHER _____

DISK MAT'L: ☐ STAINLESS STEEL ☐ COATED W/ _____

☐ MAKE: _____ ☐ MODEL: _____

○ NON-LUBE ○ LUB'D. ☐ LUBRICATION: _____

☐ SPACER LENGTH: _____ MM ☐ LIMITED END-FLOAT REQ'D

☐ CPLG. RATING: _____ kW/100 RPM @ 1.0 S.F.; ACTUAL S.F.: _____

☐ SHAFT JUNCT. RATING: @ DRIVER: _____ kW; @ INPUT SHAFT: _____ kW

☐ MOUNTING ARR'GMT: @ INPUT SHAFT: _____ DRIVER: _____

☐ MFT. MAX. BORE: _____ MM; PROPOSED BORE _____ MM (3.2.1.9)

DRIVER HALF-CPLG. MTD BY: ☐ DRIVER MFR ☐ COMP. VENDOR

☐ IDLING ADAPTER FOR DRIVER HALF-CPLG. REQ'D (3.2.1.6)

PIPING REQUIREMENTS:

☐ RECOMMENDED STRAIGHT RUN OF PIPE DIA. BEFORE SUCTION: _____

○ VENDOR TO OBSERVE FLANGE PARTING (2.1.15ba)

○ THROUGH STUDS REQUIRED FOR PIPING FLANGES (3.5.1.18)

MISCELLANEOUS:

○ VENDOR PRESENT DURING INITIAL ALIGN. CHECK (2.1.15b)

○ VENDOR CHECK ALIGN. AT OPERATING TEMP. (2.1.15c)

○ BASE DESIGNED FOR COLUMN MOUNTING (3.3.11)

○ THERMAL RELIEF VALVES PROVIDED BY VENDOR (3.4.4.6.3)

○ FOR WATER COOLED EXCHANGERS

○ FOR _____

○ PURCHASER WILL PREPARE COORDINATION MEETING AGENDA (5.1.3)

○ SHOP INSPECTIONS & TESTS: (4.1.1)

○ ADVANCE NOTIFICATION REQ'D: _____ DAYS (4.1.1)

	OBS'D	WIT.
● SHOP INSPECTION: (4.1.3)		
● HYDROSTATIC: (4.3.2)	○	○
● 4-HOUR COMBINED TEST: (4.3.4)	○	○
○ ASME, PTC-10 TEST: (4.3.5.1)	○	○
INCLUDES ○ AIR FILTER;		
○ AFTERCOOLER;		
○ _____		
○ GUIDE VANE TEST: (4.3.5.2)	○	○
○ AT _____ NON-100% POSITIONS		
○ SOUND-LEVEL TEST: (4.3.5.3)	○	○
○ SPARE ROTOR TEST: (4.3.5.4)	○	○
○ SPARE ROTOR MECH ONLY:	○	○
● IMPELLER OVERSPEED TEST: (4.3.3)	○	○
● RADIOGRAPH OF IMPELLERS: (4.2.2.6)	○	○
○ RESID. UNBALANCE CHECK: (2.2.7.4.3)	○	○
● OIL SYSTEM CLEAN: (4.3.4.2.2)	○	○
● CONTROL SYSTEM CHECK: (4.3.4.2.4)	○	○
● BRG., SEAL, GEAR CHECK: (4.3.4.5.1)	○	○
● GEAR CONTACT CHECK: (4.3.4.5.3)	○	○
● CLEANLINESS CHECK - VESSELS: (4.2.3.3)	○	○
● CLEANLINESS CHECK - PIPING: (4.2.3.3)	○	○
○ HARDNESS CHECK OF PINIONS: (4.2.3.4)	○	○
○ OF BULL-GEAR:	○	○
○ OF WELD REPAIRS:	○	○
○ NDE OF MAJOR REPAIRS: (2.2.10.4.2)	○	○
○ GEAR TOOTH MAG-PART.: (4.2.2.8)	○	○
○ FINAL INSP. PRIOR TO PAINT:	○	○
○ INSP. OF PREP. FOR SHIPMENT: (4.4)	○	○
_____	○	○
_____	○	○

○ PRIOR DOCUMENT. ON MAJOR REPAIRS NOT REQ'D (2.2.10.4.3)

○ RETAIN FINAL ASSEMBLY CLEARANCES (4.2.1.1d)

○ SUBMITTAL OF INSPECTOR'S CHECKLIST (4.1.5)

SIGNED BY REP FOR: ○ PURCHASER ○ VENDOR

IF DESIGN REQ'S DISASSEMBLY OF PINION FOR BRG. INSPECTION,

○ FORGO BEARING INSPECTION BASED ON TEST DATA; OR

○ INSPECT BEARING AND RETEST. (4.3.4.5.2)

☐ **WEIGHT, KGS.**

INTEG. GEAR/COMP: _____ DRIVER: _____

GEAR UPPER CASE: _____ BULL-GEAR: _____

1st STG. PINION: _____ 2nd STG. PINION: _____

INTERCOOLER: _____ BUNDLE: _____

AFTERCOOLER: _____ BUNDLE: _____

BASE: _____ CONTROL PANEL: _____

MAX. FOR MAINTENANCE (IDENTIFY): _____

TOTAL SHIPPING WT.: _____

☐ **SPACE REQUIREMENTS (MM)**

	L	W	H
COMPLETE UNIT:	_____	_____	_____
CONTROL PANEL: (IF SEP.)	_____	_____	_____
INLET FILTER - SILENCER:	_____	_____	_____
AFTERCOOLER(IF FURN.):	_____	_____	_____
OTHER: _____	_____	_____	_____

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LUBE OIL SYSTEM (2.2.9)

BASIC SYSTEM REQ'MTS. - NORMAL OIL FLOW (2.2.9.5.2)

LUBE OIL TO:	M³/H	(BARG)(kPA)	SSU @ 40°C
☐ COMP./GEAR:			
☐ DRIVER:			
☐ EXT. GEAR:			

☐ OIL SYSTEM PRESSURES: (BARG)(kPA)

SUPPLY: _____ PUMP RV SETTING: _____

SYSTEM DESIGN: _____ HYDROTEST: _____

OIL COOLER: (2.2.9.5.5)

	SHELL SIDE	TUBESIDE
☐ OPERATING PRESSURE: (BARG)(kPA)		
☐ MAX. ALLOW WORK PRESS: (BARG)(kPA)		
☐ MAX. ALLOW TEMP: °C		
☐ FOULING FACTOR:		

☐ SURFACE AREA: _____ M²; DUTY: _____ Kcal/H

☐ REMOVABLE BUNDLE TO BE FURNISHED

☐ ASME CODE STAMPED ○ DESIGNED TO TEMA _____

☐ TUBES: NO. _____; O.D. _____ MM; LENGTH: _____ MM

WALL THICKNESS: _____ MM ☐ AVG. ☐ MIN.

☐ MATERIALS:

CHANNELS/HEADS: _____; SHELL: _____

TUBES: _____; TUBE SHEETS: _____

CHANNEL COVERS: _____; TUBE SUPTS: _____

OIL FILTERS: (2.2.9.5.6)

☐ MICRON RATING: _____ ○ NOMINAL ○ ABSOLUTE

☐ ΔP(BAR)(kPA): CLEAN: _____; DIRTY: _____; COLLAPSE: _____

☐ ELEMENT: MAKE: _____; MODEL: _____

☐ NO. ELEMENTS: _____; ☐ MEDIA: _____

☐ CORE MAT'L: _____; ☐ HSG. MAT'L: _____

☐ HSG. MAWP: _____ (BARG)(kPA) ☐ MAX. ALLOW TEMP. _____ °C

OIL HEATER: (2.2.9.5.7)

☐ STEAM HEATER REQ'D; ☐ ELECTRIC HEATER REQ'D;

☐ RATING: _____ kW

☐ WATT DENSITY: _____ W/CM²

OIL RESERVOIR: (2.2.9.5.8)

☐ RETENTION TIME: _____ MIN.; ☐ CAPACITY: _____ M³

☐ FREE SURFACE AREA: _____ M² ☐ INT'L BAFFLES

LUBRICANT: (2.2.9.3) ☐ SYNTHETIC ☐ HYDROCARBON

☐ DESCRIPTION: _____

☐ MIN. ALLOW. OIL TEMP: (2.2.9.5.2) _____ °C, _____ SSU

☐ SYSTEM COMPONENT SUPPLIERS:

	MFR.	MODEL
MAIN PUMP:		
STANDBY PUMP:		
ELECTRIC MOTOR(S):		
STEAM TURBINE(S):		
OIL COOLER(S):		
OIL FILTERS:		
ACCUMULATOR(S):		
SUCTION STRAINERS:		
CHECK VALVES:		
TRANSFER VALVE(S):		
PUMP COUPLING:		
PUMP RELIEF VALVES:		
ELECTRIC HEATER:		
_____		

PUMPS: (2.2.9.5.2)	MAIN	STANDBY
☐ HORIZONTAL:		
☐ VERTICAL:		
☐ SUBMERGED:		
☐ MOTOR DRIVEN:		
☐ TURBINE DRIVEN:		
☐ SHAFT DRIVEN:		
☐ CENTRIFUGAL:		
☐ ROTARY:		
☐ FLANGE CONNECTED:		
☐ M³/H (RATED):		
☐ @ (BARG)(kPA) DISCH.:		
☐ SkW @ MAX. SSU:		
☐ DRIVER kW:		
☐ CASING MATERIAL:		
☐ SPEED:		
☐ COUPLING:		
☐ OSHA GUARD:		
☐ MECHANICAL SEAL:		

STANDBY PUMP CONTROL RESET:

○ MANUAL ○ AUTOMATIC ○ HOA SELECTOR SWITCH

SILENCERS

INLET AIR FILTER/SILENCER: (3.7)

☐ MFR.: _____; MODEL: _____

☐ DESCRIPTION: _____

☐ PIPING CONNECTION: _____

☐ CLEAN ΔP, AS QUOTED: _____ (BAR)(kPA)

☐ CORROSION PROTECTION: _____

○ FILTER WILL BE REMOTE MOUNTED BY PURCHASER AT A DISTANCE _____ M FROM COMPRESSOR.

○ FILTER WILL BE ELEVATED _____ M ABOVE GRADE.

DISCHARGE BLOWOFF SILENCER: (3.7)

☐ MFR.: _____; MODEL: _____

☐ DESCRIPTION: _____

☐ FLANGE CONNECTION:

MOUNTING: ☐ HORIZONTAL ☐ VERTICAL

SUPPORTED BY: ☐ PIPING ☐ OTHER: _____

☐ SPL (dBA) AT 1M FROM DISCHARGE OF SILENCER: _____

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CONTROLS AND INSTRUMENTATION (3.4)

LOCAL CONTROL PANEL: (3.4.3)

◯ ELECTRICAL AREA CLASSIFICATION: CL _____, GR _____, DIV _____ (3.4.3.2)

I.E.C. G.R: _____, ZONE: _____; TEMP. CLASS: _____

PANEL ENCLOSURE REQUIREMENT: (3.4.3.2)

☐ NEMA TYPE 4X ENCLOSURE MODEL: _____ ☐ NEMA TYPE 7

☐ IEC TYPE: ☐ "D"-FLAMEPROOF ☐ "P"-PRESSURIZED ☐ "E"-INCREASED SAFETY

PANEL FEATURES (3.4.3.2)

◯ VIBRATION ISOLATORS ☐ STRIP HEATER ☐ INTERNAL COOLING

◯ WEATHERHOOD ☐ PURGE CONNECTIONS ◯ OTHER _____

PURGE REQUIREMENT: (3.4.3.2)

☐ NONE ◯ INSTRUMENT AIR ◯ NITROGEN

☐ TYPE X - *REDUCES THE CLASSIFICATION FROM DIV.1 TO NONHAZARDOUS.*

☐ TYPE Y - *REDUCES THE CLASSIFICATION FROM DIV.1 TO DIV.2.*

☐ TYPE Z - *REDUCES THE CLASSIFICATION FROM DIV.2 TO NONHAZARDOUS.*

◯ TROPICALIZATION REQUIRED (3.4.6.6)

☐ **INSTRUMENT SUPPLIERS:**

	MFR.:	SIZE & TYPE:
PRESSURE GAUGES:		
TEMPERATURE GAUGES:		
LEVEL GAUGES:		
DIFF. PRESSURE GAUGES:		
PRESSURE SWITCHES:		
TEMPERATURE SWITCHES:		
LEVEL SWITCHES:		
PRESSURE TRANSMITTERS:		
TEMPERATURE TRANSMITTERS:		
LEVEL TRANSMITTERS:		
CONTROL VALVES:		
PRESSURE RELIEF VALVES:		
THERMAL RELIEF VALVES:		
TEMPERATURE CONTROL VALVE:		
SIGHT FLOW INDICATORS:		
PURGE FLOW INDICATORS:		
SOLENOID VALVES:		
ANNUNCIATOR:		
TUBE FITTINGS:		
_____		
_____		
_____		
_____		

SWITCH CLOSURES: (3.4.5.3.2)

ALARM CONTACTS SHALL: ◯ OPEN ◯ CLOSE TO SOUND ALARM AND BE NORMALLY ◯ ENERGIZED ◯ DE-ENERGIZED

SHUTDOWN CONTACTS SHALL: ◯ OPEN ◯ CLOSE TO TRIP AND BE NORMALLY ◯ ENERGIZED ◯ DE-ENERGIZED

(NOTE: *NORMAL* CONDITION IS WHEN COMPRESSOR IS IN OPERATION.)

◯ SHUTDOWN SYSTEMS ARE *NOT* TO BE PROVIDED WITH A MEANS TO PERMIT TESTING WITHOUT SHUTTING DOWN THE UNIT. (3.4.5.3.3)

◯ NON-SHUTDOWN DEVICES ARE *NOT* REQUIRED TO HAVE VALVING TO PERMIT REPLACEMENT DURING OPERATION. (3.4.5.4.1)

◯ ISOLATION VALVES *ARE* REQUIRED FOR SHUTDOWN SENSING DEVICES. (3.4.5.4.1)

MISCELLANEOUS INSTRUMENTATION:

◯ THROUGH FLOWING INSTRUMENT SENSING LINE REQUIRED. (3.5.2.4)

◯ LIQUID FILLED GAUGES ARE REQUIRED FOR AREAS SUBJECT TO VIBRATION. (3.4.4.4.1)

◯ RELIEF VALVES MAY HAVE BODIES IN MATERIALS OTHER THAN STEEL. (3.4.4.8.2.) ☐ RV BODY MATERIAL: _____

◯ THERMAL RELIEFE VALVES REQUIRED FOR COMPONENTS THAT CAN BE ISOLATED. (3.4.4.9.1)

◯ FLOW INDICATOR TYPE/MATERIAL IF OTHER THAN BULLS EYE TYPE WITH STEEL BODY: (3.4.4.9.1) _____

◯ PURGE REQUIRED FOR ANNUNCIATOR. (3.4.5.2) NFPA 496 PURGE TYPE: ◯ X ◯ Y ◯ Z ◯ CONNECTION ONLY

◯ COMBINATION BLOCK & BLEED VALVES MAY BE SUBSTITUTED. (3.5.3.1)

◯ _____

◯ _____

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INSTRUMENTATION SCOPE OF SUPPLY

	ELEMENT										INDICATOR							
	PROV BY		TYPE			LOCATION			INSTALL BY		PROV BY		LOCATION					
	VENDOR	PURCHASER	DIRECT READOUT	SWITCH	TRANSMITTER (1)	VENDOR PKG.	LOCAL PANEL	PURCH. PIPING	VENDOR	PURCHASER	VENDOR	PURCHASER	VENDOR PKG.	PURCH PIPING	LOCAL PANEL	ALARM	SHUTDOWN	REPEAT SIGNAL
PRESSURE:																		
COMPRESSOR SUCTION _____ STAGE																		
COMPRESSOR DISCH. _____ STAGE																		
LUBE OIL DISCHARGE																		
LUBE OIL FILTER ΔP																		
LUBE OIL SUPPLY																		
AIR FILTER/SILENCER ΔP																		
TEMPERATURE:																		
COMPRESSOR SUCTION _____ STAGE																		
COMPRESSOR DISCH. _____ STAGE																		
OIL COOLER INLET & OUTLET																		
COMP. PINION JOURNAL BRG.																		
BULL GEAR JOURNAL BRG.																		
BULL GEAR THRUST BRG.																		
DRIVER JOURNAL BRG.																		
DRIVER THRUST BRG.																		
RESERVOIR																		
LEVEL:																		
LUBE OIL RESERVOIR																		
SEPARATOR																		
VIBRATION:																		
RADIAL VIBRATION EACH STAGE																		
RADIAL VIBRATION BULL GEAR SHAFT																		
AXIAL POSITION BULL GEAR SHAFT																		
AXIAL POSITION _____ STAGE PINION																		
RADIAL VIBRATION ON DRIVER																		
AXIAL POSITION ON DRIVER SHAFT																		
ACCELEROMETER ON GEAR BOX																		
FLOW:																		
OIL RETURN																		
SEAL GAS																		
MISCELLANEOUS:																		
STANDY L.O. PUMP RUNNING																		
PANEL PURGE FAILURE																		
ANNUNCIATOR PURGE FAILURE																		
SURGE RECOGNITION																		
OIL HEATER ON																		
COMMON REMOTE ALARM INDICATION																		
COMMON REMOTE SHUTDOWN INDICATION																		

NOTES: 1) TRANSMITTERS SUPPLIED BY VENDOR SHALL INCLUDE SENSING ELEMENT.
2) SUPPLY "REPEAT SIGNAL" FOR CONTROL ROOM ALSO.

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INTEGRALLY GEARED CENTRIFUGAL AIR COMPRESSOR DATA SHEET SI UNITS

JOB NO. _____ ITEM NO. _____ PAGE _____ OF _____

PURCHASE ORDER NO. _____ DATE _____

INQUIRY NO. _____ BY _____

REVISION _____ DATE _____

(INTER-) (AFTER) COOLER(S) (3.6)

SERVICE OF UNIT: _____ ITEM NO.: _____

SIZE: _____ TYPE: _____ ☐ HOR ☐ VERT CONNECTED IN: ☐ PARALLEL ☐ SERIES

SURF/UNIT (GROSS/EFF.) _____ SQ.M: SHELLS/UNIT: _____ SURF./SHELL (GROSS/EFF.) _____ SQ.M

PERFORMANCE OF ONE UNIT

	SHELL SIDE	TUBE SIDE
○ FLUID NAME		
☐ FLUID QUANTITY, TOTAL (KG/H)		
VAPOR - IN/OUT		
LIQUID - IN/OUT		
☐ TEMPERATURE - IN/OUT (°C)		
☐ SPECIFIC GRAVITY		
☐ VISCOSITY, LIQUID (CP) (SSU)		
☐ SPECIFIC HEAT (KCAL/KG-°C)		
☐ THERMAL CONDUCTIVITY (KCAL-M/H-M²-°C)		
☐ LATENT HEAT (KCAL/KG-°C)		
☐ INLET PRESSURE (BARG)(kPA)		
◯ VELOCITY (M/SEC)		
◯ PRESSURE DROP - ALLOW./CALC. (BARG)(kPA)		
◯ FOULING RESISTANCE - MINIMUM		

☐ HEAT EXCHANGED _____ KCAL/H MTD - CORRECTED _____ °C

☐ TRANSFER RATE (KCAL/H-M²-°C): SERVICE _____ CLEAN _____

☐ CONSTRUCTION OF ONE SHELL

	SHELL SIDE	TUBE SIDE
DESIGN/TEST PRESS. (BARG)(kPA)		
DESIGN TEMPERATURE (°C)		
NO. PASSES PER SHELL		
CORROSION ALLOWANCE (MM)		
NOZZLES: _____ INLET		
SIZE & _____ OUTLET		
RATING _____ VENT-DRAIN		

SKETCH: BUNDLE, NOZZLE ORIENTATIONS

TUBE NO. _____: OD _____ (MM); THK (MIN.)(AVG.) _____ (MM); LENGTH _____ (M); PITCH _____ (MM) ◁ 30 △ 60 ☐ 90 ◇ 45

TUBE TYPE: _____; MATERIAL _____

SHELL MAT'L. _____; ID _____ (MM); OD _____ (MM); SHELL COVER MAT'L. _____ (INTEG.) (REMOV.)

CHANNEL OR BONNET MAT'L. _____ CHANNEL COVER MAT'L. _____

TUBESHEET - STATIONARY MAT'L. _____ TUBESHEET - FLOATING MAT'L. _____

FLOATING HEAD COVER MAT'L. _____ IMPINGEMENT PROTECTION. _____

BAFFLES - CROSS MAT'L. _____ TYPE _____ % CUT (DIAM.) (AREA) _____ SPACING: C/C: _____ INLET _____ (MM)

BAFFLES - LONG. MAT'L. _____ SEAL TYPE _____

SUPPORTS - TUBE _____ U-BEND _____ TYPE: _____

BYPASS SEAL ARRANGEMENT _____ TUBE - TUBESHEET JOINT _____

GASKETS - SHELL SIDE _____ - TUBE SIDE _____

- FLOATING HEAD _____

ASME SECT. VIII CODE REQUIREMENTS: ☐ DESIGN & TEST ☐ STAMP ☐ NOT APPLIC. TEMA CLASS _____

WEIGHT/SHELL _____ (KG); FILLED W/ WATER _____ (KG); BUNDLE _____ (KG)

REMARKS: _____

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NEMA FRAME INDUCTION MOTORS TO IEEE 841

MANUFACTURER: _____ MODEL: _____ SERIAL NO.: _____ FRAME: _____ (NEMA)(IEC)

DRIVEN EQUIPMENT TYPE: _____ DRIVEN EQUIPMENT ITEM NO.: _____ MOTOR ITEM NO.: _____

○ OPERATING CONDITIONS

SITE DATA: (3.1.1)

ELECTRICAL SUPPLY: VOLTS _____, PHASE _____, HERTZ _____

ELECTRICAL AREA CLASSIFICATION: ○ NON-HAZARDOUS

○ CLASS _____, GROUP _____, DIVISION _____

ATMOSPHERIC MIXTURE: _____

IGNITION TEMPERATURE: _____ °(F)(C); TEMP. CODE: _____

ALTITUDE: ○ LESS THAN 1000 M ○ _____ M

AMBIENT TEMPERATURE: ○ BETWEEN -25 & 40°C ○ _____ °C

UNUSUAL CONDITIONS: _____

DRIVE SYSTEM:

○ DIRECT CONNECTED ○ EXTERNAL GEAR

○ OTHER _____

STARTING: (3.1.3)(3.1.6)

○ FULL VOLTAGE ○ REDUCED VOLTAGE _____ %

○ LOADED ○ UNLOADED

○ VOLTAGE DIP _____ %

☐ PERFORMANCE

NO LOAD CURRENT (AMPS) _____

FULL LOAD TORQUE (N-M) _____

STARTS PER HOUR: _____ HOT, _____ COLD

ACCELERATION TIME: _____ SECONDS

LOAD	CURRENT-AMP	EFFICIENCY	POWER FACTOR
FULL			
75%			
50%			
LOCKED ROTOR			

☐ CONSTRUCTION FEATURES

☐ NAMEPLATE: _____ HP, RPM _____, S.F. 1.15

NEMA TORQUE DESIGN: ○ A ○ B ○ C ○ D

☐ NEMA LOCKED ROTOR kVA CODE LETER: _____

EFFICIENCY: ○ STANDARD ○ HIGH ○ PREMIUM

NOISE DESIGN: ○ STANDARD ○ LOW NOISE

○ MAX. SOUND PRESSURE LEVEL (dBA) @ 1 M: _____

☐ EXPECTED SPL (dBA) @ 1 M: _____

ENCLOSURE: ☐ TEFC ☐ TENV ☐ EXPLOSION PROOF

☐ "D"-FLAMEPROOF ☐ "P"-PRESSURIZED ☐ "E"-INCREASED SAFETY

MOUNTING: ☐ HORIZONTAL ☐ VERTICAL

☐ FOOT MOUNTED ☐ FLANGE MOUNTED

☐ SHAFT UP ☐ SHAFT DOWN

MAIN TERMINAL BOX MOUNTING LOCATION: ☐ F-1 ☐ F-2

FAN: ☐ REVERSIBLE ☐ UNI-DIRECTIONAL ● NON-SPARKING

BEARING TYPE: ☐ BALL ☐ ROLLER ☐ SLEEVE

BEARING LUBRICATION: ☐ GREASE ☐ RING OIL ○ OIL MIST

GREASE FITTING: ☐ PLUGGED ○ ALEMITE ○ OTHER _____

BEARING SHIELDING: ☐ SINGLE ☐ DOUBLE ○ SEALED FOR LIFE

MOTOR ROTATION: (FACING END OPPOSITE SHAFT EXTENSION)

☐ CW ☐ CCW ☐ BI-DIRECTIONAL

INSULATION CLASS: ○ B ● F ○ OTHER: _____

● NON-HYGROSCOPIC ○ TROPICALIZED

☐ TEMPERATURE RISE: *(DEFAULT IS 80°C ABOVE 40°C BY RES. @ 1.0)*

_____ °C ABOVE _____ °C BY: _____ @ _____ S.F.

☐ MOTOR TO BE "THERMALLY PROTECTED"

☐ MOTOR TO BE "OVER TEMP PROT- "

☐ TYPE "1 - "WINDING - RUNNING & LOCKED-ROTOR PROTECTED"

☐ TYPE "2 - "WINDING - RUNNING PROTECTED"

☐ TYPE "3 - "WINDING - PROTECTED, NON-SPECIFIC"

○ SPACEHEATER REQ'D. ☐ RATED AT: _____ WATTS

☐ VOLTS _____, PHASE _____, HERTZ _____

☐ MAX SHEATH TEMPERATURE: _____ °C

○ SEPARATE JUNCTION BOX FOR SPACE HEATER LEADS

MOTOR THRUST LOAD: ○ _____ N ○ NONE

DIRECTION OF THRUST: ○ TOWARD COUPLING ○ AWAY FROM COUPLING

☐ MOTOR THRUST RATING: _____ N

TESTING

IEEE TESTING: ○ OBSERVED ○ WITNESSED ○ SUBMIT CERT'D RESULTS

○ SPECIAL TESTING: _____

MISCELLANEOUS

PAINTING: ○ IEEE 841 STD. ○ OTHER: _____

○ _____

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ALLOWABLE PIPING FORCES & MOVEMENTS: (2.2.4)

	COMPRESSOR INLET		COMPRESSOR DISCHARGE		PACKAGE OUTLET	
	FORCE (N)	MOMENT (N-M)	FORCE (N)	MOMENT (N-M)	FORCE (N)	MOMENT (N-M)
AXIAL						
VERT.						
TRANS.						

ADDITIONAL DATA:

8.26.98 SHT 10 OF 10 SIDATASH.XLS

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APPENDIX B—NOMENCLATURE FOR EQUIPMENT

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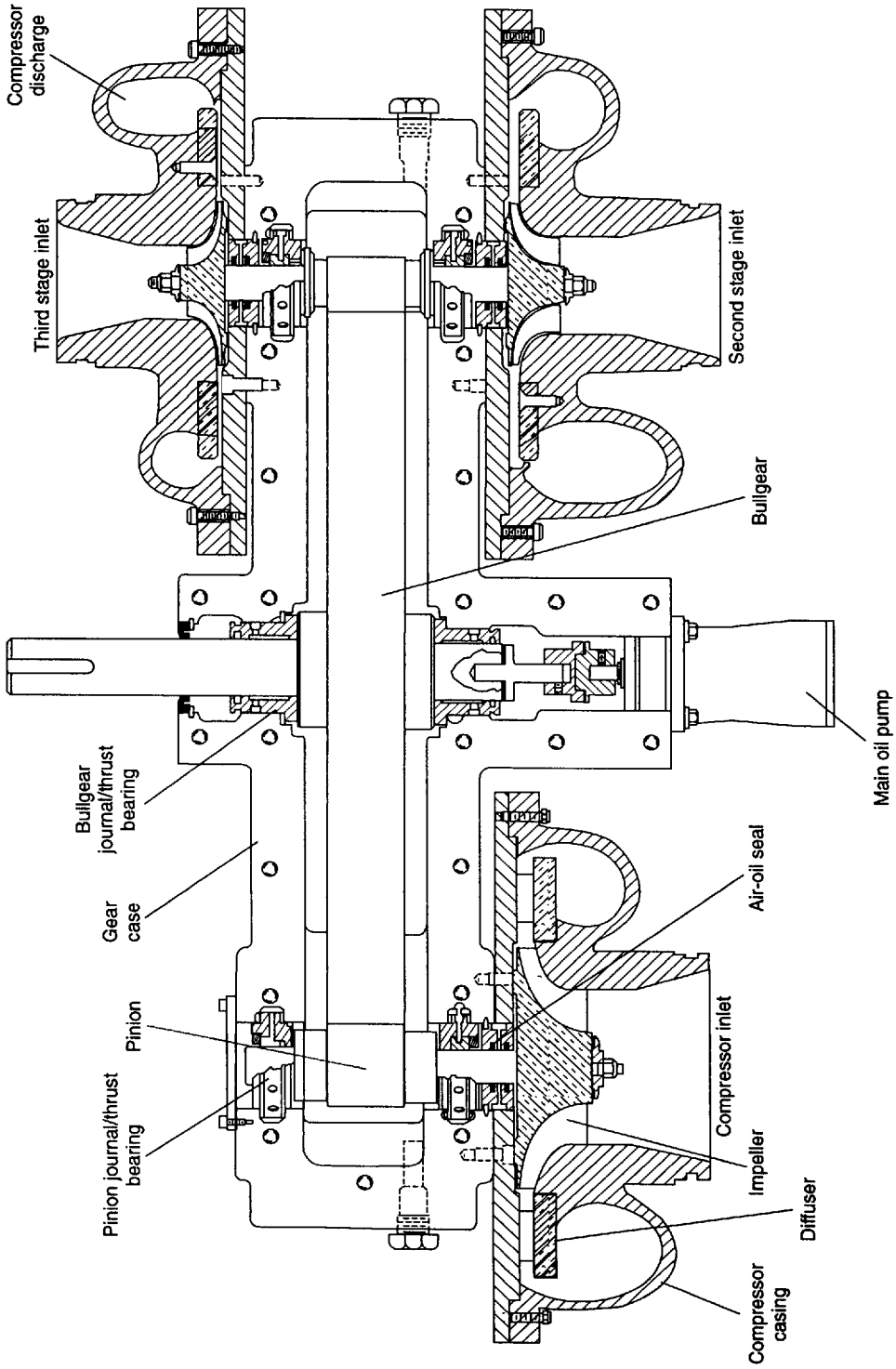


Figure B-1—Integrally Geared Centrifugal Air Compressor Nomenclature

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APPENDIX C—CRITICAL SPEEDS AND DAMPED UNBALANCED RESPONSE ANALYSIS

C.1 Critical Speeds

Note: For detailed discussion of rotordynamics refer to API Recommended Practice 684.

C.1.1 When the frequency of a periodic forcing phenomenon (exciting frequency) applied to a rotor-bearing support system coincides with a natural frequency of that system, the system may be in a state of resonance.

C.1.2 A rotor-bearing support system in resonance will have its normal vibration displacement amplified. The magnitude of amplification and the rate of phase-angle change are related to the amount of damping in the system and the mode shape taken by the rotor.

Note: The mode shapes are commonly referred to as the *first rigid (translatory or bouncing) mode*, the *second rigid (conical or rocking) mode*, and the *(first, second, third, . . ., nth) bending mode.*

C.1.3 When the rotor amplification factor (see Figure C-1), as measured at the shaft radial vibration probes, is greater than or equal to 2.5, that corresponding frequency is called *critical speed.* For the purposes of this standard, a critically damped system is one in which the amplification factor is less than 2.5.

C.1.4 An exciting frequency may be less than, equal to, or greater than the rotational speed of the rotor. Potential exciting frequencies that are to be considered in the design of rotor-bearing systems shall include, but are not limited to the following sources:

a. Unbalance in the rotor system.
b. Oil-film instabilities (whirl).
c. Internal rubs.
d. Blade, vane, nozzle, and diffuser passing frequencies.
e. Gear-tooth meshing and side bands.
f. Coupling misalignment.
g. Loose rotor-system components.
h. Hysteretic and friction whirl.
i. Boundary-layer flow separation.
j. Acoustic and aerodynamic cross-coupling forces.
k. Asynchronous whirl.

C.2 Lateral Analysis

C.2.1 When specified, the vendor shall provide a damped unbalanced-response analysis for each machine to assure acceptable amplitudes of vibration at any speed from zero to trip.

C.2.2 The damped unbalanced-response analysis shall include, but shall not be limited to the following considerations:

a. Support (base, frame, and bearing-housing) stiffness, mass, and damping characteristics, including effects of rotational speed variation. The vendor shall state the assumed support-system values and the basis for these values (for example, test of identical rotor support systems, assumed values).
b. Bearing lubricant-film stiffness and damping changes due to speed, load, preload, oil temperatures, accumulated assembly tolerances, and maximum-to-minimum clearances.
c. Rotational speed, including the various starting-speed detents, operating speed, and load ranges (including agreed-upon test conditions if different from those specified), trip speed, and coast-down conditions.
d. Rotor masses, including the mass moment of coupling halves, stiffness, and damping effects (for example, accumulated fit tolerances, fluid stiffening and damping, and frame and casing effects).
e. Asymmetrical loading (for example, partial arc admission, gear forces, and eccentric clearances).
f. The influence over the operating range of the calculated values for hydrodynamic stiffness and damping generated by the casing-end seals.

C.2.3 The damped unbalanced-response analysis shall include the following:

a. A plot and identification of the mode shape at each resonant speed (critically damped or not) from zero to trip, as well as the next mode occurring above the trip speed.
b. Frequency, phase, and response amplitude data (Bode Plots) at the vibration probe locations through the range of each critical speed, using the following arrangement of unbalance for the particular mode. This unbalance shall be sufficient to raise the displacement of the rotor at the probe locations to the vibration limit defined by the following equation:

In SI units,

$$L_v = 25.4\sqrt{\frac{12,000}{N}} \qquad \text{(C-1)}$$

In customary units:

$$L_v = \sqrt{\frac{12,000}{N}}$$

Where:

L_V = vibration limit (amplitude of unfiltered vibration), in micrometers (mils) peak to peak.
N = operating speed nearest the critical of concern, in revolutions per minute.

This unbalance shall be no less than two times the unbalanced defined by the following equation:

In SI units:

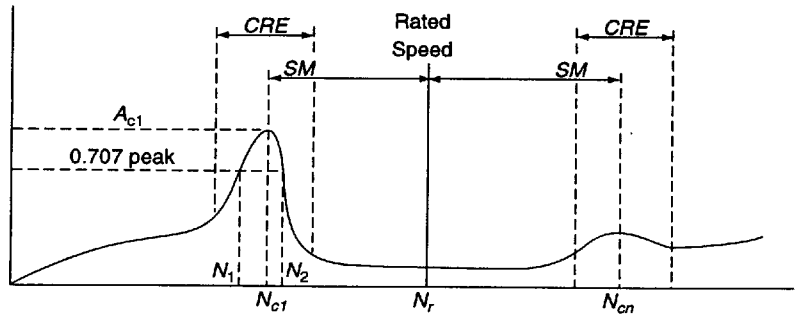
$$U = 6350W/N \qquad \text{(C-2)}$$

In customary units,

$$U = 4W/N$$

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N_{c1} = Rotor first critical, center frequency, cycles per minute.
N_{cn} = Critical speed, *n*th.
N_r = Rated speed.
N_1 = Initial (lesser) speed at 0.707 × peak amplitude (critical).
N_2 = Final (greater speed at 0.707 × peak amplitude (critical).
N_2—N_1 = Peak width at the half-power point.
AF = Amplification factor

$$= \frac{N_{c1}}{N_2 - N_1}$$

SM = Separation margin.
CRE = Critical response envelope.
A_{c1} = Amplitude at N_{c1}.
A_{cn} = Amplitude at N_{cn}.

Note: The shape of the curve is for illustration only and does not necessarily represent any actual rotor response plot.

Figure C-1—Rotor Response Plot

Where:

U = input unbalance from the rotor dynamic response analysis, in gram-millimeters (ounce-inches).

W = journal static weight load in kilograms (pounds) or for bending modes where the maximum deflection occurs at the shaft ends, the overhung weight load (that is the weight outboard of the bearing) in kilograms (pounds).

N = operating speed nearest the critical of concern, in revolutions per minute.

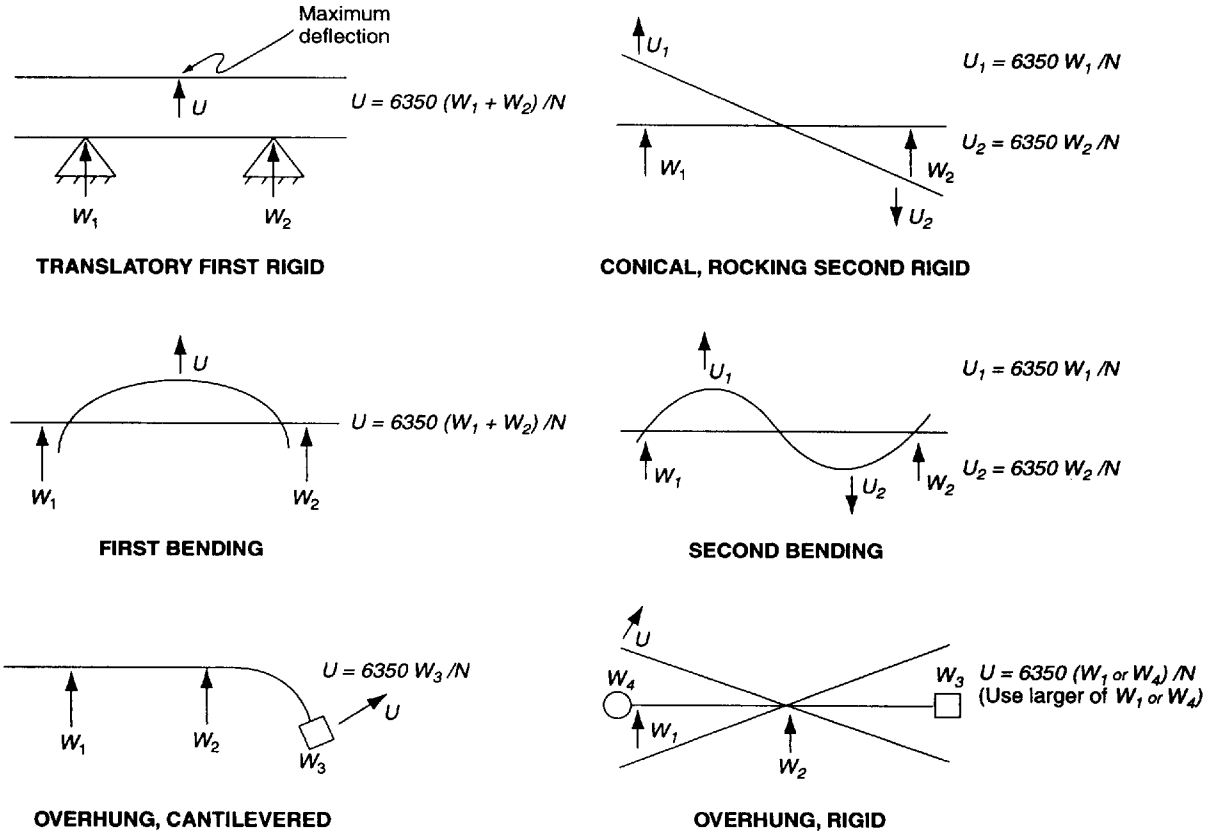
The unbalance weight or weights shall be placed at the locations that been analytically determined to affect the particular mode most adversely. For translatory modes, the unbalance shall be based on both journal static weights and shall be applied at the locations of maximum displacement. For conical modes, each unbalance shall be based on the journal weight and shall be applied at the location of maximum displacement of the mode nearest the journal used for the unbalanced calculation, 180 degrees out of phase. Figures C-2A and C-2B, show the typical mode shapes and indicate the location and definition of U for each of the shapes.

c. Modal diagrams for each response in the preceding Item b, indicating the phase and major-axis amplitude at each coupling engagement plane, the centerlines of the bearings, the locations of the vibration probes, and each seal area throughout the machine. The minimum design diametral running clearance of the seals shall also be indicated.

d. A verification test of the rotor unbalance, to establish the validity of the analytical model. A verification test of the rotor unbalance is required at the completion of the mechanical running test. Therefore, additional plots based on the actual unbalance to be used during the test shall be provided as follows: For machines that meet the requirements of C.2.3, Item b, and C.2.4, additional Bode plots, as specified in C.2.3, Item b, shall be provided. The location of the test unbalanced shall be determined by the vendor. The amount of unbalance shall be sufficient to raise the vibration levels, as measured at the vibration probes, to those specified in C.2.3, Item b. In all cases, the unbalance plots shall include the effects of any test-stand conditions (including the effects of test seals) that may be used during the verification test of the rotor unbalance (see C.3).

e. Unless otherwise specified, a stiffness map of the undamped rotor response from which the damped unbalanced-response analysis specified in Item c of this section was derived. This plot shall show frequency versus support-system stiffness with the calculated support-system stiffness curves superimposed.

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Note: See Section C.2.3 for definitions of the variables.

Figure C-2A—Typical Mode Shapes, SI Units

f. For machines whose bearing support system stiffness values are less than or equal to 3.5 times the bearing stiffness values, the calculated frequency-dependent support stiffness and damping values (impedences) or the values derived from modal testing. The results of the damped unbalanced-response analysis shall include Bode plots that compare absolute shaft motion relative to the bearing housing.

C.2.4 The damped unbalanced-response analysis shall indicate that the compressor in the unbalanced condition described in C.2.3, Item b, will meet the following acceptance criteria (see Figure C-1):

a. If the amplification factor is less than 2.5, the response is considered critically damped, and no separation margin is required.

b. If the amplification factor is 2.5–3.55, a separation margin of 15 percent above the rated speed and 5 percent below the rated speed is required.

c. If the amplification factor is greater than 3.55 and the critical response peak is below the rated speed, the required separation margin (a percentage of rated speed) is equal to the following:

$$SM = 100 - [84 + 6/(AF - 3)] \qquad \text{(C-3)}$$

Where:

SM = separation margin.

AF = amplification factor.

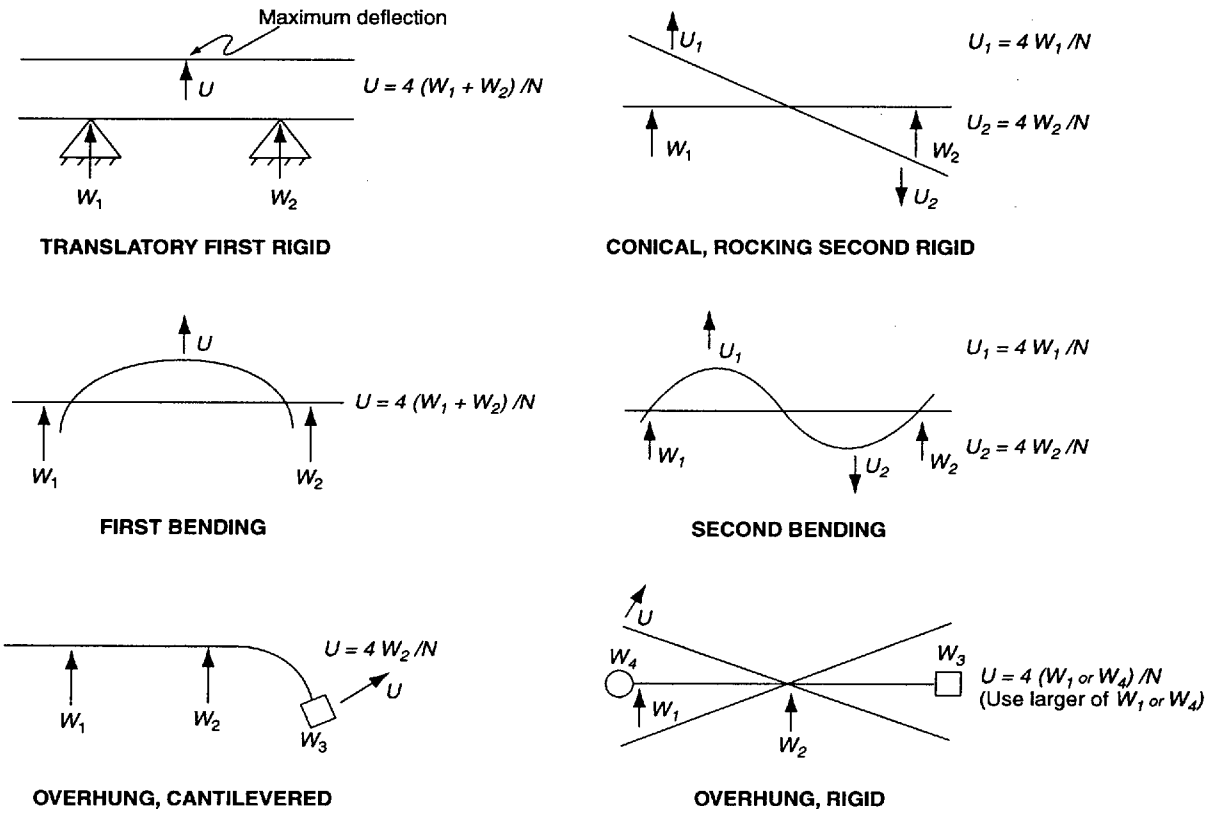
d. If the amplification factor is greater than 3.55, and the critical response peak is above the trip speed, the required separation margin (a percentage of rated speed) is equal to the following:

$$SM = [126 + 6/(AF - 3)] - 100 \qquad \text{(C-4)}$$

C.2.5 The calculated unbalanced peak-to-peak rotor amplitudes (see C.2.3, Item b) at any speed from zero to trip shall not exceed 75 percent of the minimum design diametral running clearances throughout the machine (with the exception of floating-ring seal locations).

C.2.6 If, after the purchaser and the vendor have agreed that all practical design efforts have been exhausted, the analysis indicates that the separation margins still can not be met, acceptable amplitudes shall be mutually agreed upon by the purchaser and the vendor, subject to the requirements of C.2.5.

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Note: See Section C.2.3 for definitions of the variables.

Figure C-2B—Typical Mode Shapes, U.S. Customary Units

C.3 Shop Verification of Unbalanced Response Analysis

C.3.1 A demonstration of rotor response at future unbalanced conditions is necessary because a well-balanced rotor may not be representative of future operating conditions (see C.2.3, Item d). This test shall be performed as part of the mechanical running test (see 4.3.4), and the results shall be used to verify the analytical model. Unless otherwise specified, the verification test of the rotor unbalance shall be performed only on the first rotor (normally the spare rotor, if two rotors are purchased). The actual response of the rotor on the test stand to the same unbalanced weight as was used to develop the Bode plots specified in C.2.3 shall be the criterion for determining the validity of the damped unbalanced-response analysis. To accomplish this, the following procedure shall be followed.

a. During the mechanical running test (see 4.3.4), the amplitudes and phase angle of the indicated vibration at the speed nearest the critical or criticals of concern shall be determined.

b. A trial weight, not more than one-half the amount calculated in C.2.3, Item b, shall be added to the rotor at the location specified in C.2.3, Item d, 90 degrees away from the phase of the indicated vibration at the speed closest to the critical or criticals of concern.

c. The machine shall then be brought up to the operating speed nearest the critical of concern, and the indicated vibration amplitudes and phase shall be measured. The results of this test and the corresponding indicated vibration data from Item a of the following shall be vectorially added to determine the magnitude and phase location of the final test weight required to produce the required vibration amplitudes.

d. The final test weight described in Item c of the following shall be added to the rotor, and the machine shall be brought up to the operating speed nearest the critical of concern. When more than one critical exists, additional test runs shall be performed for each, using the highest speed for the initial test run.

Note: It is recognized that the dynamic response of the unit on the test stand will be a function of the agreed-upon test conditions and that unless the test-stand results are obtained at the conditions of pressure, temperature, speed, and load expected in the field, they may not be the same as the results expected in the field.

C.3.2 The parameters to be measured during the test shall be speed and shaft synchronous (1x) vibration amplitudes

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with corresponding phase. The vibration amplitudes and phase from each pair of x-y vibration probes shall be vectorially summed at each response peak to determine the maximum amplitude of vibration. The major-axis amplitude of each response peak shall not exceed the limits specified in C.2.5. (More than one application of the unbalance weight and test run may be required to satisfy these criteria.)

The gain of the recording instrumentation used shall be predetermined and preset before the test so that the highest response peak is within 60 percent to 100 percent of the recorder's full scale on the test-unit coast-down (deceleration; see C.3.4). The major-axis amplitudes at the operating speed nearest the critical or criticals of concern shall not exceed the values predicted in accordance with C.2.3, Item d, before coast-down through the critical of concern.

C.3.2.1 Vectorial addition of slow-roll (300 revolutions per minute–600 revolutions per minute) electrical and mechanical runout is required to determine the actual vibration amplitudes and phase during the verification tests. Vectorial addition of the bearing-housing motion is required for machines that have flexible rotor supports (see C.2.3, Item f).

Note 1:The phase of each vibration signal, *x* or *y*, is the angular measure in degrees, of the phase difference (lag) between a phase reference signal (from a phase transducer sensing a once-per-revolution mark on the rotor, as described in API Standard 670) and the next positive peak in time of the synchronous (1*x*) vibration signal. (A phase change will occur through a critical or if a change in a rotor's balance condition occurs because of shifting or looseness in the assembly.)

Note 2: The major-axis amplitude is properly determined from lissajous (orbit) display on an oscilloscope or equivalent instrument. When the phase angle between the *x* and *y* signals is not 90 degrees, the major-axis amplitude can be approximated by $(x2 + y2)$ 0.5. When the phase angle between *x* and *y* signals is 90 degrees, the major-axis amplitude value is the greater of the two vibration signals.

C.3.2.2 The results of the verification test shall be compared with those from the original analytical model. The vendor shall correct the model if it fails to meet any of the following criteria:

a. The actual critical speeds shall not deviate from the predicted speeds by more than ± 5 percent.
b. The predicted amplification factors shall not deviate from the actual test-stand values by more than ± 20 percent.
c. The actual response peak amplitudes, including those that are critically damped, shall be within ± 50 percent of the predicted amplitudes.

C.3.3 Additional testing will be required if, from the test data just described or from the damped, corrected unbalanced response analysis (see C.3.2.2.), it appears that either of the following conditions exists:

a. Any critical response will fail to meet the separation margin requirements (see C.2.4).
b. The requirements of C.2.5 have not been met.

C.3.4 Rotors requiring additional testing per C.3.3 shall receive additional testing as follows: Unbalance weights shall be placed as described in C.2.3, Item b; this may require disassembly of the machine for placement of the unbalance weights. Unbalanced magnitudes shall be achieved by adjusting the indicated unbalance that exists in the rotor from the initial run to raise the displacement of the rotor at the probe location to the vibration limit defined by Equation C.1 (See C.2.3, Item b) at the rated speed; however, the unbalance used shall be no less than twice the unbalance limit specified in 2.2.7.4.2. The measurements from this test, taken in accordance with C.3.2, shall meet the following criteria:

a. At no speed outside the operating speed range, including the separation margins, shall the shaft deflections exceed 90 percent of the minimum design running clearances.
b. At no speed within the operating speed range, including the separation margins, shall the shaft exceed 55 percent of the minimum design running clearances or 150 percent of the allowable vibration limit at the probes (see C.2.3, Item b).

The internal deflection limits specified in C.3.4, Items a and b, shall be based on the calculated displacement ratios between the probe locations and the areas of concern identified in C.2.3, Item c. Actual internal displacements for these tests shall be calculated by multiplying these ratios by the peak readings from the probes. Acceptance will be based on these calculated displacements, or inspection of the seals if the machine is opened. Damage to any portion of the machine as a result of this testing shall constitute failure of the test. Minor internal rubs that do not cause clearance changes outside the vendor's new-part tolerance do not constitute damage.

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APPENDIX D—PROCEDURE FOR DETERMINATION OF RESIDUAL UNBALANCE

D.1 Scope

This appendix describes the procedure to be used to determine residual unbalance in machine rotors. Although some balancing machines may be set up to read out the exact amount of unbalance, the calibration can be in error. The only sure method of determining residual unbalance is to test the rotor with a known amount of unbalance.

D.2 Definition

Residual unbalance refers to the amount of unbalance remaining in a rotor after balancing. Unless otherwise specified, residual unbalance shall be expressed in g•mm or oz-in.

D.3 Maximum Allowable Residual Unbalance

D.3.1 The maximum allowable residual unbalance per plane shall be determined from the equation in 2.2.7.4.2 of this standard.

D.3.2 If the actual static weight load on each journal is not known, assume that the total rotor weight is equally supported by the bearings. For example, a two bearing rotor weighing 2700 kg (6000 lbs) would be assumed to impose a static weight load of 1350 kg (3000 lbs) on each journal.

D.4 Residual Unbalance Check

D.4.1 GENERAL

D.4.1.1 When the balancing machine readings indicate that the rotor has been balanced to within the specified tolerance, a residual unbalance check shall be performed before the rotor is removed from the balancing machine.

D.4.1.2 To check the residual unbalance, a known trial weight is attached to the rotor sequentially in six (or twelve, if specified by the purchaser) equally spaced radial positions, each at the same radius. The check is run in each correction plane, and the readings in each plane are plotted on a graph using the procedure specified in D.4.2.

D.4.2 PROCEDURE

D.4.2.1 Select a trial weight and radius that will be equivalent to between one and two times the maximum allowable residual unbalance [that is, if U_{max} is 1440 g•mm (2 oz-in.), the trial weight should cause 1440–2880 g•mm (2–4 oz-in.) of unbalance].

D.4.2.2 Starting at the last known heavy spot in each correction plane, mark off the specified number of radial positions (six or twelve) in equal (60 or 30 degree) increments around the rotor. Add the trial weight to the last known heavy spot in one plane. If the rotor has been balanced very precisely and the final heavy spot cannot be determined, add the trial weight to any one of the marked radial positions.

D.4.2.3 To verify that an appropriate trial weight has been selected, operate the balancing machine and note the units of unbalance indicated on the meter. If the meter pegs, a smaller trial weight should be used. If little or no meter reading results, a larger trial weight should be used. Little or no meter reading generally indicates that the rotor was not balanced correctly, the balancing machine is not sensitive enough, or a balancing machine fault exists (i.e., a faulty pickup). Whatever the error, it must be corrected before proceeding with the residual check.

D.4.2.4 Locate the weight at each of the equally spaced positions in turn, and record the amount of unbalance indicated on the meter for each position. Repeat the initial position as a check. All verification shall be performed using only one sensitivity range on the balance machine.

D.4.2.5 Plot the readings on the residual unbalance work sheet and calculate the amount of residual unbalance (see Figure D-1). The maximum meter reading occurs when the trial weight is added at the rotor's heavy spot; the minimum reading occurs when the trial weight is opposite the heavy spot. Thus, the plotted readings should form an approximate circle (see Figure D-2). An average of the maximum and minimum meter readings represents the effect of the trial weight. The distance of the circle's center from the origin of the polar plot represents the residual unbalance in that plane.

D.4.2.6 Repeat the steps described in D.4.2.1 through D.4.2.5 for each balance plane. If the specified maximum allowable residual unbalance has been exceeded in any balance plane, the rotor shall be balanced more precisely and checked again. If a correction is made in any balance plane, the residual unbalance check shall be repeated in all planes.

D.4.2.7 For stack component balanced rotors, a residual unbalance check shall be performed after the addition and balancing of the first rotor component, and at the completion of balancing of the entire rotor, as a minimum.

Note: This ensures that time is not wasted and rotor components are not subjected to unnecessary material removal in attempting to balance a multiple component rotor with a faulty balancing machine.

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Equipment (Rotor) No.: _____

Purchase Order No.: _____

Correction Plane (inlet, drive end, etc.—use sketch): _____

Balancing Speed: _____ rpm

N—Maximum Allowable Rotor Speed: _____ rpm

W—Weight of Journal (closest to this correction plane): _____ kg (lbs)

U_{max}—Maximum Allowable Residual Unbalance =
6350 *W/N* (4 *W/N*)
6350 ¥ _____ kg/_____ rpm; 4 ¥ _____lbs/_____rpm _____ g•mm (oz-in.)

Trial unbalance (2 ¥ U_{max}) _____ g•mm (oz-in.)

R—Radius (at which weight will be placed): _____ mm (in.)

Trial Unbalance Weight = Trial Unbalance/*R*

_____g•mm/_____mm/_____oz-in./_____in. _____ g (oz)

Conversion Information: 1 ounce = 28.350 grams

Test Data

Position	Trial Weight Angular Location	Balancing Machine Amplitude Readout
1		
2		
3		
4		
5		
6		
7		

Rotor Sketch

Test Date—Graphic Analysis

Step 1: Plot data on the polar chart (Figure D-1). Scale the chart so the largest and smallest amplitude will fit conveniently.

Step 2: With a compass, draw the best fit circle through the six points and mark the center of this circle.

Step 3: Measure the diameter of the circle in units of scale chosen in Step 1 and record. _____ units

Step 4: Record the trial unbalance from above. _____ g•mm (oz-in.)

Step 5: Double the trial unbalance in Step 4 (may use twice the actual residual unbalance). _____ g•mm (oz-in.)

Step 6: Divide the answer in Step 5 by the answer in Step 3. _____ Scale Factor

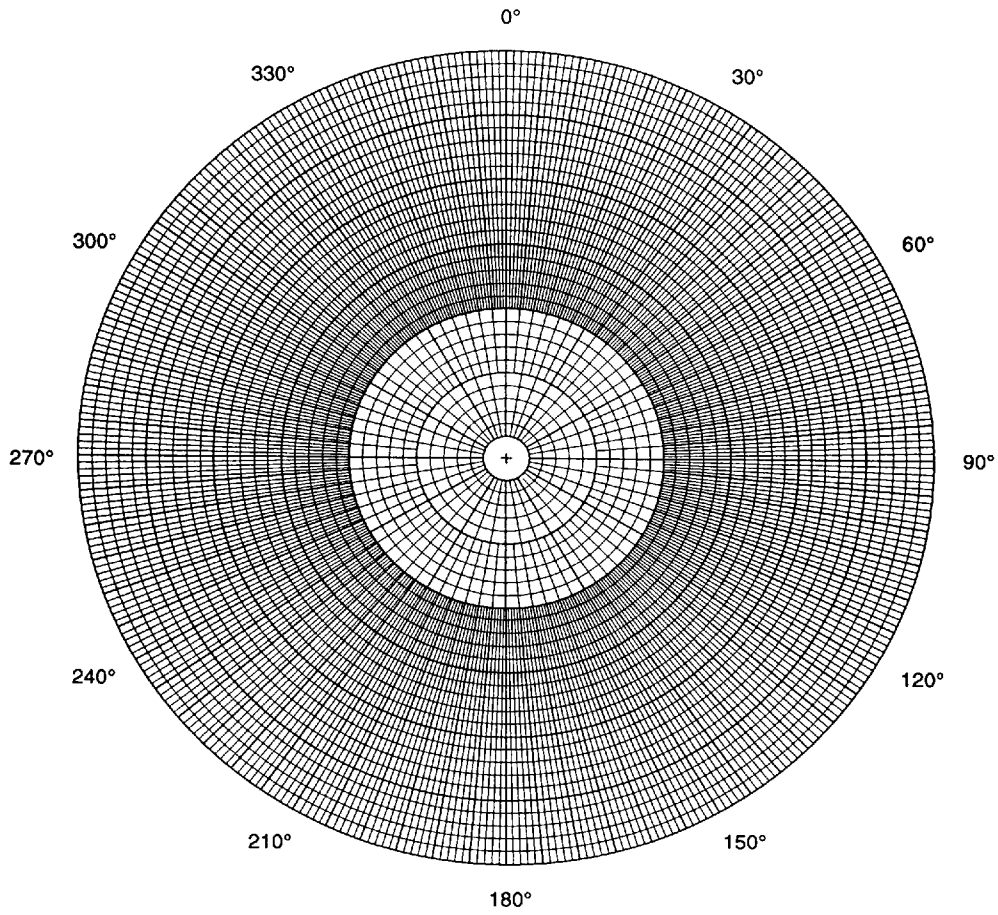
You now have a correlation between the units on the polar chart and the g•in of actual balance.

Notes:

1. The trial weight angular location should be referenced to a keyway or some other permanent marking on the rotor.
2. The balancing machine amplitude readout for position "7" should be the same as position "1," indicating repeatability. Slight variations may result from imprecise positioning of the trial weight.

Figure D-1—Residual Unbalance Work Sheet

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The circle you have drawn must contain the origin of the polar chart. If it doesn't, the residual unbalance of the rotor exceeds the applied test unbalance.

NOTE: Several possibilities for the drawn circle not including the origin of the polar chart are operator error during balancing, a faulty balancing machine pickup or cable, or the balancing machine is not sensitive enough.

If the circle does contain the origin of the polar chart, the distance between origin of the chart and the center of your circle is the actual residual unbalance present on the rotor correction plane. Measure the distance in units of scale you choose in Step 1 and multiply this number by the scale factor determined in Step 6. Distance in units of scale between origin and center of the circle times scale factor equals actual residual unbalance.

Record actual residual unbalance _____(g•mm)(oz-in.)

Record allowable residual unbalance (from Figure D-1) _____(g•mm)(oz-in.)

Correction plane _____ for Rotor No. _____ (has/has not) passed.

By _____ Date _____

Figure D-1—Residual Unbalance Work Sheet (Continued)

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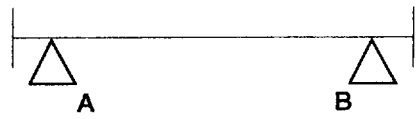
Equipment (Rotor) No.:	*C–101*	
Purchase Order No.:		
Correction Plane (inlet, drive end, etc.–use sketch):	*A*	
Balancing Speed:	*800*	rpm
N—Maximum Allowable Rotor Speed:	*10,000*	rpm
W—Weight of Journal (closet to this correction plane):	*908*	~~kg~~ (lbs)
U_{max}—Maximum Allowable Residual Unbalance = 6350 *W/N* (4 *W/N*)		
~~6350 ¥ _____ kg/_____ rpm~~; 4 ¥ *908* lbs/ *10,000* rpm	*0.36*	~~g•mm~~ (oz-in.)
Trial unbalance (2 ¥ U_{max})	*0.72*	~~g•mm~~ (oz-in.)
R—Radius (at which weight will be placed):	*6.875*	~~mm~~ (in.)
Trial Unbalance Weight = Trial Unbalance/*R*		
~~_____g•mm/_____mm~~/ *0.72* oz-in./ *6.875* in.	*0.10*	~~g~~ (oz)

Conversion Information: 1 ounce = 28.350 grams

Test Data

Position	Trial Weight Angular Location	Balancing Machine Amplitude Readout
1	*0*	*14.0*
2	*60*	*12.0*
3	*120*	*14.0*
4	*180*	*23.5*
5	*240*	*23.0*
6	*300*	*15.5*
7	*0*	*13.5*

Rotor Sketch



Test Date—Graphic Analysis

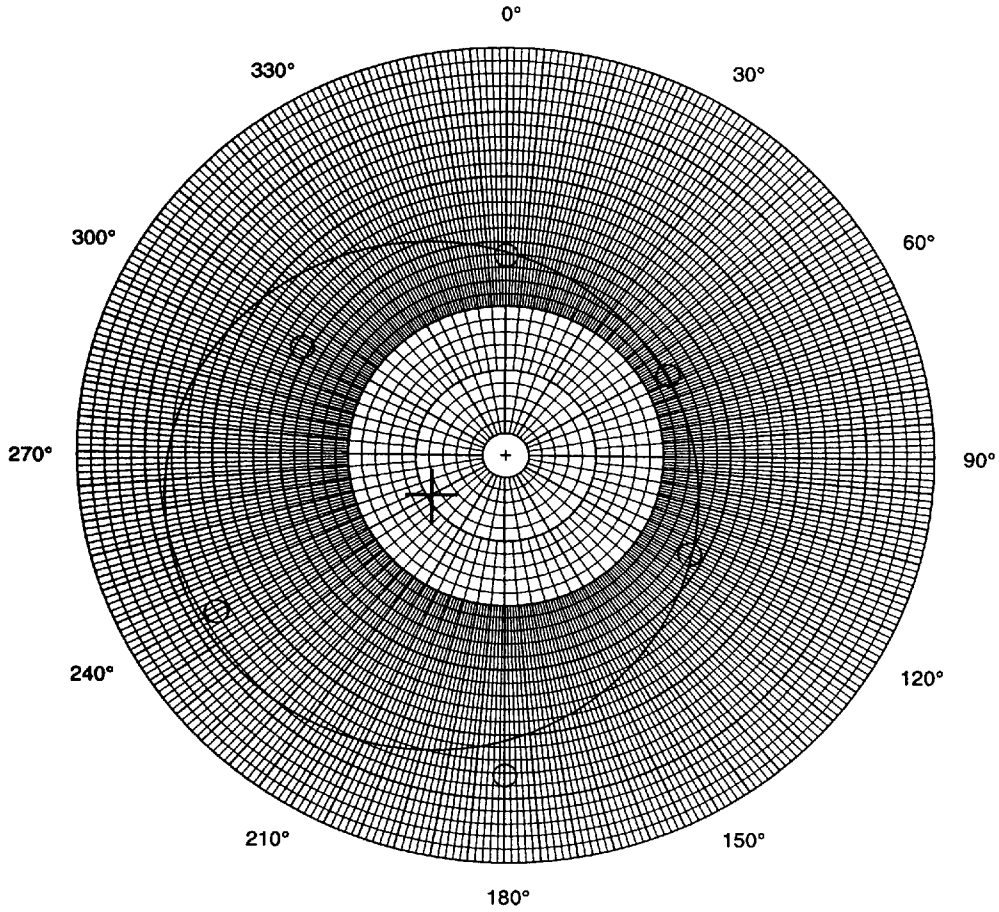
Step 1: Plot data on the polar chart (Figure D-2 continued). Scale the chart so the largest and smallest amplitude will fit conveniently.		
Step 2: With a compass, draw the best fit circle through the six points and mark the center of this circle.		
Step 3: Measure the diameter of the circle in units of scale chosen in Step 1 and record.	*35*	units
Step 4: Record the trial unbalance from above.	*0.72*	~~g•mm~~ (oz-in.)
Step 5: Double the trial unbalance in Step 4 (may use twice the actual residual unbalance).	*1.44*	~~g•mm~~ (oz-in.)
Step 6: Divide the answer in Step 5 by the answer in Step 3.	*0.041*	Scale Factor

You now have a correlation between the units on the polar chart and the g•in. of actual balance.

Notes:
1. The trial weight angular location should be referenced to a keyway or some other permanent marking on the rotor.
2. The balancing machine amplitude readout for position "7" should be the same as position "1," indicating repeatability. Slight variations may result from imprecise positioning of the trial weight.

Figure D-2—Sample Calculations for Residual Unbalance

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The circle you have drawn must contain the origin of the polar chart. If it doesn't, the residual unbalance of the rotor exceeds the applied test unbalance.

NOTE: Several possibilities for the drawn circle not including the origin of the polar chart are operator error during balancing, a faulty balancing machine pickup or cable, or the balancing machine is not sensitive enough.

If the circle does contain the origin of the polar chart, the distance between origin of the chart and the center of your circle is the actual residual unbalance present on the rotor correction plane. Measure the distance in units of scale you choose in Step 1 and multiply this number by the scale factor determined in Step 6. Distance in units of scale between origin and center of the circle times scale factor equals actual residual unbalance.

Record actual residual unbalance *6.5 (0.041) = 0.27* (g•mm)(oz-in.)

Record allowable residual unbalance (from Figure D-2) *0.36* (g•mm)(oz-in.)

Correction plane *A* for Rotor No. *C-101* (has/~~has not~~) passed.

By *John Inspector* Date *11-31-94*

Figure D-2—Sample Calculations for Residual Unbalance (Continued)

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APPENDIX E—MATERIAL SPECIFICATIONS FOR MAJOR COMPONENT PARTS

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Table E-1—Material Specifications for Major Component Parts

Part	Material	Specification[a]	Form	Temperature			
				Degrees Fahrenheit		Degrees Celsius	
				Minimum	Maximum	Minimum	Maximum
Compressor casings, scrolls (volutes), diaphragms	Cast iron	ASTM A 278	Cast	-20	500	-28.89	260
	Cast iron	ATSM A 48	Cast	-20	500	-28.89	260
	Ductile iron	ATSM A 395	Cast	-20	500	-28.89	260
	Ductile cast iron	ATSM A 536, Grade 60-40-18	Cast				
	Steel	ATSM A 27	Cast	-20	650	-28.89	343.33
Diffuser	Ductile cast iron	ATSM A 536, Grade 60-40-18	Cast				
	Carbon steel	ATSM A 516, Grade 60	Plate				
	Steel	Lead bearing, free machining	Plate	-20	500	-28.89	260
	Aluminum	ATSM B 26, Alloy 355.0	Cast	-320	400	-195.56	204.44
	Aluminum	ATSM B 209, Alloy 6061-T6	Plate				
	Aluminum	ATSM Alloy 443.0	Cast				
Stage inlet adapter	Cast iron	ATSM A 48, Class 30	Cast	-20	500	-28.89	260
	Cast iron	Meehanite GA	Cast	-20	500	-28.89	260
	Steel	ATSM A 53, Grade B, Type S	Wrought	-20	650	-28.89	343.33
	Steel	ATSM A 106	Wrought	-20	750	-28.89	398.89
	Steel	ATSM A 355, Grade B, Type S	Wrought				
Interstage piping	Cast iron	ATSM A 278, Class 30	Cast				
	Cast iron	ATSM A 278, Class 40	Cast	-20	500	-28.89	260
	Steel	ATSM A 53, Grade B, Type S	Wrought	-20	650	-28.89	343.33
	Steel	ATSM A 106	Wrought	-20	750	-28.89	398.89
Impellers	Stainless steel	ATSM A 564, Type 630	Bar/shape		750		398.89
	Stainless steel	AISI Type 410	Cast		750		398.89
	Stainless steel	AISI Type 403	Forged		750		398.89
	Stainless steel	SAE AMS 5353B	Cast		600		315.56
Pinion gears	Steel	AISI Type 4340	Bar stock		750		398.89
	Steel	AISI Type 8620	Forged		750		398.89
	Steel	AISI Type 9310	Forged		750		398.89
	Steel	AISI Type 4140	Forged		750		398.89
	Steel	AISI Type 4340	Forged		750		398.89
Bull-gear shaft	Steel	AISI Type 4140	Forged		750		398.89
	Steel	AISI Type 4340	Forged				
Bull-gear rim	Steel	AISI Type 4140	Forging				
	Steel	AISI Type 4340	Forging				
Gear casing	Cast iron	ATSM A 498, Class 40	Cast	-20	500	-28.89	260
	Cast iron	ATSM A 48, Class 30	Cast	-20	500	-28.89	260
	Cast iron	ATSM A 278, Class 40	Cast				
	Steel	AISI Type 1010 or 1020	Bar stock	-20	650	-28.89	343.33
Backplate	Steel	ASTM A 516, Grade 60	Plate				
Baseplate	Cast iron	ATSM A 278, Class 40	Cast	-20	500	-28.89	260
	Steel	AISI Type 1010 or 1020	Plate	-20	650	-28.89	343.33
	Steel	ASTM A 283, Grade D	Plate	-20	650	-28.89	343.33
Bolts[b]	Steel	ATSM A 193, Grade B7	Bar/forging				
	Steel	ATSM A 449, Grade 5	Forged	-20	650	-28.89	343.33
	Steel	ATSM A 307, Grade B	Forged	-20	500	-28.89	260
	Steel	ATSM A 307, Grade B	Bar stock				
	Steel	ATSM A 574	Bar stock				
	Steel	AISI Type 4140	Bar stock				
	Steel	SAE J429, Grade 8	Forged	-20	450	-28.89	232.22
Nuts[b]	Steel	ASTM A 108	Forged	-20	500	-28.89	260
	Steel	ASTM A 194, Grade 2H	Bar/forging				
	Steel	ASTM A 307, Grade B	Forged	-20	500	-28.89	260

[a]Description of AISI types can be found in ASTM E 527.

[b]For pressure joints.

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APPENDIX F—INSPECTOR'S CHECKLIST

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Appendix F—Inspector's Checklist

Item	Standard 672 Reference	Date Inspected	Inspected By	Status
Review of welder qualifications	2.2.10.4.1 3.5.1.15			
Package arrangement	2.1.12			
Piping	3.5			
Control panel	3.4.3			
Baseplate	3.3			
Casing connections	2.2.3			
Nameplates and rotation arrows	2.2.11			
Shaft runout	2.2.5.1.3 2.2.7.4.5			
Hardness checks	4.2.3.4			
Dimensional checks	4.2.3.5			
Dynamic balancing	2.2.7.4			
Impact testing	2.2.10.5.3			
Nondestructive testing	2.2.10.4.2 2.2.10.4.5.1 2.2.10.4.5.2 2.2.10.4.6.1 4.2.2			
Impeller overspeed test	4.3.3			
Hydrostatic test	4.3.2			
Mechanical/performance test	4.3.4			
Optional tests	4.3.5			
Preparation for shipment	4.4			

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APPENDIX G—PACKAGED INTEGRALLY GEARED CENTRIFUGAL AIR COMPRESSORS VENDOR DRAWING AND DATA REQUIREMENTS

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PACKAGED, INTEGRALLY GEARED CENTRIFUGAL AIR COMPRESSORS VENDOR DRAWING AND DATA REQUIREMENTS

JOB NO. _____ ITEM NO. _____
PURCHASE ORDER NO. _____ DATE _____
REQUISITION NO. _____ DATE _____
INQUIRY NO. _____ DATE _____
PAGE 1 OF 2 BY _____
FOR _____ REVISION _____
SITE _____ UNIT _____
SERVICE _____ NO. REQUIRED _____

Proposal[a] Bidder shall furnish _____ copies of data for all items indicated by an X.

Review[b] Vendor shall furnish _____ copies and _____ reproducibles of drawings and data indicated.

Final[c] Vendor shall furnish _____ copies and _____ reproducibles of drawings and data indicated.
Vendor shall furnish _____ installation, operating, and maintenance or technical data manuals.

DISTRIBUTION RECORD

Final—Received from vendor
Final—Due from vendor[c]
Review—Returned to vendor
Review—Received from vendor
Review—Due from vendor[c]

			DESCRIPTION						
			1. Certified dimensional outline drawing and list of connections.						
			2. Cross-sectional drawing and bill of materials.						
			3. Vendor recommendation for piping.						
			4. Control, alarm, and trip settings.						
			5. Grouting details.						
			6. Shaft-coupling assembly drawings and bills of materials.						
			7. Sealing air system schematics and bills of materials.						
			8. Foundation loading diagram including dimensions of baseplates.						
			9. Dimensional outline drawing for all major auxiliary equipment.						
			10. Lube-oil schematics and bills of materials.						
			11. Lube-oil system assembly and arrangement drawings.						
			12. Electrical, instrumentation, and control schematics and bills of materials..						
			13. Electrical and instrumentation arrangement drawings.						
			14. ISA data sheets for all instruments.						
			15. Control valve, relief valve, and orifice plate sizing calculations.						
			16. Tabulation of utility requirements.						
			17. Curves showing performance at rated conditions						
			18. Curves showing performance characteristics at other specified inlet conditions.						
			19. Curves showing the effects of inlet guide vanes at off-design conditions.						
			20. Lateral critical speed analysis report.						
			21. Torsional critical speed analysis report.						
			22. Transient torsional analysis for synchronous motors.						
			23. Allowable flange loadings for all customer connections.						
			24. Coupling alignment requirements.						
			25. Certified hydrostatic test logs						
			26. Mechanical running test logs.						
			27. Performance test logs.						
			28. Procedures for any special or optional tests.						

[a]Proposal drawings and data do not have to be certified or as built. Typical data shall be clearly identified as such.
[b]Purchaser will indicate in this column the time frame for submission of materials using the nomenclature given at the end of this form.
[c]Bidder shall complete these two columns to reflect his actual distribution schedule and include this form with his proposal.

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TYPICAL, INTEGRALLY GEARED CENTRIFUGAL AIR COMPRESSORS VENDOR DRAWING AND DATA REQUIREMENTS

JOB NO. _____ ITEM NO. _____
PAGE 2 OF 2 BY _____
DATE _____ REVISION _____

Proposal[a] Bidder shall furnish _____ copies of data for all items indicated by an X.

Review[b] Vendor shall furnish _____ copies and _____ reproducibles of drawings and data indicated.

Final[c] Vendor shall furnish _____ copies and _____ reproducibles of drawings and data indicated.
Vendor shall furnish _____ installation, operating, and maintenance or technical data manuals.

DISTRIBUTION RECORD

Final—Received from vendor
Final—Due from vendor[c]
Review—Returned to vendor
Review—Received from vendor
Review—Due from vendor[c]

			DESCRIPTION					
			29. Material certificates of compliance.					
			30. Rotor balancing logs.					
			31. Rotor combined mechanical and electrical runout.					
			32. As-built data sheets.					
			33. Installation material.					
			34. Operating and maintenance manuals.					
			35. Spare parts recommendations.					
			36. Progress reports and delivery schedule.					
			37. List of drawings.					
			38. Shipping list.					
			39. List of special tools furnished for maintenance.					
			40. Material Safety Data Sheets.					
			41. Preservation, packaging, and shipping procedures					
			42. Technical data manual.					

[a]Proposal drawings and data do not have to be certified or as-built. Typical data shall be clearly identified as such.
[b]Purchaser will indicate in this column the time frame for submission of materials using the nomenclature given at the end of this form.
[c]Bidder shall complete these two columns to reflect his actual distribution schedule and include this form with his proposal.

Notes:
1. Send all drawings and data to _____
2. All drawings and data must show project, appropriation, purchase order, and item numbers in addition to the plant location and unit. In addition to the copies specified above, one set of the drawings/instructions necessary for field installation must be forwarded with the shipment.

Nomenclature:
_____ S—number of weeks prior to shipment.
_____ F—number of weeks after firm order.
_____ D—number of weeks after receipt of approved drawings.

Vendor _____
Date _____ Vendor Reference _____
Signature _____
(Signature acknowledges receipt of all instructions)

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DESCRIPTION

The following numbered items correspond to the "Description" portion of the preceding "Packaged, Integrally Geared Centrifugal Air Compressors Vendor Drawing and Data Requirements."

1. Certified dimensional outline drawing(s) and list of connections, including the following:
   a. Size, type, rating, location, and identification of all customer connections.
   b. The weight of the package and approximate overall erection and maintenance handling weights of equipment and subassemblies that weigh more than 130 kilograms (300 pounds).
   c. Principal dimensions including overall package, maintenance clearances, dismantling clearances, and those required for the piping design.
   d. Shaft centerline height.
   e. Direction of rotation for the bull-gear shaft.
   f. The location of the center of gravity.
2. Cross-sectional drawing and bill of materials, including the following:
   a. Journal-bearing clearances and tolerances.
   b. Axial rotor float for all rotors.
   c. Axial position of impellers and tolerance allowed.
3. Vendor recommendation for piping, including requirements for straight length of air inlet piping or for straightening vanes where applicable.
4. Control, alarm, and trip settings (pressures and recommended temperatures).
5. Grouting details.
6. Shaft-coupling assembly drawings and bills of materials, including the following:
   a. The make, size, and type of the couplings.
   b. Mounting procedure.
   c. Shaft-end gap and tolerance.
   d. Coupling guards
7. Sealing air system schematics and bills of materials, including the following:
   a. Gas flows and control-valve (regulator) settings.
   b. Pipe and valve sizes.
   e. Instrumentation, safety devices, and control schemes.
   f. List of purchaser connections (if any).
8. Foundation loading diagram including dimensions of baseplates complete with the following:
   a. Diameter, number, and locations of bolt holes; thickness of the metal through which the bolts must pass; and recommended clearance.
   b. Weights and centers of gravity for major components.
9. Dimensioned outline drawing for all major auxiliary equipment to include all requirements in Item 1.
10. Lube-oil schematics and bills of materials, including the following:
   a. Oil flows and pressures at each level.
   b. Pipe and valve sizes.
   c. Instrumentation, safety devices, and control schemes.
11. Lube-oil system assembly and arrangement drawings, including size, rating, and location of all customer connections.
12. Electrical, instrumentation, and control schematics and bills of materials for all systems. The schematics shall show all control settings, alarm, and shutdown limits (set points). Drawings shall include, but not be limited to the following:
   a. Electrical one-line diagram.
   b. Elementary (schematic) wiring diagram.
   c. Interconnecting wiring/tubing diagrams.
   d. Conduit/wiring installation plans and details.
13. Electrical and instrumentation arrangement drawings, including junction box location drawing and lists of connections.
14. ISA data sheets for all instruments.
15. Control valve, relief valve, and orifice plate sizing calculations.
16. Tabulation of utility requirements (may be on as built purchaser data sheets).
17. Curves showing discharge pressure and brake horsepower plotted against delivered inlet flow at rated conditions. Performance curves shall indicate surge and rated capacity.
18. Curves showing performance characteristics at other specified inlet conditions.
19. Curve showing the effects of inlet quide vanes at off-design inlet conditions.
20. Lateral critical speed analysis report, including but not limited to the following:
   a. Complete description of the method used.
   b. Graphic display of critical speeds versus operating speed.
   c. Graphic display of bearing and support stiffness and its effect on critical speeds.
   d. Graphic display of rotor response to unbalance (including damping).
   e. Journal static loads.
   f. Stiffness and damping coefficients.
   g. Tilting-pad bearing geometry and configuration, including the following:
      1. Pad angle (arc) and number of pads.
      2. Pivot offset.
      3. Pad clearance (with journal radius, pad bore radius, and bearing-set bore radius).
      4. Preload.
21. Torsional critical speed analysis report, including but not limited to the following:

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a. Complete description of the method used.
b. Graphic display of the mass elastic system.
c. Tabulation identifying the mass moment and torsional stiffness of each component identified in the mass elastic system.
d. Graphic display of exciting forces versus speed and frequency.
e. Graphic display of torsional critical speeds and deflections (mode-shape diagram).
f. Effects of alternative coupling on analysis.

22. Transient torsional analysis of all units using synchronous motors.
23. Allowable flange loadings for all customer connections, referenced to a defined point.
24. Coupling alignment requirements, including recommended coupling limits during operation. Note all shaft-end position changes and support growth from a reference ambient temperature of 15°C (59°F) or another temperature specified by the purchaser. Include the recommended alignment method and cold setting targets.
25. Certified hydrostatic test logs.
26. Mechanical running test logs, including but not limited to the following:
a. Oil pressures and temperatures.
b. Vibration, including (where applicable) an x-y plot of amplitude versus revolutions per minute during start-up and coast-down.
27. Performance test logs and report.
28. Procedures for any special or optional tests (see 4.3.5).
29. Material certificates of compliance or mill test reports of items as agreed upon in the precommitment or pre inspection meetings.
30. Rotor balancing logs, including a residual unbalance report in accordance with Appendix D.
31. Rotor combined mechanical and electrical runout in accordance with 2.2.5.1.3.
32. As-built data sheets.
33. Installation manual describing the following (see 5.3.6.2):
a. Storage procedures.
b. Foundation plan.
c. Grouting details.
d. Setting equipment, rigging procedures, component weights, and lifting diagrams.
e. Dismantling clearances.
f. Written sequence for final tests and checks prior to initial start-up.
34. Operating and maintenance manuals describing the following (see 5.3.6.3):
a. Start-up.
b. Normal shutdown.
c. Emergency shutdown.
d. Operating limitations, other restrictions, and a list of undesirable speeds (see 2.2.7.1.3).
e. Lube-oil recommendations and specifications.
f. Routine operational procedures, including recommended inspection schedules and procedures.
g. Instructions for the following:
1. Disassembly and reassembly of rotor in casing.
2. Disassembly and reassembly of journal bearings.
4. Disassembly and reassembly of thrust bearing.
5. Disassembly and reassembly of seals (including maximum and minimum clearances).
6. Torquing procedures and torque values.
h. Performance data, including the following:
1. Curve showing certified test performance corrected for rated conditions.
2. Curve showing expected performance at other specified conditions including the effects of inlet guide vanes.
i. Vibration analysis data, per Items 20–22 of the preceding.
j. As-built data, including the following
1. As-built data sheets.
2. Hydrostatic test logs, per Item 25.
3. Mechanical running test logs, per item 27 above.
4. Rotor balancing logs, per Item 30.
5. Rotor mechanical and electrical runout at each journal, per Item 31.
6. Test logs of all specified optional tests.
k. Drawings and data, including the following:
1. Certified dimensional outline drawing and list of connections.
2. Cross-sectional drawing and bill of materials.
3. Lube-oil schematics and bills of materials.
4. Electrical and instrumentation schematics and bills of materials.
5. Control- and trip-system data.

35. Spare parts list with stocking level recommendations in accordance with 5.3.5.
36. Progress reports and delivery schedule, including vendor buy-outs and milestones (refer to 5.3.4).
37. List of drawings, including latest revision numbers and dates.
38. Shipping list, including all major components that will ship separately.
39. List of special tools furnished for maintenance (see 3.9).
40. Material Safety Data Sheets (OSHA Form 20).
41. Preservation, packaging and shipping procedures.
42. Technical data manual, including the following (see 5.3.6.4):
a. As-built purchaser data sheets, per Item 32.
b. Certified performance curves, per Items 17–19.
c. Drawings, in accordance with 5.3.2.
d. Spare parts list in accordance with 5.3.5.
e. Utility data, per Item 16.
f. Applicable reports, per Items 20, 21, 22.

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